

Supplement to “Group recovery after trimming, and level-free flagging, in robust clusterwise regression”

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S0 Contents

Throughout, *ESF* is the exact-subsample flagging procedure of Algorithm 1 of the paper, followed by the recovery step of Algorithm 2 unless “ESF alone” is written; “reweighted” or “rw” is the reweighting rule of the paper (Section 3 of the paper, applied twice, $c = 2.5$, a median absolute deviation per group); TCLUST-REG* denotes TCLUST-REG with 5% second trimming in the covariates. The analysis plans, the code, the raw results and the exact job scripts are in the repository <https://github.com/sorujov/eesp>.

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Tables S4–S11 give, for every design, the mean accuracy on clean units of every method at every fraction of outliers, with Monte Carlo standard errors. Columns 0.00–0.20 come from the first part of the study (50 data sets per cell), columns 0.25–0.35 from the second part (100 data sets per cell, new data); the column “0.30 (part 1)” repeats $\varepsilon = 0.30$ from the first part, which lay outside the criteria of Plan 1. The Laplace, contaminated normal and noise-component mixtures, added after Plan 1, appear in Tables S4–S11 but not in Table S1. “Trimmed alternation” is a simple trimmed least-squares alternation with the reweighting step, given the computing time of ESF alone without the covariate screen; it is reported for completeness only. A failed fit counts as accuracy 0; CTLE failed on 63 of the 2,000 data sets of the first part and 154 of the 2,400 of the second; TLE failed in 16 and 22 fits over all its levels, and the bisquare mixture in 8 and 2.

Analysis plans. Each plan was written as a plain-text file before the data it concerns were analysed (Plans 1 and 2) or generated (Plans 3 to 5), and served to fix the comparison and its thresholds before the run: Plan 1 for the first part, Plan 2 for the second, Plans 3 and 4 for the recovery step, on the stored fits and on 600 fresh data sets, and Plan 5 for three designs outside the Gaussian family; the third part is descriptive (Table S47). Table S34 gives the SHA-256 digest of each file as frozen; the files in the repository reproduce these digests. The file of Plan 5 carries one line added after freezing, which records that the code differs from that of Plan 4 only by an optional argument not used by the plan; the digest is that of the file without this line. Plans 3 to 5 also record the digests of the code they fix. The plans document what was fixed when; the main rival pipeline of the paper (long-search TCLUST-REG with reweighting and the recovery step), the three mixtures and sequential RANSAC, the best-of-five rule and the guard in the scoring scale were decided after all of them.

S1 Software and settings of the competitors

TCLUST-REG is the function `tclustreg` of the FSDA toolbox, a MATLAB toolbox. We ran it unmodified (GitHub commit 37f998e of 24 September 2026) under GNU Octave 9.4 with the `statistics` package (1.6.7 for the simulations, 1.6.3 for the data of Section 5), with three one-line substitutes for MATLAB functions that Octave lacks: a version check that returns false, `coder.target`, which returns true for “MATLAB”, and `convertStringsToChars`, which returns its arguments; they are in the repository. The runs with adaptive second-level trimming (Section S6) needed a fourth, for `any(x, 'all')`, which Octave’s `any` does not accept. We had no MATLAB licence and could not check that MATLAB gives the same results. Octave is slower than MATLAB, so the running times of TCLUST-REG are indicative only.

The trimmed cluster-weighted model, TLE, CTLE, the bisquare mixture and the Laplace mixture come from the R package `RobMixReg` 1.1.3; the trimmed cluster-weighted model was given 50 random starts and 20 concentration steps, more than its defaults, and the Laplace mixture is `mixLp` with `nit`= 20. The bisquare mixture `mixlinrb_bi` raised an error in our R session, because its single-start routine fits `lm(formula, data, weights = w)` with a formula whose environment does not contain `w`; we ran a copy of the two package functions in which only this line sets the formula environment first (`bisq_patched` in `r_methods.R`). The computation is otherwise the package’s. No package offers the contaminated normal and the noise-component mixtures of regressions, so the implementations are ours. The contaminated normal mixture is fitted by an ECM algorithm from 50 starts, each from K random elemental lines with the units far from all of them starting as bad points, with the inflation started at 20 and the ratio of the group variances bounded by 12; starts from random partitions, which we tried first, were captured by the outliers. The noise-component mixture is fitted by EM from 10 random partitions, the best continued after 10 iterations, with the ratio of the component variances bounded by 12, as in TCLUST-REG. Table S21 repeats the three mixtures with more starts.

A simple trimmed alternation of our own at the same five levels, with the reweighting step, completes the set that the comparison of Plan 1 used; it is a home-made baseline, so it appears in the Supplement only, although it is the best competitor in some cells of Table S1.

S2 Proofs

This section proves the results of Section 3 of the paper: Proposition 1 and the guarantee for a vote within one stage (S2.0, S2.1), Proposition 2 and its general version (S2.2), and Proposition 3 together with the result on the local peak test (S2.3). Each result is for a fixed input or fixed parameters. None covers the passage from one stage of ESF to the next, the weighted vote, or the search for the candidate line of the recovery step; the remarks that follow each proof say what it covers and what it does not.

S2.0 Proposition 1: precise statement and proof

Let $\mathcal{F}_{\ell-1}$ be the flagged set after stage $\ell - 1$, with $\mathcal{F}_0$ the covariate screen, $E_\ell = \{1, \dots, n\} \setminus \mathcal{F}_{\ell-1}$ the set from which stage ℓ draws, $N_\ell = |E_\ell|$, $\varepsilon_\ell = |E_\ell \cap O|/N_\ell$, and $\mathcal{H}_{\ell-1}$ the data together with everything drawn before stage ℓ . A replicate of stage ℓ is obtained by drawing uniform subsamples of size m from E_ℓ until an admissible one is found; let q_ℓ^c be the fraction of admissible subsamples among the clean subsamples of E_ℓ , and $q_\ell > 0$ the fraction among all subsamples of E_ℓ .

Proposition 1 (precise statement). Conditionally on $\mathcal{H}_{\ell-1}$, the subsamples of stage ℓ are independent, and each is uniform on the admissible subsamples of size m of E_ℓ , however many units two of them share. Let $p_\ell = \binom{N_\ell - |E_\ell \cap O|}{m} / \binom{N_\ell}{m}$ be the probability that a uniform subsample of E_ℓ is clean, and $a_\ell = q_\ell^c / q_\ell$. Conditionally on $\mathcal{H}_{\ell-1}$, each replicate of stage ℓ is clean with probability $\pi_\ell = p_\ell a_\ell$, and the stage contains a clean replicate with probability $1 - (1 - p_\ell a_\ell)^{B/L}$. Moreover $a_\ell \geq q_\ell^c$ and, when $|E_\ell \cap O| \leq N_\ell - m$, $\{1 - \varepsilon_\ell N_\ell / (N_\ell - m + 1)\}^m \leq p_\ell \leq (1 - \varepsilon_\ell)^m$.

The first statement. Fix a replicate of stage ℓ . Conditionally on $\mathcal{H}_{\ell-1}$ its proposals are independent uniform subsamples of size m of E_ℓ , because E_ℓ is determined by $\mathcal{H}_{\ell-1}$ and the draws use fresh randomness. (The implementation draws m units without replacement with equal weights on E_ℓ and zero weight elsewhere, which gives a uniform subsample of E_ℓ .) Whether a proposal is admissible is determined by the proposal and the data. The first admissible proposal in a sequence of independent proposals has the law of one proposal conditioned on admissibility, which is the uniform law on the admissible subsamples; the sequence stops almost surely because $q_\ell > 0$. Different replicates use disjoint randomness, so they are conditionally independent. $\square$

The rest. By the first statement, a replicate of stage ℓ is clean with conditional probability equal to the number of admissible clean subsamples divided by the number of admissible subsamples. Dividing both by $\binom{N_\ell}{m}$ gives $\pi_\ell = p_\ell q_\ell^c / q_\ell = p_\ell a_\ell$. The replicates of the stage are conditionally independent, which gives the probability that at least one is clean. Next, $a_\ell \geq q_\ell^c$ because $q_\ell \leq 1$. Finally $p_\ell = \prod_{j=0}^{m-1} (N_\ell - |E_\ell \cap O| - j) / (N_\ell - j)$; each factor equals $1 - |E_\ell \cap O| / (N_\ell - j)$, which lies between $1 - |E_\ell \cap O| / (N_\ell - m + 1)$ and $1 - |E_\ell \cap O| / N_\ell = 1 - \varepsilon_\ell$. $\square$

Remarks. The proposition is exact and elementary. What matters for stage ℓ is the contamination among the units still eligible: the flagged set lowers ε_ℓ when it contains outliers, and the probability of a clean replicate rises geometrically in m as it does so; the proposition does not say that earlier stages lower ε_ℓ . It does say where ESF breaks down: the first stage draws from the whole sample, less the covariate screen, and with $m = 8$ its probability of containing a clean replicate is about 0.84 at $\varepsilon = 0.20$, 0.45 at 0.30 and 0.28 at 0.35, and with $m = 12$ about 0.51, 0.13 and 0.06 (taking $a_\ell = 1$ and $p_\ell \approx (1 - \varepsilon)^m$); if it finds no clean replicate, its flags need not remove the outliers, and the breakdown observed from 25% to 30% of outliers in some designs is of this order.

S2.1 A vote within one stage

A vote within one stage. The same conditional independence gives a guarantee for a vote within one stage. Suppose a clean replicate gives unit i its correct label with probability at least $\bar{q}$, where the label refers to the least-squares partition of the clean eligible units. If $\pi_\ell \bar{q} > \frac{1}{2}$, a plurality vote of the replicates of stage ℓ labels every such unit correctly with conditional probability at least $1 - n \exp\{-2(B/L)(\pi_\ell \bar{q} - \frac{1}{2})^2\}$, whatever the contaminated replicates do (proved below). Since $\pi_\ell \approx (1 - \varepsilon_\ell)^m$, this requires

$$m < \frac{\log(2\bar{q})}{-\log(1 - \varepsilon_\ell)} \approx \frac{\log(2\bar{q})}{\varepsilon_\ell}, \quad (\text{S1})$$

while admissibility and the accuracy of a clean replicate require m of order at least $(d+1)/\pi_{\min}$. The subsample size must lie between the two bounds. As the contamination among the eligible units grows, the range of admissible m shrinks; each stage of flagging lowers that contamination and so widens the range again. The window can be empty: with $\bar{q}$ close to one, (S1) requires $\varepsilon_\ell < 1 - 2^{-1/m}$, which is 0.083 for $m = 8$ and 0.056 for $m = 12$, so at the contamination levels of our experiments the guarantee applies only at stages where flagging has already removed most of the outliers. This guarantee covers a single stage, not the weighted vote across stages that Algorithm 1 uses; it motivates the design but says nothing about the final output.

The vote within one stage. Let t_i be the label of unit i in the least-squares partition of the clean eligible units, labelled by the same pilot as the replicates, and suppose a clean replicate gives unit i the label t_i with conditional probability at least $\bar{q}$. Then a replicate of stage ℓ gives unit i the label t_i with conditional probability $Q_i \geq \pi_\ell \bar{q}$, whatever the contaminated replicates do, since the event that the replicate is clean and labels i correctly has probability π_ℓ times the conditional probability of a correct label given cleanness. By the first statement of Proposition 1, the number of the B/L replicates that give unit i the label t_i is binomial with success probability Q_i . If $Q_i \geq \frac{1}{2} + \gamma$ with $\gamma > 0$, the plurality differs from t_i only if at most half of the replicates vote for t_i , which by Hoeffding's inequality has probability at most $\exp\{-2(B/L)\gamma^2\}$. A union bound over at most n units, with $\gamma = \pi_\ell \bar{q} - \frac{1}{2}$, gives the statement above. The window (S1) is $(1 - \varepsilon_\ell)^m \bar{q} > \frac{1}{2}$ solved for m , with $a_\ell = 1$. $\square$

S2.2 The final reweighting step

Proposition 2 of Section 3 collects parts of Theorems S1 and S2 below.

The last two lines of Algorithm 1 apply a reweighting step to the parameter produced by the concentration steps. We study the step as a map on its own, applied to a fixed parameter $\theta = (\theta_1, \dots, \theta_K) \in \mathbb{R}^{Kd}$, and let the sample size grow. The step, as implemented, is the following. For unit i let $z_i(\theta) = \arg \min_k |y_i - x_i^\top \theta_k|$ be its nearest surface, the smallest index in case of a tie, and $r_i(\theta) = \min_k |y_i - x_i^\top \theta_k|$ its absolute residual from that surface. Let $n_k = \#\{i : z_i(\theta) = k\}$. The scale of group k is $\hat{s}_k = 1.4826 \hat{m}_k$, where $\hat{m}_k$ is a sample median of $\{r_i(\theta) : z_i(\theta) = k\}$, by which we mean any value between the two middle order statistics of that set; the implementation takes the square root of the median of the squared residuals, which lies between them. When $n_k < 10$ the median over all n residuals is used instead. The flagged set and the flagged fraction are

$$\mathcal{F}_n(\theta) = \{i : r_i(\theta) > c \hat{s}_{z_i(\theta)}\}, \quad \hat{\alpha}_n(\theta) = |\mathcal{F}_n(\theta)|/n.$$

The refitted parameter $\hat{\theta}_k$ is the least-squares fit of y on x over $\{i : z_i(\theta) = k, i \notin \mathcal{F}_n(\theta)\}$ when this set has at least d units, and $\hat{\theta}_k = \theta_k$ otherwise.

The population counterparts are defined for a probability measure P on $\mathcal{U} = \mathbb{R}^d \times \mathbb{R}$ and a unit $u = (x, y)$:

$$\begin{aligned} h_\theta(u) &= \arg \min_k |y - x^\top \theta_k|, & r_\theta(u) &= \min_k |y - x^\top \theta_k|, & A_k(\theta) &= \{u : h_\theta(u) = k\}, \\ m_k(\theta) &= \inf\{t \geq 0 : P(r_\theta \leq t \mid A_k(\theta)) \geq \tfrac{1}{2}\}, & s_k(\theta) &= 1.4826 m_k(\theta), \\ W_k(\theta) &= A_k(\theta) \cap \{r_\theta \leq c s_k(\theta)\}, & \alpha(\theta) &= P(r_\theta > c s_{h_\theta(\theta)}) = 1 - \sum_k P(W_k(\theta)), \\ M_k(\theta) &= \mathbb{E}[xx^\top \mathbf{1}_{W_k(\theta)}], & v_k(\theta) &= \mathbb{E}[xy \mathbf{1}_{W_k(\theta)}], & \theta_k^\infty(\theta) &= M_k(\theta)^{-1} v_k(\theta). \end{aligned}$$

Here $m_k(\theta)$ is the median of the residual r_θ among the units assigned to surface k , $s_k(\theta)$ the corresponding scale, $W_k(\theta)$ the set of units assigned to k and not flagged, $\alpha(\theta)$ the population flagged fraction, and $\theta^\infty(\theta)$ the population refit. The dependence on θ is dropped when θ is fixed.

Assumption S1 (Clean law). $P = (1 - \varepsilon)P_0 + \varepsilon C$ with $\varepsilon \in [0, 1)$. Under P_0 a unit is generated by a label $z \in \{1, \dots, K\}$ with $\mathbb{P}(z = k) = \pi_k > 0$, a covariate x whose conditional law given $z = k$ has $\mathbb{E}_0[\|x\|^2] < \infty$, and an error $e \sim \mathcal{N}(0, 1)$ independent of (x, z) , through $y = x^\top \theta_z^0 + \sigma_z e$ with $\sigma_k > 0$. The contaminating law C is arbitrary.

Assumption S2 (Contamination beyond the cut-off). At the given θ , there is $c' > c$ such that $C(r_\theta \geq c' s_{h_\theta(\theta)}) = 1$: every contaminated unit lies beyond c' scales from every surface of θ , the scale being that of the surface nearest to it.

The scale $s_k(\theta)$ in Assumption S2 is computed under P , so the assumption is a condition on P ; it is the form of “beyond the cut-off” that refers to the scale the step actually uses. In the designs of the study the gross outliers are shifted by 15 to 30 noise scales from their line (moderate outliers by 4 to 8), and uniform noise is redrawn until it lies more than three noise scales from both true lines, so at θ^0 the assumption holds for all or nearly all contaminated units whenever $s_k(\theta^0)$ is close to σ_k ; a shifted unit can still fall near another group’s line, and the assumption then fails for that unit.

Assumption S3 (Regularity at θ). *For every k : $P(A_k(\theta)) > 0$; the median of r_θ on $A_k(\theta)$ is positive and unique, that is, $m_k(\theta) > 0$, $P(r_\theta \leq t \mid A_k) < \frac{1}{2}$ for $t < m_k$ and $> \frac{1}{2}$ for $t > m_k$; and $M_k(\theta)$ is nonsingular.*

Lemma S1 (Plug-in cut-off). *Let $u_1, u_2, \dots$ be independent with law P , let k be fixed, let $\hat{s}_k$ be random with $\hat{s}_k \rightarrow s_k$ almost surely, and let $g : \mathcal{U} \rightarrow \mathbb{R}^p$ be measurable with $P(A_k, r_\theta = cs_k) = 0$ and $\mathbb{E}[\|g\| \mathbf{1}\{A_k, r_\theta \leq cs_k + \delta_0\}] < \infty$ for some $\delta_0 > 0$. Then, almost surely,*

$$\frac{1}{n} \sum_{i=1}^n g(u_i) \mathbf{1}\{z_i(\theta) = k, r_i(\theta) \leq c\hat{s}_k\} \rightarrow \mathbb{E}[g \mathbf{1}_{W_k}].$$

Proof. Fix $\delta \in (0, \delta_0)$. Almost surely $|c\hat{s}_k - cs_k| \leq \delta$ for all large n , and then the average on the left differs from $n^{-1} \sum_i g(u_i) \mathbf{1}\{z_i = k, r_i \leq cs_k\}$ by at most $n^{-1} \sum_i \|g(u_i)\| \mathbf{1}\{z_i = k, cs_k - \delta < r_i \leq cs_k + \delta\}$ in norm. By the strong law of large numbers the second average converges to $\mathbb{E}[g \mathbf{1}_{W_k}]$ and the bound to $\mathbb{E}[\|g\| \mathbf{1}\{A_k, cs_k - \delta < r_\theta \leq cs_k + \delta\}]$, both integrable by hypothesis. Letting $\delta \downarrow 0$ along a sequence, the last expectation tends to $\mathbb{E}[\|g\| \mathbf{1}\{A_k, r_\theta = cs_k\}] = 0$ by dominated convergence. $\square$

Theorem S1 (One reweighting step at a fixed input). *Let $u_1, \dots, u_n$ be independent with law P , fix θ , and let Assumptions S1–S3 hold at θ . Then, almost surely as $n \rightarrow \infty$:*

- (i) $\hat{s}_k \rightarrow s_k(\theta)$ for every k ;
- (ii) $\hat{\alpha}_n(\theta) \rightarrow \alpha(\theta) = \varepsilon + (1 - \varepsilon)\kappa(\theta)$, where $\kappa(\theta) = P_0(r_\theta > cs_{h_\theta}(\theta))$ is the flagged fraction among clean units;
- (iii) $\hat{\theta} \rightarrow \theta^\infty(\theta)$, and $\theta_k^\infty(\theta) = \mathbb{E}_0[xx^\top \mathbf{1}_{W_k}]^{-1} \mathbb{E}_0[xy \mathbf{1}_{W_k}]$: the limit of the refit is an average over the clean law alone, and the contamination enters it only through the scales $s_k(\theta)$ that define W_k .

Proof. *Step 1: no mass at the cut-off.* Under P_0 , y has a density conditionally on (x, z) , so $P_0(|y - x^\top \theta_j| = t) = 0$ for every j and t , and $P_0(r_\theta = t) \leq \sum_j P_0(|y - x^\top \theta_j| = t) = 0$. Under C , Assumption S2 gives $C(A_k, r_\theta < c's_k) = 0$. Hence $P(A_k, r_\theta = cs_k) = 0$, and for $\delta_0 = \frac{1}{2}(c' - c) \min_k s_k$, which is positive by Assumption S3, and every $g \geq 0$,

$$\mathbb{E}[g \mathbf{1}\{A_k, r_\theta \leq cs_k + \delta_0\}] = (1 - \varepsilon) \mathbb{E}_0[g \mathbf{1}\{A_k, r_\theta \leq cs_k + \delta_0\}]. \quad (\text{S2})$$

In particular $P(W_k) = (1 - \varepsilon)P_0(W_k)$, $M_k = (1 - \varepsilon)\mathbb{E}_0[xx^\top \mathbf{1}_{W_k}]$ and $v_k = (1 - \varepsilon)\mathbb{E}_0[xy \mathbf{1}_{W_k}]$; the factors $1 - \varepsilon$ cancel in θ_k^∞ , which proves the second claim of (iii).

Step 2: scales. By the strong law, $n_k/n \rightarrow P(A_k) > 0$, so $n_k \geq 10$ for all large n and the pooled fallback is inactive. For $t \geq 0$ put $\hat{F}_k(t) = n_k^{-1} \#\{i : z_i = k, r_i \leq t\}$ and $F_k(t) = P(r_\theta \leq t \mid A_k)$; then $\hat{F}_k(t) \rightarrow F_k(t)$ almost surely for each fixed t , as a ratio of two averages. Fix $\delta > 0$. By the uniqueness of the median, $F_k(m_k - \delta) < \frac{1}{2} < F_k(m_k + \delta)$, so for all large n , $\hat{F}_k(m_k - \delta) < \frac{1}{2} < \hat{F}_k(m_k + \delta)$: fewer than half of the residuals of group k are at most $m_k - \delta$ and more than half are at most $m_k + \delta$. Both middle order statistics, and hence $\hat{m}_k$, then lie in $(m_k - \delta, m_k + \delta]$. Taking $\delta = 1/j$, $j \geq 1$, gives $\hat{m}_k \rightarrow m_k$ almost surely, which is (i).

Step 3: flagged fraction. Apply Lemma S1 with $g \equiv 1$, the hypotheses holding by Step 1 and (i): $n^{-1} \#\{i : z_i = k, r_i \leq c\hat{s}_k\} \rightarrow P(W_k)$ for each k . Summing over k , $1 - \hat{\alpha}_n(\theta) \rightarrow \sum_k P(W_k) = 1 - \alpha(\theta)$. By Step 1, $\sum_k P(W_k) = (1 - \varepsilon) \sum_k P_0(W_k) = (1 - \varepsilon)(1 - \kappa(\theta))$, which rearranges to (ii).

Step 4: refit. Apply Lemma S1 with $g = xx^\top$ and with $g = xy$; the integrability hypothesis holds by (S2), since $\mathbb{E}_0\|x\|^2 < \infty$ and $\mathbb{E}_0 y^2 < \infty$ under Assumption S1. Thus $n^{-1} \sum_i x_i x_i^\top \mathbf{1}\{z_i = k, i \notin \mathcal{F}_n\} \rightarrow M_k$ and $n^{-1} \sum_i x_i y_i \mathbf{1}\{z_i = k, i \notin \mathcal{F}_n\} \rightarrow v_k$ almost surely. Since M_k is nonsingular,

$P(W_k) > 0$, so the number of unflagged units of group k exceeds d for all large n , the fallback is inactive, and $\hat{\theta}_k$ is the image of the two averages under the map $(M, v) \mapsto M^{-1}v$, which is continuous at (M_k, v_k) . This is (iii). $\square$

Theorem S1 identifies the limit of the step at any input satisfying the assumptions. It does not say that the input of the step in Algorithm 1 is close to θ^0 . The next theorem evaluates the limits at $\theta = \theta^0$, where they can be made explicit, and quantifies the effect of overlap between the groups. Write

$$\eta_k = P_0(h_{\theta^0} \neq k \mid z = k), \quad \eta = \sum_k \pi_k \eta_k = P_0(h_{\theta^0} \neq z),$$

for the probability that a clean unit of group k , or a clean unit, is nearer to another true surface than to its own. Let $p_k = P(A_k(\theta^0))$ and

$$\delta_k = 1 - \frac{(1 - \varepsilon)\pi_k(1 - \eta_k)}{p_k}, \quad \nu_k = \delta_k + \frac{1}{2}\eta_k,$$

so that δ_k is the fraction of the units assigned to surface k that are not clean units of group k ; it satisfies $\delta_k p_k \leq (1 - \varepsilon)\eta + \varepsilon$, since the clean units of other groups assigned to k have mass at most $(1 - \varepsilon)\eta$. Finally let

$$q(u) = 1.4826 \Phi^{-1}\left(\frac{1+u}{2}\right), \quad 0 < u < 1,$$

an increasing function with $q(\frac{1}{2}) = 1.4826 \Phi^{-1}(\frac{3}{4}) = 1.0000$ to four decimals; $q(u)\sigma$ is the scale the step assigns to a $\mathcal{N}(0, \sigma^2)$ residual whose u -quantile in absolute value is used in place of the median.

Theorem S2 (The step at the true parameter). *Let Assumption S1 hold, let $\theta = \theta^0$, and suppose that $P_0(x^\top(\theta_k^0 - \theta_j^0) = 0) = 0$ for $j \neq k$ (no two surfaces coincide on a set of covariates of positive mass), that $\nu_k < \frac{1}{2}$ for every k , and that Assumption S2 holds at θ^0 .*

(i) *For every k , $p_k > 0$ and the median of r_{θ^0} on $A_k(\theta^0)$ is positive and unique, and*

$$q(\tfrac{1}{2} - \nu_k) \leq \frac{s_k(\theta^0)}{\sigma_k} \leq q(\tfrac{1}{2} + \nu_k).$$

(ii) *The clean flagged fraction satisfies*

$$\sum_k \pi_k 2\Phi(-c q(\tfrac{1}{2} + \nu_k)) - \eta \leq \kappa(\theta^0) \leq \sum_k \pi_k 2\Phi(-c q(\tfrac{1}{2} - \nu_k)) + \eta,$$

$$\text{and } \alpha(\theta^0) = \varepsilon + (1 - \varepsilon)\kappa(\theta^0).$$

(iii) *Suppose in addition that $\|x\| \leq M_x$ P_0 -almost surely and that*

$$\lambda_k := \{2\Phi(c q(\tfrac{1}{2} - \nu_k)) - 1\} \lambda_{\min}(\mathbb{E}_0[xx^\top \mid z = k]) - M_x^2 \eta_k > 0.$$

Then $M_k(\theta^0)$ is nonsingular with $\lambda_{\min}(M_k(\theta^0)) \geq (1 - \varepsilon)\pi_k \lambda_k$, Assumption S3 holds at θ^0 , Theorem S1 applies, and

$$\|\theta_k^\infty(\theta^0) - \theta_k^0\| \leq \frac{2c s_k(\theta^0) M_x}{\pi_k \lambda_k} \eta.$$

In particular, if $\varepsilon = 0$, $\eta = 0$ and every $\mathbb{E}_0[xx^\top \mid z = k]$ is nonsingular, as in (iii), then $s_k(\theta^0) = q(\frac{1}{2})\sigma_k$, $\kappa(\theta^0) = 2\Phi(-c q(\frac{1}{2}))$ and $\theta^\infty(\theta^0) = \theta^0$: the true parameter is a fixed point of the population reweighting step, and the flagged fraction is $2\Phi(-c)$ up to the rounding in 1.4826.

Proof. Throughout, $h = h_{\theta^0}$, $r = r_{\theta^0}$, $A_k = A_k(\theta^0)$, $s_k = s_k(\theta^0)$ and $G_k(t) = \mathbb{P}(\sigma_k |e| \leq t) = 2\Phi(t/\sigma_k) - 1$.

Geometry of the assignment. For $j \neq k$ put $\Delta_j(x) = x^\top(\theta_k^0 - \theta_j^0)$, which is nonzero P_0 -almost surely. On $\{z = k\}$ we have $y - x^\top \theta_k^0 = \sigma_k e$ and $y - x^\top \theta_j^0 = \sigma_k e + \Delta_j$, and $|a| < |a + \Delta|$ if and only if $\Delta(2a + \Delta) > 0$. Hence, given x , the event $\{h = k\}$ is $\{-L(x) < \sigma_k e < U(x)\}$ with

$L(x) = \min\{\Delta_j/2 : \Delta_j > 0\}$ and $U(x) = \min\{-\Delta_j/2 : \Delta_j < 0\}$, the minimum over an empty set being $+\infty$; both are positive. Ties have probability zero. On $\{z = k, h = k\}$ the residual is $r = \sigma_k|e|$.

Composition of A_k . The mass of A_k splits into the clean units of group k with $h = k$, of mass $(1 - \varepsilon)\pi_k(1 - \eta_k)$, and the rest, of mass $\delta_k p_k$; the rest consists of clean units of other groups, of mass at most $(1 - \varepsilon)\sum_j \pi_j P_0(h \neq j \mid z = j) = (1 - \varepsilon)\eta$, and of contaminated units. Since $\delta_k \leq \nu_k < \frac{1}{2}$, $p_k > 0$ and $\eta_k < 1$. The conditional distribution function of r on A_k is therefore

$$F_k(t) = (1 - \delta_k)F_k^{\text{own}}(t) + \delta_k F_k^{\text{oth}}(t), \quad F_k^{\text{own}}(t) = \frac{P_0(\sigma_k|e| \leq t, h = k \mid z = k)}{1 - \eta_k},$$

with F_k^{oth} some distribution function. Since e is independent of (x, z) , $P_0(\sigma_k|e| \leq t \mid z = k) = G_k(t)$, and $P_0(\sigma_k|e| \leq t, h = k \mid z = k)$ lies between $G_k(t) - \eta_k$ and $G_k(t)$. Hence

$$(1 - \delta_k) \frac{G_k(t) - \eta_k}{1 - \eta_k} \leq F_k(t) \leq (1 - \delta_k) \frac{G_k(t)}{1 - \eta_k} + \delta_k. \quad (\text{S3})$$

(i) *Uniqueness.* By the geometry above, F_k^{own} has density

$$f_k^{\text{own}}(t) = \frac{\varphi(t/\sigma_k)}{\sigma_k(1 - \eta_k)} \mathbb{E}_0[\mathbf{1}\{t < U(x)\} + \mathbf{1}\{t < L(x)\} \mid z = k],$$

which is positive exactly when $P_0(\max\{L(x), U(x)\} > t \mid z = k) > 0$; this property is monotone in t , so $f_k^{\text{own}} > 0$ on $[0, T_k)$ and $f_k^{\text{own}} = 0$ on $[T_k, \infty)$ for some $T_k \in (0, \infty]$. If the median of F_k were not unique, F_k would equal $\frac{1}{2}$ on an interval of positive length. On $[0, T_k)$ the function F_k is strictly increasing, so the interval lies in $[T_k, \infty)$, where $F_k^{\text{own}} = 1$ and $F_k \geq 1 - \delta_k > \frac{1}{2}$, a contradiction. The median is positive because $F_k^{\text{own}}(0) = 0$ gives $F_k(0) \leq \delta_k < \frac{1}{2}$.

(i) *Bounds.* Let m_k be the median. If $F_k(t) \geq \frac{1}{2}$ then, by the upper bound in (S3), $G_k(t) \geq u_k^- := (\frac{1}{2} - \delta_k)(1 - \eta_k)/(1 - \delta_k)$, so $m_k \geq G_k^{-1}(u_k^-)$. If $G_k(t) \geq u_k^+ := \eta_k + (1 - \eta_k)/(2(1 - \delta_k))$ then, by the lower bound, $F_k(t) \geq \frac{1}{2}$, so $m_k \leq G_k^{-1}(u_k^+)$; here $0 < u_k^-$ and $u_k^+ < 1$ because $\delta_k < \frac{1}{2}$ and $\eta_k < 1$. Now $(\frac{1}{2} - \delta_k)/(1 - \delta_k) \geq \frac{1}{2} - \delta_k$ and $1/(2(1 - \delta_k)) \leq \frac{1}{2} + \delta_k$ for $\delta_k \leq \frac{1}{2}$, whence $u_k^- \geq (\frac{1}{2} - \delta_k)(1 - \eta_k) \geq \frac{1}{2} - \nu_k > 0$ and $u_k^+ \leq \eta_k + (1 - \eta_k)(\frac{1}{2} + \delta_k) \leq \frac{1}{2} + \nu_k < 1$. Since $G_k^{-1}(u) = \sigma_k \Phi^{-1}((1 + u)/2)$ is increasing and $s_k = 1.4826 m_k$, the bounds follow.

(ii). Split $\kappa(\theta^0) = P_0(r > cs_h)$ according to whether $h = z$. On $\{h = z = k\}$, $r = \sigma_k|e|$ and $s_h = s_k$, so $P_0(h = z, r > cs_h) = \sum_k \pi_k P_0(\sigma_k|e| > cs_k, h = k \mid z = k)$, which lies between $\sum_k \pi_k \{G'_k - \eta_k\}$ and $\sum_k \pi_k G'_k$ with $G'_k = \mathbb{P}(\sigma_k|e| > cs_k) = 2\Phi(-cs_k/\sigma_k)$. The event $\{h \neq z\}$ contributes between 0 and η . Inserting the bounds of (i) on s_k/σ_k into G'_k gives (ii); the expression for $\alpha(\theta^0)$ is Theorem S1(ii), whose Assumption S3 is verified in (iii) below, or directly from (S2).

(iii) *Nonsingularity.* By (S2), $M_k = (1 - \varepsilon)\mathbb{E}_0[xx^\top \mathbf{1}_{W_k}]$, and $W_k \supseteq \{z = k, h = k, \sigma_k|e| \leq cs_k\}$. In the positive semidefinite order,

$$\begin{aligned} \mathbb{E}_0[xx^\top \mathbf{1}\{z = k, h = k, \sigma_k|e| \leq cs_k\}] \\ = \mathbb{E}_0[xx^\top \mathbf{1}\{z = k, \sigma_k|e| \leq cs_k\}] - \mathbb{E}_0[xx^\top \mathbf{1}\{z = k, h \neq k, \sigma_k|e| \leq cs_k\}], \end{aligned}$$

where the first term equals $\pi_k G_k(cs_k)\mathbb{E}_0[xx^\top \mid z = k]$ by the independence of e and (x, z) , and the second is at most $M_x^\top \pi_k \eta_k I$ because $xx^\top \preceq \|x\|^2 I$. With $G_k(cs_k) = 2\Phi(cs_k/\sigma_k) - 1 \geq 2\Phi(cq(\frac{1}{2} - \nu_k)) - 1$ by (i), $\lambda_{\min}(M_k) \geq (1 - \varepsilon)\pi_k \lambda_k > 0$. The other two parts of Assumption S3 were shown in (i).

(iii) *Bias.* $\theta_k^\infty - \theta_k^0 = M_k^{-1}\mathbb{E}[x(y - x^\top \theta_k^0)\mathbf{1}_{W_k}]$, and by (S2) the expectation is $(1 - \varepsilon)$ times its P_0 -version. Split the latter by z . On $\{z = k\}$, $y - x^\top \theta_k^0 = \sigma_k e$ and $W_k \cap \{z = k\} = \{z = k, h = k, |e| \leq cs_k/\sigma_k\}$, so

$$\begin{aligned} \mathbb{E}_0[xe \mathbf{1}\{z = k, h = k, |e| \leq cs_k/\sigma_k\}] \\ = \mathbb{E}_0[x \mathbf{1}\{z = k\}] \mathbb{E}[e \mathbf{1}\{|e| \leq cs_k/\sigma_k\}] - \mathbb{E}_0[xe \mathbf{1}\{z = k, h \neq k, |e| \leq cs_k/\sigma_k\}]. \end{aligned}$$

The first term vanishes by the symmetry of e ; the second has norm at most $(cs_k/\sigma_k)M_x \pi_k \eta_k$, so the contribution of $\{z = k\}$ is at most $cs_k M_x \pi_k \eta_k$ in norm. On $\{z = j\}$, $j \neq k$, the factor $|y - x^\top \theta_k^0| = r \leq$

cs_k on W_k , so the contribution is at most $cs_k M_x P_0(z = j, h = k)$ in norm, and summing over $j \neq k$ gives at most $cs_k M_x \eta$. Altogether $\|\mathbb{E}[x(y - x^\top \theta_k^0) \mathbf{1}_{W_k}]\| \leq (1 - \varepsilon)cs_k M_x (\pi_k \eta_k + \eta) \leq 2(1 - \varepsilon)cs_k M_x \eta$, and dividing by $\lambda_{\min}(M_k) \geq (1 - \varepsilon)\pi_k \lambda_k$ gives the bound.

The special case. If $\varepsilon = 0$ and $\eta = 0$ then $\delta_k = \nu_k = 0$, so $s_k = q(\frac{1}{2})\sigma_k$ by (i), $\kappa = 2\Phi(-cq(\frac{1}{2}))$ by (ii), and the bias bound is zero. $\square$

Three comments. First, the constant in the limit of the flagged fraction is $2\Phi(-cs_k/\sigma_k)$ and not $2\Phi(-c)$. On the units assigned to surface k the minimum-over-surfaces residual is the residual from surface k itself, so the per-group median is a median of own-surface residuals; it is pulled down because the clean units of group k with the largest errors are nearer to another surface and are assigned there, and pulled up by the units of other groups and by the contamination assigned to surface k . Theorem S2(i) bounds the net effect by $\nu_k = \delta_k + \eta_k/2$, which is of the order of $\eta + \varepsilon$. The first effect is intrinsic to any scale computed within an assigned group and vanishes only when the groups do not overlap. Second, the refit is exactly unbiased at θ^0 when the groups do not overlap and moves by $O(\eta)$ when they do; the source is the asymmetric truncation of the error on $\{h = k\}$, not the contamination, which the step removes entirely under Assumption S2. Third, Algorithm 1 reports the flagged fraction recomputed at the refitted parameter, that is $\hat{\alpha}_n(\hat{\theta})$ rather than $\hat{\alpha}_n(\theta)$; Theorem S1 concerns the flag set the refit uses. The limit of the reported fraction is $\alpha(\theta^\infty(\theta))$ whenever α is continuous at $\theta^\infty(\theta)$ and the convergence of $\hat{\alpha}_n$ is uniform near it, which we do not prove.

Scope of Proposition 2 of the paper. The proposition is a law of large numbers for one application of the map at a fixed input. It says what the flagged fraction estimates under Gaussian errors, ε plus the fraction κ of clean units flagged, which is close to $2\Phi(-c) \approx 0.012$ when the groups overlap little, and that the contamination is removed entirely once it lies beyond the cut-off, leaving a bias of order η from the overlap of the groups. Two restrictions of Theorem S2 should be stated. First, the constant is $\nu_k = \delta_k + \eta_k/2$, where δ_k is the fraction of the units assigned to surface k that are not clean units of group k and η_k the overlap of group k ; it satisfies $\nu_k \leq \{(1 - \varepsilon)\eta + \varepsilon\}/p_k + \eta_k/2$ with p_k the mass assigned to surface k , so it is of order $\eta + \varepsilon$ divided by p_k and is not small for a small group. Second, the theorem requires $\nu_k < \frac{1}{2}$ for every group, a breakdown condition per group, and this fails for a small group with moderate contamination: with $\pi_k = 0.15$, $\varepsilon = 0.20$ and the contamination split evenly between the two assignment regions of two groups, δ_k is about 0.45 without overlap and ν_k exceeds $\frac{1}{2}$ once about a tenth of the group's clean units lie nearer to the other line. The theorem does not cover that regime. Three further things it does not cover should be stated. It does not show that ESF delivers an input close to θ^0 . The code applies the map more than once: once at the end of ESF, twice after TCLUS-REG in the reweighted rows of Section 4 of the paper, and three times to each candidate of the recovery step. And the reported flagged fraction $\hat{\alpha}$ is evaluated at the refitted parameter $\hat{\theta}$, whereas the proposition concerns the flags computed at the input θ ; the two limits agree under a continuity condition that we do not prove (the third comment above).

S2.3 The group-recovery step

Proposition 3 of the paper is for fixed parameters; the step reads them off the data, and the reweighting refit of the candidate is not covered. The local peak test guards against a slice of a band of gross outliers, which the likelihood would value like a group; for a residual density fixed in advance it is a concave maximisation whose solution is zero for a band that fills the window and the Gaussian share for a Gaussian group over a uniform background, as follows. Let $W = 3cs$, $u = 1/(2W)$, g_W the $\mathcal{N}(0, s^2)$ density truncated to $[-W, W]$, and for a residual density p on the window let $w^*(p)$ maximise $L_p(w) = \int p \log\{wg_W + (1 - w)u\}$ over $[0, 1]$.

Proposition S1. (i) L_p is strictly concave, so $w^*(p)$ is unique, and the EM iteration in w started at $\frac{1}{2}$ converges monotonically to it; the same holds for the sample version, whose maximiser converges to $w^*(p)$ almost surely for independent residuals of density p .

(ii) If $|r|$ is stochastically smaller under $\tilde{p}$ than under p , then $w^*(\tilde{p}) \geq w^*(p)$.

(iii) If $p = u$, then $w^*(p) = 0$; if p is uniform on $[-V, V]$, $V \leq W$, then w^* is nonincreasing in V .

(iv) If $p = w_0 g_W + (1 - w_0)u$, then $w^*(p) = w_0$.

The test thus passes a group whose units are at least half of those in the window and fails a slice of a band that fills it. Between the two, solving $L'_p(w) = 0$ for $c = 2.5$, a uniform band of half-width V has $w^* \geq \frac{1}{2}$ exactly when $V \leq 3.44s$: a band narrower than about seven noise scales passes the test and is left to the scale restriction and the likelihood. The implementation uses the untruncated normal density, which differs from g_W by the mass $2\Phi(-3c) \approx 6 \times 10^{-14}$ outside the window; (ii) and (iii) hold for it as stated. The proposition concerns a given residual density: the effect of the RANSAC selection on p is not analysed, and the fifty EM iterations of the implementation approximate the limit in (i).

Proposition 3 of the paper (precise statement). Let the units have law P .

- (i) Let G be an event with $P(G) = \pi_*$, on which $y = x^\top \theta_* + \sigma_* e$ with $e \sim \mathcal{N}(0, 1)$ independent of x given G . Let F have noise weight $\pi_0 > \pi_*$, and let F' be F with the line $(\theta_*, \sigma_*, \pi_*)$ added and the noise weight lowered to $\pi_0 - \pi_*$. Then (3) of the paper holds with $\xi = \mathbb{E}[\log\{(R/\pi_0)f_F(u)\} \mid G] \geq 0$, the extent to which F already explains G , and $\nu = \mathbb{E}[(\pi_0/R)f_F(u)^{-1}\mathbf{1}_{G^c}] \leq 1 - \pi_*$, the noise mass that F attributes to the units outside G ; a bound on ξ in terms of the distance of G from the lines of F is given below.
- (ii) Let S be an event with $P(S) = \pi_S$, on which $y = x^\top \theta_S + \sigma_S e$ as in (i). Let F contain the lines $\theta_S \pm \frac{\delta}{2}e_1$, e_1 the intercept coordinate, each with scale σ_S and proportion $\pi_S/2$, and let F'' replace them by the line $(\theta_S, \sigma_S, \pi_S)$, all else equal. With $a = \delta/(2\sigma_S)$, $-\varphi(1)a^2\pi_S \leq L(F'') - L(F) \leq (\varphi(1)a^2 + \frac{1}{4}a^4)\pi_S$. If the density r of the other components on S is replaced by tr and $t \downarrow 0$, the difference tends to π_S times the Kullback–Leibler divergence from $\mathcal{N}(0, 1)$ to $\frac{1}{2}\mathcal{N}(-a, 1) + \frac{1}{2}\mathcal{N}(a, 1)$, which lies in $[0, a^4/4]$.

Hence, if a candidate F_1 arises from the original F_0 by the two operations and the right side of (3) exceeds $\varphi(1)a^2\pi_S$, then almost surely $\ell_n(F_1) - \ell_n(F_0) > \frac{d+1}{2} \log n$ for all large n , and the candidate is accepted if its scales are admissible.

Proposition 3(i). Write $f = f_F$ and $f' = f_{F'}$, so that $f' = f - \pi_*/R + (\pi_*/\sigma_*)\varphi((y - x^\top \theta_*)/\sigma_*)$, and split $L(F') - L(F) = \mathbb{E}[(\log f' - \log f)\mathbf{1}_G] + \mathbb{E}[(\log f' - \log f)\mathbf{1}_{G^c}]$.

On G , $y - x^\top \theta_* = \sigma_* e$, and dropping the lines of F from f' ,

$$f' \geq \frac{\pi_*}{\sigma_*}\varphi(e) + \frac{\pi_0 - \pi_*}{R} = \frac{\pi_*}{\sigma_*}\varphi(e)(1 + \tau e^{e^2/2}), \quad \tau = \frac{(\pi_0 - \pi_*)\sigma_*\sqrt{2\pi}}{\pi_*R}.$$

Since $\mathbb{E} \log \varphi(e) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \mathbb{E} e^2 = -\frac{1}{2} \log(2\pi e)$,

$$\mathbb{E}[\log f' \mid G] \geq \log \frac{\pi_*}{\sigma_*} - \frac{1}{2} \log(2\pi e) + \rho, \quad \rho = \mathbb{E} \log(1 + \tau e^{e^2/2}) \geq 0.$$

For $v \geq 0$, $\log(1 + v) \leq 2\sqrt{v}$ (the difference vanishes at 0 and has derivative $v^{-1/2} - (1 + v)^{-1} > 0$), and $\mathbb{E} e^{e^2/4} = (1 - \frac{1}{2})^{-1/2} = \sqrt{2}$; hence $\rho \leq 2\sqrt{\tau} \mathbb{E} e^{e^2/4} = 2\sqrt{2\tau}$. Next, $f = (\pi_0/R)\{1 + (R/\pi_0) \sum_k (\pi_k/s_k) \varphi((y - x^\top \theta_k)/s_k)\}$, so $\mathbb{E}[\log f \mid G] = \log(\pi_0/R) + \xi$. By Jensen's inequality, $\xi \leq \log\{1 + (R/\pi_0) \sum_k \pi_k \mathbb{E}[s_k^{-1} \varphi((y - x^\top \theta_k)/s_k) \mid G]\}$, and on G , $y - x^\top \theta_k = x^\top (\theta_* - \theta_k) + \sigma_* e$ with e independent of x ; integrating over e first, the convolution of two centred normal densities gives $\mathbb{E}[s_k^{-1} \varphi((y - x^\top \theta_k)/s_k) \mid x, G] = \varphi_k(x^\top (\theta_* - \theta_k))$, where φ_k is the density of $N(0, s_k^2 + \sigma_*^2)$. Hence

$$\xi \leq \log\left\{1 + \frac{R}{\pi_0} \sum_k \pi_k \mathbb{E}[\varphi_k(x^\top (\theta_* - \theta_k)) \mid G]\right\},$$

the bound on ξ referred to in Proposition 3(i) of the paper: it is small when the lines of F are many scales $(s_k^2 + \sigma_*^2)^{1/2}$ away from G . Multiplying by $P(G) = \pi_*$, the first expectation is at least π_* times the brace in (3) plus $\pi_*\rho$; the statement drops $\rho \geq 0$, and the bound $\rho \leq 2\sqrt{2\tau}$ shows that the leading term is exact up to ξ and the two dropped nonnegative terms, ρ and the density of the lines of F on G in f' .

On G^c , $f' \geq f - \pi_*/R > 0$ because $f \geq \pi_0/R > \pi_*/R$, so $\log f' - \log f \geq -\log\{f/(f - \pi_*/R)\}$ and the second expectation is at least $-\Lambda$, $\Lambda = \mathbb{E}[\log\{f/(f - \pi_*/R)\}\mathbf{1}_{G^c}]$. With $t = (\pi_0/R)/f \in (0, 1]$ and $b = \pi_*/\pi_0 < 1$, $\log\{f/(f - \pi_*/R)\} = -\log(1 - bt)$; the function $t \mapsto -\log(1 - bt)$ is convex on $[0, 1]$ and vanishes at 0, so it is at most t times its value at 1, namely $t \log\{\pi_0/(\pi_0 - \pi_*)\}$. Taking expectations over G^c gives $\Lambda \leq \nu \log\{\pi_0/(\pi_0 - \pi_*)\}$, and $\nu \leq P(G^c)$ since $t \leq 1$. $\square$

Proposition 3(ii). On S , the residuals from the two lines are $\sigma_S e \mp \delta/2$, so their joint density is

$$A = \frac{\pi_S}{2\sigma_S} \{\varphi(e - a) + \varphi(e + a)\} = \frac{\pi_S}{\sigma_S} \varphi(e) e^{-a^2/2} \cosh(ae), \quad \text{against} \quad B = \frac{\pi_S}{\sigma_S} \varphi(e)$$

for the single line, and $\log(B/A) = a^2/2 - \log \cosh(ae)$. Let $r \geq 0$ be the density of the other components, the same in both fits. Off S the two densities coincide, and on S , $\log f_{F''} - \log f_F = \psi$ with $\psi = \log\{(B + r)/(A + r)\}$. Since $t \mapsto \log\{(B + t)/(A + t)\}$ is monotone on $[0, \infty)$ with limit 0, ψ lies between $\min\{\log(B/A), 0\}$ and $\max\{\log(B/A), 0\}$. Hence

$$-\pi_S D_-(a) \leq L(F'') - L(F) = \pi_S \mathbb{E}[\psi | S] \leq \pi_S D_+(a),$$

where $D_{\pm}(a) = \mathbb{E}[(\pm\{\frac{1}{2}a^2 - \log \cosh(ae)\})^+]$. Two inequalities for $x \in \mathbb{R}$: $\log \cosh x \leq x^2/2$, because both sides vanish at 0 and $\tanh x \leq x$ for $x \geq 0$; and $\log \cosh x \geq x^2/2 - x^4/12$, because the difference vanishes at 0 and its derivative $\tanh x - x + x^3/3$ is nondecreasing on $[0, \infty)$, its own derivative being $x^2 - \tanh^2 x \geq 0$. By the first, $D_-(a) \leq \mathbb{E}[(a^2 e^2/2 - a^2/2)^+] = \frac{1}{2} a^2 \mathbb{E}(e^2 - 1)^+ = \varphi(1)a^2$, since $\mathbb{E}(e^2 - 1)^+ = 2 \int_1^\infty (z^2 - 1)\varphi(z) dz = 2\varphi(1)$. Further $D_+(a) - D_-(a) = \mathbb{E} \log(B/A) = \mathbb{E} \log\{\varphi(e)/\frac{1}{2}(\varphi(e - a) + \varphi(e + a))\}$ is the Kullback–Leibler divergence stated, hence nonnegative, and by the second inequality it is at most $\mathbb{E}[a^4 e^4]/12 = a^4/4$. So $D_+(a) \leq \varphi(1)a^2 + a^4/4$, which gives the two bounds. Finally, if r is replaced by tr , then $\psi \rightarrow \log(B/A)$ pointwise as $t \downarrow 0$, dominated by $|\log(B/A)| \leq a^2 e^2/2 + a^2/2$, which is integrable; by dominated convergence the difference tends to $\pi_S \mathbb{E} \log(B/A)$. $\square$

Acceptance. If F_1 arises from F_0 by (i) and then (ii), then $L(F_1) - L(F_0)$ is at least the right side of (3) minus $\varphi(1)a^2\pi_S$, which is positive by assumption. By the strong law, applied to the bounded summands $\log f_{F_1}(u_i)$ and $\log f_{F_0}(u_i)$, $\{\ell_n(F_1) - \ell_n(F_0)\}/n \rightarrow L(F_1) - L(F_0) > 0$ almost surely, while $\frac{d+1}{2} \log n = o(n)$. $\square$

Proposition S1. On the window, g_W lies between $\varphi(3c)/(s\kappa)$ and $\varphi(0)/(s\kappa)$ with $\kappa = 1 - 2\Phi(-3c)$, and $u > 0$, so $\log\{wg_W + (1 - w)u\}$ is bounded on $[-W, W] \times [0, 1]$ and L_p is finite. Write $q_w = wg_W + (1 - w)u$.

(i). For fixed r , $w \mapsto \log q_w(r)$ is concave, and strictly so unless $g_W(r) = u$, which happens for at most two values of r , a p -null set. Hence L_p is strictly concave and has a unique maximiser. Its derivative $L'_p(w) = \int p(g_W - u)/q_w$ is continuous and strictly decreasing, so $w^* = \sup\{w \in [0, 1] : L'_p(w) \geq 0\}$, the supremum of the empty set being 0. The EM map is $M(w) = \int p w g_W / q_w$, and

$$M(w) - w = \int p \frac{w g_W - w q_w}{q_w} = w(1 - w)L'_p(w), \quad M'(w) = \int p \frac{g_W u}{q_w^2} > 0.$$

Thus M is increasing, its fixed points are 0, 1 and the zero of L'_p in $(0, 1)$ when there is one, and $M(w) > w$ exactly when $L'_p(w) > 0$, that is when $w < w^*$. Start at $w_0 = \frac{1}{2}$. If $w_0 < w^*$, then $w_0 < M(w_0) \leq M(w^*) = w^*$ when $w^* < 1$, and $M(w_0) \leq 1$ otherwise; by induction the iterates increase and stay at most w^* , so they converge to a fixed point in $(\frac{1}{2}, w^*]$, which is w^* . If $w_0 > w^*$, they decrease and stay at least w^* , and converge to a fixed point in $[w^*, \frac{1}{2})$, which is w^* when $w^* > 0$ and 0 when $w^* = 0$. The sample criterion $L_N(w) = N^{-1} \sum_{i \leq N} \log q_w(r_i)$ has the same properties, with p replaced by the empirical law, as soon as one r_i has $g_W(r_i) \neq u$, which is almost sure when the r_i have a density; the implemented iteration is its EM map, so the same argument applies to its maximiser $\hat{w}_N$. If the r_i are independent with density p , then $L_N(w) \rightarrow L_p(w)$ almost surely for each w in a countable dense subset of $[0, 1]$, by the strong law for bounded summands; the functions $w \mapsto \log q_w(r)$ have derivative bounded by $\max(g_W, u)/\min(g_W, u)$ uniformly in r , so the L_N are equi-Lipschitz and the convergence is uniform on $[0, 1]$; uniform convergence to a continuous function with a unique maximiser on a compact set gives $\hat{w}_N \rightarrow w^*(p)$.

(ii). For $t \geq 0$ put $h_w(t) = (g_W(t) - u)/(wg_W(t) + (1 - w)u)$. As a function of $g \in (0, \infty)$, $(g - u)/(wg + (1 - w)u)$ has derivative $u/(wg + (1 - w)u)^2 > 0$, and $g_W(t)$ is decreasing in t , so h_w is decreasing and bounded. Since $L'_p(w) = \mathbb{E}_p[h_w(|r|)]$, the stochastic order gives $L'_{\tilde{p}}(w) \geq L'_p(w)$ for every w , hence $\{w : L'_{\tilde{p}}(w) \geq 0\} \supseteq \{w : L'_p(w) \geq 0\}$ and $w^*(\tilde{p}) \geq w^*(p)$ by the characterisation in (i).

(iii). If $p = u$, then $L'_p(0) = \int u(g_W - u)/u = \int_{-W}^W g_W - 1 = 0$, so $L'_p(w) < 0$ for $w > 0$ and $w^* = 0$. With the untruncated normal density g in place of g_W the same computation gives $L'_p(0) = \kappa - 1 < 0$, and again $w^* = 0$. If p is uniform on $[-V, V]$, then $|r|$ is stochastically smaller for smaller V , and (ii) applies; the proof of (ii) uses only that the peak density is decreasing in $|r|$, so it holds with g in place of g_W as well.

(iv). If $p = w_0 g_W + (1 - w_0)u = q_{w_0}$, then $L'_p(w_0) = \int (g_W - u) = 0$, so w_0 is the zero of L'_p and $w^* = w_0$. $\square$

Numerical values. The constants quoted in Section 3 were obtained by solving $L'_p(w) = 0$ by quadrature and bisection with $s = 1$ and $c = 2.5$. For the uniform band of half-width V , w^* is 0, 0.08, 0.18, 0.34, 0.71 and 1 at $V = 7.5, 6, 5, 4, 3$ and 2.5 , and equals $\frac{1}{2}$ at $V = 3.44$. For a slice whose density on $|r| \leq cs$ exceeds that on the rest of the window by a factor $1 + \beta$, as the RANSAC selection tends to produce, w^* reaches $\frac{1}{2}$ at $\beta = 5.2$, that is when the central third of the window holds 76% of its units against 33% for a uniform slice. For a mixture of a Gaussian share $w_0 \in \{0.3, 0.5, 0.7\}$ and a uniform background the computed maximiser equals w_0 to six decimals, with g_W and with the untruncated density alike. The bounds of Proposition 3 were checked by quadrature and by Monte Carlo with 2×10^6 units: with $R/\sigma_* = 30$, $\pi_* = 0.1$, $\pi_0 = 0.15$ and two lines seven scales away, the mean gain per unit of G was 1.6695 against a leading term of 1.5768, $\rho = 0.0928$ and $\xi \leq 1.2 \times 10^{-4}$; with the lines three scales away the gain fell to 0.90 and the Jensen bound on ξ was 1.85 against a true value of 1.47. For the un-splitting, $D_+ - D_-$ and D_- were within their bounds at $a \in \{0.1, 0.25, 0.5, 1\}$, and with a noise floor present the mean change per unit of S at $a = 0.5$ was 0.016, inside $[-0.049, 0.061]$, against 0.012 for the divergence.

S3 Tables

Table S1: Comparison of Plan 1 (first part of the study): in each response design and level, the competitor with the highest mean accuracy, chosen afterwards, and the paired difference ESF alone minus that competitor with its standard error (50 data sets).

Design	ε	ESF alone	Best competitor		Difference	SE
D1	0.00	0.911	Bisquare mixture	0.912	-0.001	0.001
D1	0.05	0.907	TCLUST-REG, $\alpha = 0.10$, reweighted	0.908	-0.001	0.000
D1	0.10	0.913	CTLE	0.914	-0.001	0.000
D1	0.20	0.910	TCLUST-REG, $\alpha = 0.40$, reweighted	0.911	-0.001	0.000
D2	0.00	0.904	Least squares (50 starts)	0.904	-0.000	0.001
D2	0.05	0.901	Trimmed alternation (supplement), $\alpha = 0.10$	0.905	-0.003	0.005
D2	0.10	0.889	Trimmed alternation (supplement), $\alpha = 0.15$	0.902	-0.014	0.009
D2	0.20	0.901	Trimmed alternation (supplement), $\alpha = 0.25$	0.903	-0.002	0.008
D3	0.00	0.898	TLE, $\alpha = 0.10$	0.912	-0.014	0.009
D3	0.05	0.913	Trimmed alternation (supplement), $\alpha = 0.10$	0.913	+0.000	0.000
D3	0.10	0.901	TLE, $\alpha = 0.15$	0.908	-0.007	0.007
D3	0.20	0.885	TCLUST-REG, $\alpha = 0.25$	0.915	-0.029	0.014
D4	0.00	0.911	TLE, $\alpha = 0.05$	0.913	-0.002	0.001
D4	0.05	0.909	TLE, $\alpha = 0.25$	0.912	-0.003	0.001
D4	0.10	0.910	Bisquare mixture	0.912	-0.002	0.001
D4	0.20	0.904	TLE, $\alpha = 0.40$	0.908	-0.004	0.002
D6	0.00	0.912	TLE, $\alpha = 0.05$	0.913	-0.001	0.001
D6	0.05	0.910	Trimmed alternation (supplement), $\alpha = 0.05$	0.910	-0.000	0.000
D6	0.10	0.912	TCLUST-REG, $\alpha = 0.10$	0.912	-0.000	0.001
D6	0.20	0.895	TCLUST-REG, $\alpha = 0.25$	0.907	-0.012	0.009
D7	0.00	0.908	Bisquare mixture	0.909	-0.001	0.001
D7	0.05	0.910	Bisquare mixture	0.911	-0.001	0.001
D7	0.10	0.907	Trimmed alternation (supplement), $\alpha = 0.40$	0.909	-0.002	0.001
D7	0.20	0.908	Trimmed alternation (supplement), $\alpha = 0.25$	0.908	+0.000	0.001
D8	0.00	0.999	TCLUST-REG, $\alpha = 0.10$, reweighted	0.999	-0.000	0.000
D8	0.05	0.999	TCLUST-REG, $\alpha = 0.25$, reweighted	0.999	+0.000	0.000
D8	0.10	0.999	TCLUST-REG, $\alpha = 0.15$	0.999	-0.000	0.000
D8	0.20	0.999	TCLUST-REG, $\alpha = 0.25$, reweighted	0.999	+0.000	0.000

Table S2: Preliminary study (earlier version of ESF without the covariate screen, other data): mean accuracy on clean units with vote weight $\lambda = 3$, $\lambda = \infty$ (the best replicate alone) and $\lambda = 0$ (equal votes).

Design	ε	$\lambda = 3$	$\lambda = \infty$	$\lambda = 0$
K2-cross	0.00	0.909	0.909	0.907
K2-cross	0.05	0.912	0.912	0.910
K2-cross	0.10	0.912	0.912	0.912
K2-cross	0.20	0.909	0.911	0.907
K2-cross	0.30	0.901	0.915	0.882
K2-cross-hetero	0.00	0.911	0.911	0.910
K2-cross-hetero	0.05	0.907	0.907	0.907
K2-cross-hetero	0.10	0.910	0.910	0.910
K2-cross-hetero	0.20	0.911	0.911	0.903
K2-cross-hetero	0.30	0.912	0.912	0.874
K2-cross-leverage	0.00	0.913	0.914	0.909
K2-cross-leverage	0.05	0.911	0.911	0.907
K2-cross-leverage	0.10	0.850	0.837	0.855
K2-cross-leverage	0.20	0.532	0.531	0.551
K2-cross-leverage	0.30	0.529	0.529	0.529
K2-cross-unequal	0.00	0.904	0.912	0.853
K2-cross-unequal	0.05	0.902	0.910	0.857
K2-cross-unequal	0.10	0.908	0.911	0.873
K2-cross-unequal	0.20	0.901	0.908	0.880
K2-cross-unequal	0.30	0.898	0.904	0.861
K3-par	0.00	0.907	0.906	0.900
K3-par	0.05	0.902	0.903	0.901
K3-par	0.10	0.905	0.904	0.893
K3-par	0.20	0.887	0.887	0.878
K3-par	0.30	0.688	0.688	0.673

Table S3: Sensitivity of ESF alone to the flagging cut-off c (with the covariate screen at the 0.975 quantile) and to the quantile q of the covariate screen (with $c = 2.5$): mean accuracy on clean units on the data of the first part (50 data sets per cell). These runs use a different random seed from those of Tables S4–S11, so the column $c = 2.5$ differs from them by the seed-to-seed variation of ESF (up to 0.03 in D3). The values used in the paper are $c = 2.5$ and $q = 0.975$.

Design	ε	$c = 2$	$c = 2.5$	$c = 3$	$q = 0.95$	$q = 0.99$
D1	0.10	0.913	0.913	0.914	0.913	0.913
D1	0.20	0.910	0.903	0.902	0.910	0.903
D1	0.30	0.907	0.907	0.907	0.907	0.907
D3	0.10	0.902	0.903	0.908	0.903	0.903
D3	0.20	0.899	0.914	0.914	0.914	0.914
D3	0.30	0.845	0.877	0.888	0.887	0.877
D5	0.10	0.912	0.912	0.911	0.912	0.911
D5	0.20	0.835	0.843	0.859	0.913	0.698
D5	0.30	0.537	0.536	0.537	0.595	0.530
D6	0.10	0.911	0.912	0.911	0.912	0.912
D6	0.20	0.907	0.904	0.905	0.905	0.904
D6	0.30	0.699	0.682	0.668	0.668	0.686

Table S4: Design D1: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.910 (0.003)	0.910 (0.002)	0.911 (0.002)	0.907 (0.003)	0.910 (0.004)
ESF alone	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.910 (0.003)	0.910 (0.002)	0.911 (0.002)	0.907 (0.003)	0.910 (0.004)
Earlier version, no covariate screen (ablation)	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.903 (0.008)	0.910 (0.002)	0.903 (0.006)	0.892 (0.011)	0.870 (0.013)
TCLUST-REG, $\alpha = 0.05$	0.911 (0.003)	0.852 (0.019)	0.529 (0.003)	0.544 (0.005)	0.544 (0.003)	0.560 (0.004)	0.552 (0.006)	0.551 (0.004)
TCLUST-REG, $\alpha = 0.10$	0.910 (0.003)	0.907 (0.002)	0.852 (0.020)	0.529 (0.003)	0.530 (0.002)	0.540 (0.003)	0.541 (0.004)	0.539 (0.003)
TCLUST-REG, $\alpha = 0.15$	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.537 (0.008)	0.528 (0.002)	0.531 (0.002)	0.532 (0.004)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.25$	0.910 (0.002)	0.905 (0.002)	0.912 (0.002)	0.911 (0.003)	0.847 (0.014)	0.561 (0.010)	0.586 (0.019)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.40$	0.909 (0.002)	0.898 (0.006)	0.909 (0.002)	0.907 (0.003)	0.908 (0.002)	0.913 (0.002)	0.907 (0.002)	0.910 (0.005)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.910 (0.003)	0.848 (0.014)	0.561 (0.011)	0.586 (0.019)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.911 (0.002)	0.902 (0.006)	0.914 (0.002)	0.911 (0.003)	0.910 (0.002)	0.911 (0.002)	0.907 (0.003)	0.910 (0.005)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.911 (0.002)	0.907 (0.002)	0.914 (0.002)	0.911 (0.003)	0.910 (0.002)	0.911 (0.002)	0.907 (0.003)	0.910 (0.005)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.911 (0.003)	0.907 (0.002)	0.914 (0.002)	0.910 (0.003)	0.910 (0.002)	0.911 (0.002)	0.907 (0.003)	0.910 (0.005)
Trimmed CWM, $\alpha = 0.05$	0.884 (0.014)	0.839 (0.019)	0.611 (0.019)	0.562 (0.008)	0.585 (0.008)	0.617 (0.010)	0.605 (0.014)	0.634 (0.012)
Trimmed CWM, $\alpha = 0.10$	0.884 (0.014)	0.878 (0.012)	0.807 (0.022)	0.540 (0.004)	0.551 (0.006)	0.572 (0.009)	0.556 (0.006)	0.582 (0.008)
Trimmed CWM, $\alpha = 0.15$	0.891 (0.012)	0.884 (0.010)	0.883 (0.011)	0.580 (0.013)	0.542 (0.004)	0.547 (0.005)	0.541 (0.006)	0.548 (0.006)
Trimmed CWM, $\alpha = 0.25$	0.876 (0.013)	0.879 (0.012)	0.884 (0.011)	0.866 (0.015)	0.794 (0.016)	0.594 (0.011)	0.615 (0.017)	0.546 (0.005)
Trimmed CWM, $\alpha = 0.40$	0.762 (0.024)	0.740 (0.022)	0.798 (0.022)	0.823 (0.018)	0.845 (0.012)	0.853 (0.011)	0.871 (0.012)	0.859 (0.012)
TLE, $\alpha = 0.05$	0.911 (0.002)	0.869 (0.016)	0.548 (0.011)	0.574 (0.008)	0.584 (0.007)	0.610 (0.008)	0.587 (0.011)	0.621 (0.010)
TLE, $\alpha = 0.10$	0.911 (0.002)	0.899 (0.008)	0.875 (0.017)	0.532 (0.004)	0.543 (0.004)	0.558 (0.004)	0.563 (0.007)	0.573 (0.006)
TLE, $\alpha = 0.15$	0.911 (0.002)	0.908 (0.002)	0.913 (0.002)	0.561 (0.013)	0.538 (0.005)	0.534 (0.002)	0.543 (0.009)	0.545 (0.004)
TLE, $\alpha = 0.25$	0.911 (0.002)	0.905 (0.002)	0.914 (0.002)	0.910 (0.003)	0.861 (0.013)	0.592 (0.012)	0.615 (0.021)	0.533 (0.003)
TLE, $\alpha = 0.40$	0.910 (0.002)	0.897 (0.008)	0.910 (0.002)	0.906 (0.004)	0.909 (0.002)	0.906 (0.004)	0.906 (0.002)	0.912 (0.003)
CTLE	0.912 (0.003)	0.907 (0.002)	0.914 (0.002)	0.909 (0.003)	0.902 (0.006)	0.774 (0.023)	0.805 (0.028)	0.650 (0.025)
Bisquare mixture	0.912 (0.003)	0.907 (0.002)	0.895 (0.018)	0.901 (0.004)	0.879 (0.008)	0.761 (0.015)	0.781 (0.020)	0.727 (0.015)
Laplace mixture	0.912 (0.002)	0.906 (0.002)	0.914 (0.002)	0.906 (0.003)	0.908 (0.002)	0.910 (0.002)	0.903 (0.003)	0.913 (0.002)
Contaminated normal mixture	0.912 (0.002)	0.907 (0.002)	0.914 (0.002)	0.910 (0.003)	0.910 (0.002)	0.908 (0.004)	0.906 (0.003)	0.903 (0.007)
Noise-component mixture	0.912 (0.002)	0.907 (0.002)	0.914 (0.002)	0.910 (0.003)	0.845 (0.015)	0.639 (0.017)	0.610 (0.022)	0.551 (0.009)
Sequential RANSAC (true noise scale given)	0.911 (0.003)	0.907 (0.002)	0.913 (0.002)	0.910 (0.003)	0.909 (0.002)	0.911 (0.002)	0.905 (0.003)	0.914 (0.002)
Least squares (50 starts)	0.911 (0.002)	0.590 (0.013)	0.544 (0.005)	0.529 (0.003)	0.529 (0.002)	0.531 (0.002)	0.533 (0.004)	0.529 (0.002)
ESF, best of five runs	0.911 (0.002)	0.907 (0.002)	0.914 (0.002)	0.910 (0.003)	0.910 (0.002)	0.911 (0.002)	0.906 (0.003)	0.914 (0.002)
Trimmed alternation (supplement), $\alpha = 0.05$	0.911 (0.002)	0.896 (0.009)	0.607 (0.017)	0.529 (0.003)	0.528 (0.002)	0.531 (0.002)	0.533 (0.004)	0.527 (0.002)
Trimmed alternation (supplement), $\alpha = 0.10$	0.911 (0.002)	0.908 (0.002)	0.696 (0.010)	0.530 (0.003)	0.528 (0.002)	0.531 (0.002)	0.532 (0.004)	0.528 (0.002)
Trimmed alternation (supplement), $\alpha = 0.15$	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.600 (0.012)	0.538 (0.003)	0.531 (0.002)	0.532 (0.004)	0.528 (0.002)
Trimmed alternation (supplement), $\alpha = 0.25$	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.910 (0.003)	0.880 (0.010)	0.604 (0.013)	0.618 (0.019)	0.539 (0.003)
Trimmed alternation (supplement), $\alpha = 0.40$	0.911 (0.003)	0.907 (0.002)	0.914 (0.002)	0.910 (0.003)	0.910 (0.002)	0.911 (0.002)	0.907 (0.003)	0.912 (0.004)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.911 (0.002)	0.853 (0.019)	0.530 (0.004)	0.553 (0.006)	0.548 (0.004)	0.555 (0.004)	0.542 (0.004)	0.541 (0.003)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.911 (0.002)	0.908 (0.002)	0.851 (0.020)	0.529 (0.003)	0.529 (0.002)	0.534 (0.003)	0.532 (0.004)	0.529 (0.002)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.911 (0.002)	0.907 (0.002)	0.913 (0.002)	0.537 (0.008)	0.528 (0.002)	0.531 (0.002)	0.532 (0.004)	0.527 (0.002)

Table S5: Design D2: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.904 (0.003)	0.901 (0.005)	0.889 (0.009)	0.901 (0.007)	0.862 (0.010)	0.765 (0.018)	0.740 (0.024)	0.552 (0.019)
ESF alone	0.904 (0.003)	0.901 (0.005)	0.889 (0.009)	0.901 (0.007)	0.865 (0.009)	0.765 (0.018)	0.747 (0.023)	0.555 (0.019)
Earlier version, no covariate screen (ablation)	0.900 (0.005)	0.906 (0.002)	0.901 (0.006)	0.892 (0.010)	0.798 (0.013)	0.658 (0.018)	0.691 (0.023)	0.522 (0.016)
TCLUST-REG, $\alpha = 0.05$	0.882 (0.005)	0.797 (0.021)	0.529 (0.023)	0.369 (0.003)	0.369 (0.002)	0.367 (0.002)	0.369 (0.003)	0.368 (0.002)
TCLUST-REG, $\alpha = 0.10$	0.869 (0.007)	0.882 (0.008)	0.794 (0.017)	0.461 (0.021)	0.389 (0.008)	0.374 (0.005)	0.369 (0.003)	0.368 (0.002)
TCLUST-REG, $\alpha = 0.15$	0.874 (0.006)	0.882 (0.008)	0.890 (0.007)	0.597 (0.020)	0.496 (0.015)	0.437 (0.013)	0.403 (0.014)	0.384 (0.008)
TCLUST-REG, $\alpha = 0.25$	0.843 (0.010)	0.872 (0.010)	0.871 (0.011)	0.858 (0.013)	0.734 (0.012)	0.618 (0.012)	0.625 (0.015)	0.538 (0.015)
TCLUST-REG, $\alpha = 0.40$	0.762 (0.014)	0.770 (0.013)	0.795 (0.016)	0.776 (0.015)	0.780 (0.011)	0.782 (0.012)	0.761 (0.017)	0.726 (0.013)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.871 (0.009)	0.888 (0.008)	0.886 (0.008)	0.879 (0.010)	0.744 (0.012)	0.619 (0.012)	0.622 (0.015)	0.537 (0.015)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.799 (0.015)	0.796 (0.015)	0.833 (0.014)	0.818 (0.015)	0.816 (0.011)	0.788 (0.013)	0.764 (0.018)	0.703 (0.014)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.799 (0.015)	0.798 (0.015)	0.829 (0.015)	0.819 (0.014)	0.813 (0.011)	0.788 (0.013)	0.764 (0.018)	0.702 (0.014)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.867 (0.009)	0.874 (0.010)	0.888 (0.006)	0.888 (0.007)	0.887 (0.004)	0.885 (0.006)	0.876 (0.010)	0.812 (0.012)
Trimmed CWM, $\alpha = 0.05$	0.721 (0.016)	0.676 (0.018)	0.512 (0.016)	0.432 (0.012)	0.416 (0.008)	0.412 (0.008)	0.427 (0.011)	0.412 (0.008)
Trimmed CWM, $\alpha = 0.10$	0.722 (0.015)	0.697 (0.017)	0.633 (0.018)	0.515 (0.016)	0.452 (0.011)	0.426 (0.008)	0.441 (0.015)	0.406 (0.007)
Trimmed CWM, $\alpha = 0.15$	0.700 (0.016)	0.687 (0.015)	0.663 (0.016)	0.536 (0.015)	0.493 (0.012)	0.453 (0.011)	0.462 (0.016)	0.438 (0.010)
Trimmed CWM, $\alpha = 0.25$	0.700 (0.014)	0.660 (0.014)	0.668 (0.015)	0.651 (0.017)	0.584 (0.012)	0.543 (0.010)	0.519 (0.014)	0.529 (0.011)
Trimmed CWM, $\alpha = 0.40$	0.648 (0.013)	0.668 (0.015)	0.650 (0.012)	0.624 (0.014)	0.630 (0.010)	0.616 (0.010)	0.602 (0.014)	0.618 (0.009)
TLE, $\alpha = 0.05$	0.770 (0.022)	0.631 (0.032)	0.556 (0.017)	0.460 (0.017)	0.406 (0.009)	0.380 (0.004)	0.396 (0.011)	0.387 (0.004)
TLE, $\alpha = 0.10$	0.797 (0.019)	0.774 (0.021)	0.617 (0.030)	0.550 (0.018)	0.447 (0.014)	0.397 (0.008)	0.391 (0.009)	0.386 (0.004)
TLE, $\alpha = 0.15$	0.726 (0.022)	0.758 (0.021)	0.747 (0.022)	0.572 (0.017)	0.536 (0.013)	0.460 (0.014)	0.442 (0.015)	0.386 (0.008)
TLE, $\alpha = 0.25$	0.724 (0.022)	0.738 (0.021)	0.722 (0.021)	0.754 (0.026)	0.627 (0.022)	0.552 (0.013)	0.537 (0.019)	0.499 (0.012)
TLE, $\alpha = 0.40$	0.731 (0.019)	0.702 (0.019)	0.696 (0.021)	0.672 (0.018)	0.725 (0.015)	0.759 (0.017)	0.741 (0.023)	0.710 (0.023)
CTLE	0.779 (0.026)	0.691 (0.045)	0.782 (0.029)	0.588 (0.025)	0.503 (0.016)	0.450 (0.013)	0.454 (0.018)	0.403 (0.014)
Bisquare mixture	0.723 (0.028)	0.742 (0.030)	0.637 (0.020)	0.563 (0.020)	0.523 (0.014)	0.465 (0.012)	0.488 (0.018)	0.418 (0.009)
Laplace mixture	0.661 (0.016)	0.605 (0.017)	0.633 (0.017)	0.607 (0.023)	0.560 (0.014)	0.532 (0.014)	0.513 (0.022)	0.481 (0.012)
Contaminated normal mixture	0.862 (0.013)	0.824 (0.016)	0.720 (0.020)	0.523 (0.023)	0.412 (0.011)	0.388 (0.009)	0.376 (0.007)	0.375 (0.005)
Noise-component mixture	0.836 (0.016)	0.843 (0.016)	0.790 (0.019)	0.748 (0.025)	0.585 (0.019)	0.455 (0.011)	0.472 (0.016)	0.399 (0.006)
Sequential RANSAC (true noise scale given)	0.733 (0.013)	0.720 (0.013)	0.732 (0.012)	0.693 (0.016)	0.692 (0.011)	0.688 (0.012)	0.667 (0.015)	0.639 (0.012)
Least squares (50 starts)	0.904 (0.003)	0.466 (0.017)	0.369 (0.004)	0.369 (0.003)	0.369 (0.002)	0.367 (0.002)	0.369 (0.003)	0.368 (0.002)
ESF, best of five runs	0.905 (0.003)	0.906 (0.002)	0.904 (0.003)	0.911 (0.003)	0.871 (0.009)	0.756 (0.022)	0.724 (0.032)	0.450 (0.018)
Trimmed alternation (supplement), $\alpha = 0.05$	0.901 (0.003)	0.854 (0.016)	0.664 (0.008)	0.396 (0.011)	0.370 (0.002)	0.367 (0.002)	0.369 (0.003)	0.368 (0.002)
Trimmed alternation (supplement), $\alpha = 0.10$	0.899 (0.003)	0.905 (0.002)	0.816 (0.017)	0.633 (0.014)	0.468 (0.013)	0.388 (0.007)	0.373 (0.003)	0.368 (0.002)
Trimmed alternation (supplement), $\alpha = 0.15$	0.899 (0.003)	0.904 (0.002)	0.902 (0.003)	0.667 (0.007)	0.652 (0.006)	0.589 (0.013)	0.606 (0.018)	0.422 (0.012)
Trimmed alternation (supplement), $\alpha = 0.25$	0.895 (0.004)	0.898 (0.004)	0.899 (0.004)	0.903 (0.006)	0.784 (0.012)	0.667 (0.005)	0.674 (0.008)	0.664 (0.003)
Trimmed alternation (supplement), $\alpha = 0.40$	0.856 (0.011)	0.874 (0.010)	0.888 (0.006)	0.888 (0.007)	0.887 (0.004)	0.885 (0.006)	0.876 (0.010)	0.812 (0.012)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.894 (0.004)	0.813 (0.020)	0.530 (0.023)	0.369 (0.003)	0.369 (0.002)	0.367 (0.002)	0.369 (0.003)	0.368 (0.002)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.890 (0.004)	0.893 (0.007)	0.804 (0.017)	0.462 (0.021)	0.388 (0.008)	0.374 (0.005)	0.369 (0.003)	0.368 (0.002)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.890 (0.004)	0.894 (0.006)	0.898 (0.006)	0.598 (0.020)	0.497 (0.015)	0.437 (0.013)	0.403 (0.014)	0.384 (0.008)

Table S6: Design D3: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.903 (0.008)	0.913 (0.002)	0.901 (0.007)	0.900 (0.010)	0.901 (0.006)	0.881 (0.008)	0.873 (0.015)	0.840 (0.011)
ESF alone	0.898 (0.009)	0.913 (0.002)	0.901 (0.007)	0.885 (0.014)	0.888 (0.008)	0.881 (0.008)	0.869 (0.015)	0.840 (0.011)
Earlier version, no covariate screen (ablation)	0.895 (0.011)	0.905 (0.007)	0.908 (0.003)	0.915 (0.003)	0.906 (0.004)	0.888 (0.007)	0.893 (0.010)	0.807 (0.012)
TCLUST-REG, $\alpha = 0.05$	0.911 (0.002)	0.893 (0.009)	0.739 (0.012)	0.687 (0.008)	0.685 (0.006)	0.679 (0.006)	0.685 (0.008)	0.685 (0.006)
TCLUST-REG, $\alpha = 0.10$	0.911 (0.003)	0.911 (0.002)	0.885 (0.010)	0.699 (0.005)	0.705 (0.004)	0.692 (0.004)	0.694 (0.007)	0.696 (0.006)
TCLUST-REG, $\alpha = 0.15$	0.911 (0.003)	0.911 (0.002)	0.907 (0.003)	0.780 (0.015)	0.707 (0.004)	0.696 (0.004)	0.705 (0.006)	0.702 (0.004)
TCLUST-REG, $\alpha = 0.25$	0.891 (0.011)	0.911 (0.002)	0.907 (0.002)	0.915 (0.002)	0.886 (0.007)	0.763 (0.009)	0.778 (0.013)	0.710 (0.004)
TCLUST-REG, $\alpha = 0.40$	0.677 (0.021)	0.723 (0.023)	0.771 (0.023)	0.871 (0.014)	0.906 (0.003)	0.907 (0.002)	0.910 (0.002)	0.910 (0.004)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.888 (0.012)	0.912 (0.002)	0.907 (0.002)	0.915 (0.002)	0.882 (0.007)	0.751 (0.009)	0.765 (0.013)	0.703 (0.004)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.666 (0.022)	0.708 (0.025)	0.762 (0.024)	0.868 (0.017)	0.908 (0.002)	0.905 (0.003)	0.909 (0.004)	0.871 (0.008)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.910 (0.002)	0.912 (0.002)	0.907 (0.002)	0.910 (0.006)	0.908 (0.002)	0.905 (0.003)	0.909 (0.004)	0.871 (0.008)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.911 (0.002)	0.912 (0.002)	0.908 (0.002)	0.834 (0.022)	0.845 (0.014)	0.883 (0.009)	0.887 (0.013)	0.875 (0.008)
Trimmed CWM, $\alpha = 0.05$	0.896 (0.007)	0.864 (0.014)	0.778 (0.014)	0.699 (0.011)	0.704 (0.008)	0.685 (0.011)	0.691 (0.013)	0.679 (0.012)
Trimmed CWM, $\alpha = 0.10$	0.873 (0.012)	0.892 (0.009)	0.865 (0.012)	0.705 (0.007)	0.707 (0.005)	0.702 (0.006)	0.688 (0.008)	0.679 (0.008)
Trimmed CWM, $\alpha = 0.15$	0.771 (0.018)	0.862 (0.014)	0.860 (0.013)	0.791 (0.012)	0.713 (0.005)	0.703 (0.005)	0.702 (0.006)	0.703 (0.005)
Trimmed CWM, $\alpha = 0.25$	0.613 (0.018)	0.662 (0.018)	0.756 (0.022)	0.888 (0.010)	0.842 (0.010)	0.777 (0.009)	0.782 (0.013)	0.713 (0.005)
Trimmed CWM, $\alpha = 0.40$	0.585 (0.012)	0.601 (0.011)	0.591 (0.012)	0.592 (0.013)	0.642 (0.012)	0.743 (0.015)	0.763 (0.019)	0.865 (0.009)
TLE, $\alpha = 0.05$	0.910 (0.002)	0.909 (0.002)	0.880 (0.005)	0.827 (0.011)	0.811 (0.007)	0.809 (0.008)	0.821 (0.009)	0.807 (0.007)
TLE, $\alpha = 0.10$	0.912 (0.002)	0.912 (0.002)	0.903 (0.003)	0.824 (0.011)	0.792 (0.007)	0.774 (0.007)	0.774 (0.008)	0.781 (0.007)
TLE, $\alpha = 0.15$	0.911 (0.002)	0.906 (0.005)	0.908 (0.002)	0.875 (0.008)	0.810 (0.007)	0.771 (0.007)	0.770 (0.009)	0.757 (0.006)
TLE, $\alpha = 0.25$	0.904 (0.008)	0.656 (0.022)	0.792 (0.022)	0.913 (0.002)	0.908 (0.002)	0.865 (0.006)	0.874 (0.006)	0.802 (0.007)
TLE, $\alpha = 0.40$	0.659 (0.023)	0.610 (0.015)	0.581 (0.011)	0.585 (0.012)	0.681 (0.015)	0.828 (0.014)	0.826 (0.019)	0.909 (0.003)
CTLE	0.893 (0.018)	0.894 (0.018)	0.907 (0.003)	0.865 (0.019)	0.792 (0.021)	0.770 (0.021)	0.793 (0.025)	0.651 (0.033)
Bisquare mixture	0.911 (0.002)	0.894 (0.018)	0.903 (0.003)	0.875 (0.007)	0.842 (0.007)	0.829 (0.007)	0.830 (0.010)	0.819 (0.009)
Laplace mixture	0.912 (0.002)	0.910 (0.002)	0.907 (0.003)	0.914 (0.002)	0.900 (0.004)	0.904 (0.002)	0.901 (0.005)	0.903 (0.003)
Contaminated normal mixture	0.911 (0.002)	0.913 (0.002)	0.907 (0.003)	0.915 (0.002)	0.910 (0.002)	0.909 (0.002)	0.912 (0.002)	0.910 (0.004)
Noise-component mixture	0.911 (0.002)	0.913 (0.002)	0.907 (0.003)	0.915 (0.002)	0.877 (0.008)	0.787 (0.010)	0.795 (0.015)	0.720 (0.005)
Sequential RANSAC (true noise scale given)	0.911 (0.002)	0.912 (0.002)	0.909 (0.002)	0.916 (0.002)	0.911 (0.002)	0.909 (0.002)	0.912 (0.002)	0.915 (0.002)
Least squares (50 starts)	0.910 (0.002)	0.731 (0.008)	0.711 (0.005)	0.698 (0.005)	0.699 (0.003)	0.696 (0.003)	0.701 (0.005)	0.702 (0.004)
ESF, best of five runs	0.911 (0.002)	0.912 (0.002)	0.908 (0.002)	0.915 (0.003)	0.906 (0.005)	0.903 (0.004)	0.909 (0.004)	0.878 (0.007)
Trimmed alternation (supplement), $\alpha = 0.05$	0.910 (0.002)	0.890 (0.008)	0.730 (0.009)	0.697 (0.004)	0.698 (0.003)	0.695 (0.003)	0.701 (0.005)	0.701 (0.004)
Trimmed alternation (supplement), $\alpha = 0.10$	0.910 (0.002)	0.913 (0.002)	0.882 (0.009)	0.701 (0.005)	0.698 (0.003)	0.695 (0.003)	0.701 (0.005)	0.701 (0.004)
Trimmed alternation (supplement), $\alpha = 0.15$	0.904 (0.008)	0.912 (0.002)	0.907 (0.003)	0.745 (0.009)	0.704 (0.004)	0.695 (0.003)	0.701 (0.005)	0.701 (0.004)
Trimmed alternation (supplement), $\alpha = 0.25$	0.740 (0.027)	0.790 (0.026)	0.862 (0.017)	0.914 (0.002)	0.876 (0.008)	0.727 (0.007)	0.736 (0.011)	0.704 (0.004)
Trimmed alternation (supplement), $\alpha = 0.40$	0.564 (0.013)	0.560 (0.014)	0.575 (0.016)	0.694 (0.026)	0.821 (0.016)	0.883 (0.009)	0.887 (0.013)	0.875 (0.008)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.912 (0.002)	0.895 (0.009)	0.739 (0.012)	0.691 (0.008)	0.693 (0.005)	0.694 (0.005)	0.700 (0.007)	0.698 (0.005)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.911 (0.002)	0.912 (0.002)	0.886 (0.010)	0.698 (0.005)	0.700 (0.004)	0.692 (0.003)	0.699 (0.005)	0.703 (0.004)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.911 (0.002)	0.912 (0.002)	0.907 (0.003)	0.778 (0.015)	0.702 (0.004)	0.695 (0.003)	0.702 (0.005)	0.701 (0.004)

Table S7: Design D4: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.911 (0.003)	0.909 (0.003)	0.910 (0.002)	0.904 (0.003)	0.911 (0.002)	0.907 (0.004)	0.908 (0.003)	0.899 (0.006)
ESF alone	0.911 (0.003)	0.909 (0.003)	0.910 (0.002)	0.904 (0.003)	0.911 (0.002)	0.907 (0.004)	0.908 (0.003)	0.899 (0.006)
Earlier version, no covariate screen (ablation)	0.911 (0.003)	0.910 (0.003)	0.910 (0.002)	0.904 (0.003)	0.907 (0.004)	0.906 (0.004)	0.900 (0.008)	0.880 (0.011)
TCLUST-REG, $\alpha = 0.05$	0.912 (0.002)	0.901 (0.008)	0.565 (0.017)	0.547 (0.005)	0.549 (0.005)	0.556 (0.004)	0.556 (0.006)	0.544 (0.004)
TCLUST-REG, $\alpha = 0.10$	0.911 (0.002)	0.908 (0.003)	0.894 (0.011)	0.525 (0.003)	0.532 (0.002)	0.533 (0.003)	0.542 (0.005)	0.533 (0.003)
TCLUST-REG, $\alpha = 0.15$	0.912 (0.002)	0.909 (0.003)	0.910 (0.002)	0.647 (0.025)	0.532 (0.004)	0.526 (0.002)	0.532 (0.003)	0.527 (0.002)
TCLUST-REG, $\alpha = 0.25$	0.911 (0.002)	0.908 (0.002)	0.909 (0.002)	0.907 (0.003)	0.875 (0.011)	0.617 (0.016)	0.602 (0.021)	0.530 (0.004)
TCLUST-REG, $\alpha = 0.40$	0.907 (0.003)	0.905 (0.002)	0.906 (0.002)	0.905 (0.003)	0.911 (0.002)	0.910 (0.002)	0.909 (0.003)	0.912 (0.002)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.911 (0.003)	0.909 (0.003)	0.910 (0.002)	0.904 (0.003)	0.877 (0.011)	0.619 (0.017)	0.605 (0.022)	0.531 (0.004)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.911 (0.002)	0.908 (0.003)	0.910 (0.002)	0.903 (0.003)	0.912 (0.002)	0.910 (0.002)	0.908 (0.003)	0.905 (0.005)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.911 (0.002)	0.908 (0.003)	0.910 (0.002)	0.903 (0.003)	0.912 (0.002)	0.910 (0.002)	0.908 (0.003)	0.905 (0.005)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.863 (0.018)	0.879 (0.015)	0.909 (0.002)	0.904 (0.003)	0.912 (0.002)	0.910 (0.002)	0.908 (0.003)	0.905 (0.005)
Trimmed CWM, $\alpha = 0.05$	0.871 (0.015)	0.813 (0.022)	0.647 (0.022)	0.565 (0.009)	0.583 (0.008)	0.612 (0.010)	0.614 (0.014)	0.632 (0.010)
Trimmed CWM, $\alpha = 0.10$	0.885 (0.011)	0.854 (0.017)	0.820 (0.021)	0.539 (0.005)	0.546 (0.005)	0.550 (0.005)	0.572 (0.010)	0.586 (0.008)
Trimmed CWM, $\alpha = 0.15$	0.891 (0.011)	0.894 (0.009)	0.878 (0.012)	0.631 (0.022)	0.542 (0.004)	0.540 (0.005)	0.544 (0.005)	0.552 (0.005)
Trimmed CWM, $\alpha = 0.25$	0.880 (0.012)	0.891 (0.009)	0.886 (0.012)	0.844 (0.017)	0.799 (0.015)	0.601 (0.012)	0.600 (0.017)	0.547 (0.005)
Trimmed CWM, $\alpha = 0.40$	0.727 (0.023)	0.796 (0.020)	0.853 (0.016)	0.873 (0.012)	0.877 (0.009)	0.875 (0.009)	0.867 (0.014)	0.834 (0.013)
TLE, $\alpha = 0.05$	0.913 (0.003)	0.887 (0.013)	0.577 (0.017)	0.584 (0.009)	0.596 (0.008)	0.648 (0.012)	0.633 (0.018)	0.670 (0.013)
TLE, $\alpha = 0.10$	0.911 (0.002)	0.911 (0.002)	0.888 (0.013)	0.534 (0.004)	0.554 (0.006)	0.589 (0.010)	0.606 (0.015)	0.598 (0.009)
TLE, $\alpha = 0.15$	0.913 (0.002)	0.909 (0.003)	0.910 (0.002)	0.651 (0.025)	0.551 (0.008)	0.549 (0.007)	0.552 (0.008)	0.559 (0.008)
TLE, $\alpha = 0.25$	0.912 (0.002)	0.912 (0.002)	0.910 (0.002)	0.906 (0.003)	0.867 (0.012)	0.658 (0.016)	0.660 (0.023)	0.555 (0.008)
TLE, $\alpha = 0.40$	0.903 (0.007)	0.890 (0.014)	0.905 (0.007)	0.908 (0.003)	0.909 (0.004)	0.911 (0.002)	0.911 (0.003)	0.913 (0.002)
CTLE	0.874 (0.026)	0.911 (0.002)	0.874 (0.026)	0.851 (0.031)	0.833 (0.025)	0.637 (0.036)	0.596 (0.056)	0.466 (0.034)
Bisquare mixture	0.912 (0.003)	0.911 (0.002)	0.912 (0.002)	0.903 (0.004)	0.879 (0.009)	0.789 (0.014)	0.808 (0.017)	0.757 (0.015)
Laplace mixture	0.911 (0.003)	0.911 (0.002)	0.911 (0.002)	0.906 (0.003)	0.913 (0.002)	0.911 (0.002)	0.909 (0.003)	0.911 (0.002)
Contaminated normal mixture	0.911 (0.002)	0.911 (0.002)	0.911 (0.002)	0.906 (0.003)	0.913 (0.002)	0.911 (0.002)	0.910 (0.003)	0.897 (0.007)
Noise-component mixture	0.911 (0.003)	0.911 (0.002)	0.911 (0.002)	0.896 (0.009)	0.830 (0.016)	0.651 (0.018)	0.701 (0.027)	0.552 (0.009)
Sequential RANSAC (true noise scale given)	0.908 (0.003)	0.906 (0.003)	0.908 (0.002)	0.904 (0.003)	0.910 (0.002)	0.908 (0.002)	0.906 (0.003)	0.908 (0.002)
Least squares (50 starts)	0.910 (0.003)	0.603 (0.017)	0.537 (0.005)	0.525 (0.003)	0.531 (0.003)	0.527 (0.002)	0.532 (0.003)	0.527 (0.002)
ESF, best of five runs	0.911 (0.003)	0.910 (0.003)	0.910 (0.002)	0.904 (0.003)	0.912 (0.002)	0.910 (0.002)	0.908 (0.003)	0.906 (0.004)
Trimmed alternation (supplement), $\alpha = 0.05$	0.911 (0.003)	0.902 (0.007)	0.579 (0.014)	0.525 (0.003)	0.528 (0.002)	0.526 (0.002)	0.532 (0.003)	0.527 (0.002)
Trimmed alternation (supplement), $\alpha = 0.10$	0.911 (0.003)	0.909 (0.002)	0.891 (0.011)	0.530 (0.003)	0.528 (0.002)	0.526 (0.002)	0.532 (0.003)	0.527 (0.002)
Trimmed alternation (supplement), $\alpha = 0.15$	0.911 (0.003)	0.909 (0.003)	0.910 (0.002)	0.675 (0.024)	0.534 (0.003)	0.526 (0.002)	0.532 (0.003)	0.527 (0.002)
Trimmed alternation (supplement), $\alpha = 0.25$	0.911 (0.003)	0.909 (0.003)	0.910 (0.002)	0.904 (0.003)	0.864 (0.012)	0.615 (0.014)	0.603 (0.018)	0.532 (0.003)
Trimmed alternation (supplement), $\alpha = 0.40$	0.863 (0.018)	0.879 (0.015)	0.909 (0.002)	0.904 (0.003)	0.912 (0.002)	0.910 (0.002)	0.908 (0.003)	0.905 (0.005)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.911 (0.003)	0.901 (0.008)	0.565 (0.017)	0.543 (0.005)	0.543 (0.003)	0.549 (0.004)	0.559 (0.006)	0.533 (0.003)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.911 (0.003)	0.909 (0.003)	0.895 (0.011)	0.525 (0.003)	0.528 (0.002)	0.531 (0.002)	0.534 (0.004)	0.530 (0.002)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.911 (0.003)	0.908 (0.003)	0.910 (0.002)	0.648 (0.025)	0.532 (0.004)	0.526 (0.002)	0.532 (0.003)	0.527 (0.002)

Table S8: Design D6: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.912 (0.002)	0.910 (0.002)	0.912 (0.002)	0.891 (0.011)	0.828 (0.015)	0.677 (0.019)	0.701 (0.027)	0.576 (0.013)
ESF alone	0.912 (0.002)	0.910 (0.002)	0.912 (0.002)	0.895 (0.009)	0.828 (0.015)	0.684 (0.019)	0.701 (0.027)	0.585 (0.014)
Earlier version, no covariate screen (ablation)	0.912 (0.002)	0.909 (0.003)	0.911 (0.002)	0.869 (0.017)	0.826 (0.016)	0.646 (0.018)	0.619 (0.023)	0.559 (0.011)
TCLUST-REG, $\alpha = 0.05$	0.912 (0.002)	0.909 (0.003)	0.910 (0.002)	0.856 (0.018)	0.626 (0.015)	0.555 (0.005)	0.560 (0.006)	0.560 (0.004)
TCLUST-REG, $\alpha = 0.10$	0.912 (0.002)	0.909 (0.002)	0.912 (0.002)	0.906 (0.003)	0.835 (0.014)	0.637 (0.014)	0.630 (0.020)	0.578 (0.007)
TCLUST-REG, $\alpha = 0.15$	0.911 (0.002)	0.909 (0.002)	0.912 (0.002)	0.905 (0.003)	0.890 (0.007)	0.779 (0.016)	0.789 (0.023)	0.638 (0.013)
TCLUST-REG, $\alpha = 0.25$	0.910 (0.002)	0.908 (0.003)	0.911 (0.002)	0.907 (0.003)	0.907 (0.002)	0.893 (0.007)	0.893 (0.011)	0.794 (0.014)
TCLUST-REG, $\alpha = 0.40$	0.906 (0.002)	0.907 (0.002)	0.908 (0.002)	0.905 (0.003)	0.908 (0.002)	0.912 (0.002)	0.912 (0.005)	0.903 (0.005)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.912 (0.002)	0.909 (0.002)	0.912 (0.002)	0.906 (0.003)	0.907 (0.002)	0.891 (0.008)	0.886 (0.014)	0.778 (0.016)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.912 (0.002)	0.909 (0.003)	0.912 (0.002)	0.905 (0.003)	0.907 (0.002)	0.908 (0.002)	0.910 (0.006)	0.898 (0.005)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.912 (0.002)	0.909 (0.003)	0.912 (0.002)	0.905 (0.003)	0.907 (0.002)	0.876 (0.011)	0.873 (0.017)	0.740 (0.019)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.913 (0.002)	0.909 (0.002)	0.911 (0.002)	0.906 (0.003)	0.907 (0.002)	0.876 (0.011)	0.872 (0.017)	0.748 (0.018)
TCLUST-REG*, $\alpha = 0.10$, reweighted	0.912 (0.002)	0.909 (0.002)	0.911 (0.002)	0.905 (0.003)	0.867 (0.012)	0.710 (0.018)	0.717 (0.025)	0.577 (0.011)
TCLUST-REG*, $\alpha = 0.25$, reweighted	0.912 (0.002)	0.909 (0.003)	0.911 (0.002)	0.906 (0.003)	0.908 (0.002)	0.904 (0.004)	0.904 (0.008)	0.804 (0.015)
Trimmed CWM, $\alpha = 0.05$	0.878 (0.014)	0.867 (0.016)	0.868 (0.016)	0.816 (0.020)	0.770 (0.016)	0.675 (0.016)	0.683 (0.022)	0.668 (0.015)
Trimmed CWM, $\alpha = 0.10$	0.887 (0.011)	0.880 (0.013)	0.883 (0.013)	0.850 (0.015)	0.787 (0.015)	0.691 (0.016)	0.705 (0.022)	0.649 (0.014)
Trimmed CWM, $\alpha = 0.15$	0.902 (0.007)	0.895 (0.009)	0.885 (0.012)	0.870 (0.011)	0.808 (0.014)	0.729 (0.014)	0.720 (0.021)	0.668 (0.015)
Trimmed CWM, $\alpha = 0.25$	0.883 (0.011)	0.881 (0.011)	0.877 (0.012)	0.859 (0.015)	0.816 (0.013)	0.753 (0.015)	0.762 (0.020)	0.680 (0.014)
Trimmed CWM, $\alpha = 0.40$	0.784 (0.023)	0.782 (0.023)	0.782 (0.023)	0.839 (0.017)	0.841 (0.012)	0.818 (0.013)	0.790 (0.020)	0.718 (0.015)
TLE, $\alpha = 0.05$	0.913 (0.002)	0.908 (0.003)	0.909 (0.002)	0.893 (0.008)	0.879 (0.008)	0.811 (0.015)	0.811 (0.021)	0.727 (0.017)
TLE, $\alpha = 0.10$	0.912 (0.002)	0.909 (0.002)	0.911 (0.002)	0.904 (0.003)	0.901 (0.002)	0.866 (0.011)	0.847 (0.018)	0.804 (0.015)
TLE, $\alpha = 0.15$	0.911 (0.002)	0.909 (0.002)	0.912 (0.002)	0.906 (0.003)	0.902 (0.002)	0.894 (0.006)	0.877 (0.013)	0.814 (0.015)
TLE, $\alpha = 0.25$	0.910 (0.002)	0.909 (0.002)	0.910 (0.002)	0.907 (0.003)	0.907 (0.002)	0.908 (0.002)	0.913 (0.003)	0.888 (0.005)
TLE, $\alpha = 0.40$	0.911 (0.002)	0.859 (0.016)	0.867 (0.015)	0.889 (0.007)	0.895 (0.003)	0.894 (0.005)	0.902 (0.006)	0.884 (0.007)
CTLE	0.876 (0.026)	0.872 (0.026)	0.911 (0.002)	0.903 (0.003)	0.866 (0.016)	0.674 (0.030)	0.744 (0.032)	0.694 (0.023)
Bisquare mixture	0.912 (0.002)	0.909 (0.002)	0.911 (0.002)	0.900 (0.003)	0.895 (0.002)	0.859 (0.013)	0.834 (0.019)	0.827 (0.013)
Laplace mixture	0.912 (0.002)	0.909 (0.003)	0.910 (0.002)	0.906 (0.003)	0.908 (0.002)	0.906 (0.004)	0.913 (0.003)	0.903 (0.004)
Contaminated normal mixture	0.912 (0.002)	0.910 (0.002)	0.911 (0.002)	0.906 (0.003)	0.904 (0.002)	0.828 (0.015)	0.833 (0.021)	0.674 (0.017)
Noise-component mixture	0.912 (0.002)	0.909 (0.002)	0.911 (0.002)	0.907 (0.003)	0.907 (0.002)	0.908 (0.005)	0.900 (0.011)	0.871 (0.011)
Sequential RANSAC (true noise scale given)	0.911 (0.002)	0.908 (0.003)	0.911 (0.002)	0.906 (0.003)	0.894 (0.006)	0.818 (0.013)	0.848 (0.017)	0.734 (0.013)
Least squares (50 starts)	0.912 (0.002)	0.909 (0.002)	0.901 (0.008)	0.579 (0.019)	0.532 (0.006)	0.529 (0.004)	0.536 (0.008)	0.526 (0.002)
ESF, best of five runs	0.912 (0.002)	0.909 (0.003)	0.912 (0.002)	0.905 (0.003)	0.887 (0.009)	0.752 (0.019)	0.753 (0.027)	0.582 (0.014)
Trimmed alternation (supplement), $\alpha = 0.05$	0.913 (0.002)	0.910 (0.002)	0.912 (0.002)	0.816 (0.023)	0.648 (0.018)	0.546 (0.009)	0.544 (0.011)	0.529 (0.002)
Trimmed alternation (supplement), $\alpha = 0.10$	0.913 (0.002)	0.910 (0.002)	0.912 (0.002)	0.898 (0.008)	0.775 (0.018)	0.604 (0.015)	0.756 (0.018)	0.554 (0.009)
Trimmed alternation (supplement), $\alpha = 0.15$	0.912 (0.002)	0.909 (0.002)	0.912 (0.002)	0.906 (0.003)	0.853 (0.013)	0.682 (0.019)	0.681 (0.026)	0.590 (0.013)
Trimmed alternation (supplement), $\alpha = 0.25$	0.912 (0.002)	0.909 (0.003)	0.912 (0.002)	0.905 (0.003)	0.906 (0.002)	0.888 (0.008)	0.859 (0.018)	0.746 (0.017)
Trimmed alternation (supplement), $\alpha = 0.40$	0.904 (0.008)	0.909 (0.002)	0.911 (0.002)	0.906 (0.003)	0.907 (0.002)	0.909 (0.002)	0.915 (0.003)	0.901 (0.004)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.913 (0.002)	0.910 (0.002)	0.912 (0.002)	0.857 (0.019)	0.615 (0.015)	0.539 (0.004)	0.540 (0.004)	0.538 (0.002)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.912 (0.002)	0.910 (0.002)	0.912 (0.002)	0.907 (0.003)	0.828 (0.015)	0.615 (0.015)	0.612 (0.021)	0.549 (0.006)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.912 (0.002)	0.909 (0.002)	0.912 (0.002)	0.906 (0.003)	0.889 (0.008)	0.763 (0.018)	0.783 (0.025)	0.605 (0.014)
TCLUST-REG*, $\alpha = 0.10$	0.911 (0.002)	0.910 (0.002)	0.911 (0.002)	0.904 (0.003)	0.870 (0.011)	0.727 (0.017)	0.734 (0.023)	0.610 (0.011)
TCLUST-REG*, $\alpha = 0.25$	0.910 (0.002)	0.909 (0.003)	0.910 (0.002)	0.907 (0.003)	0.907 (0.002)	0.905 (0.004)	0.907 (0.007)	0.819 (0.013)

Table S9: Design D7: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.908 (0.002)	0.910 (0.002)	0.907 (0.002)	0.908 (0.003)	0.910 (0.002)	0.906 (0.004)	0.893 (0.011)	0.877 (0.011)
ESF alone	0.908 (0.002)	0.910 (0.002)	0.907 (0.002)	0.908 (0.003)	0.910 (0.002)	0.906 (0.004)	0.893 (0.011)	0.877 (0.011)
Earlier version, no covariate screen (ablation)	0.908 (0.002)	0.910 (0.002)	0.908 (0.002)	0.901 (0.009)	0.902 (0.006)	0.867 (0.012)	0.879 (0.015)	0.748 (0.019)
TCLUST-REG, $\alpha = 0.05$	0.909 (0.002)	0.863 (0.018)	0.522 (0.002)	0.537 (0.004)	0.531 (0.002)	0.536 (0.003)	0.540 (0.004)	0.533 (0.003)
TCLUST-REG, $\alpha = 0.10$	0.907 (0.002)	0.911 (0.002)	0.837 (0.022)	0.531 (0.003)	0.527 (0.002)	0.534 (0.002)	0.535 (0.004)	0.535 (0.002)
TCLUST-REG, $\alpha = 0.15$	0.906 (0.002)	0.910 (0.002)	0.908 (0.002)	0.547 (0.011)	0.528 (0.004)	0.529 (0.002)	0.530 (0.004)	0.532 (0.002)
TCLUST-REG, $\alpha = 0.25$	0.905 (0.002)	0.910 (0.002)	0.907 (0.003)	0.908 (0.003)	0.805 (0.017)	0.564 (0.011)	0.570 (0.016)	0.530 (0.002)
TCLUST-REG, $\alpha = 0.40$	0.895 (0.003)	0.899 (0.003)	0.902 (0.003)	0.903 (0.003)	0.907 (0.002)	0.905 (0.002)	0.906 (0.003)	0.858 (0.012)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.907 (0.002)	0.911 (0.002)	0.909 (0.003)	0.908 (0.003)	0.806 (0.017)	0.564 (0.011)	0.569 (0.016)	0.530 (0.002)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.907 (0.002)	0.910 (0.002)	0.908 (0.003)	0.908 (0.003)	0.910 (0.002)	0.909 (0.002)	0.909 (0.003)	0.856 (0.013)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.907 (0.002)	0.910 (0.002)	0.908 (0.003)	0.908 (0.003)	0.910 (0.002)	0.909 (0.002)	0.909 (0.003)	0.855 (0.013)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.907 (0.002)	0.911 (0.002)	0.909 (0.003)	0.908 (0.003)	0.909 (0.002)	0.909 (0.002)	0.908 (0.003)	0.908 (0.002)
Trimmed CWM, $\alpha = 0.05$	0.876 (0.014)	0.828 (0.020)	0.546 (0.011)	0.542 (0.005)	0.554 (0.005)	0.570 (0.005)	0.576 (0.010)	0.573 (0.006)
Trimmed CWM, $\alpha = 0.10$	0.887 (0.011)	0.885 (0.013)	0.778 (0.025)	0.536 (0.004)	0.538 (0.003)	0.542 (0.003)	0.553 (0.007)	0.553 (0.004)
Trimmed CWM, $\alpha = 0.15$	0.892 (0.007)	0.879 (0.014)	0.829 (0.020)	0.555 (0.013)	0.539 (0.004)	0.538 (0.003)	0.534 (0.004)	0.546 (0.004)
Trimmed CWM, $\alpha = 0.25$	0.864 (0.016)	0.871 (0.015)	0.839 (0.020)	0.805 (0.019)	0.661 (0.017)	0.557 (0.008)	0.558 (0.011)	0.535 (0.003)
Trimmed CWM, $\alpha = 0.40$	0.813 (0.019)	0.846 (0.016)	0.827 (0.018)	0.850 (0.017)	0.799 (0.016)	0.771 (0.016)	0.768 (0.022)	0.683 (0.016)
TLE, $\alpha = 0.05$	0.909 (0.002)	0.864 (0.018)	0.530 (0.007)	0.540 (0.004)	0.547 (0.004)	0.558 (0.005)	0.562 (0.009)	0.566 (0.007)
TLE, $\alpha = 0.10$	0.908 (0.002)	0.903 (0.008)	0.858 (0.018)	0.539 (0.004)	0.534 (0.003)	0.542 (0.004)	0.539 (0.006)	0.549 (0.005)
TLE, $\alpha = 0.15$	0.908 (0.002)	0.884 (0.020)	0.908 (0.002)	0.591 (0.018)	0.535 (0.005)	0.537 (0.003)	0.533 (0.005)	0.538 (0.004)
TLE, $\alpha = 0.25$	0.907 (0.002)	0.885 (0.019)	0.909 (0.003)	0.908 (0.003)	0.844 (0.014)	0.594 (0.014)	0.601 (0.019)	0.538 (0.003)
TLE, $\alpha = 0.40$	0.904 (0.002)	0.830 (0.035)	0.904 (0.004)	0.905 (0.003)	0.908 (0.002)	0.908 (0.002)	0.905 (0.003)	0.892 (0.008)
CTLE	0.909 (0.002)	0.911 (0.002)	0.907 (0.003)	0.891 (0.018)	0.885 (0.012)	0.775 (0.025)	0.763 (0.038)	0.545 (0.026)
Bisquare mixture	0.909 (0.002)	0.911 (0.002)	0.908 (0.003)	0.904 (0.003)	0.815 (0.014)	0.680 (0.013)	0.652 (0.018)	0.633 (0.011)
Laplace mixture	0.908 (0.002)	0.911 (0.002)	0.909 (0.003)	0.907 (0.003)	0.907 (0.002)	0.905 (0.002)	0.904 (0.004)	0.902 (0.002)
Contaminated normal mixture	0.909 (0.002)	0.911 (0.002)	0.907 (0.002)	0.909 (0.003)	0.820 (0.016)	0.741 (0.019)	0.729 (0.027)	0.653 (0.017)
Noise-component mixture	0.908 (0.002)	0.911 (0.002)	0.907 (0.002)	0.901 (0.008)	0.767 (0.018)	0.620 (0.016)	0.595 (0.020)	0.541 (0.006)
Sequential RANSAC (true noise scale given)	0.908 (0.002)	0.909 (0.002)	0.907 (0.002)	0.906 (0.003)	0.908 (0.002)	0.908 (0.002)	0.906 (0.003)	0.910 (0.002)
Least squares (50 starts)	0.908 (0.002)	0.564 (0.015)	0.533 (0.004)	0.533 (0.004)	0.528 (0.003)	0.533 (0.003)	0.533 (0.004)	0.532 (0.003)
ESF, best of five runs	0.907 (0.002)	0.910 (0.002)	0.908 (0.003)	0.908 (0.003)	0.910 (0.002)	0.910 (0.002)	0.909 (0.003)	0.909 (0.002)
Trimmed alternation (supplement), $\alpha = 0.05$	0.907 (0.002)	0.883 (0.013)	0.546 (0.007)	0.531 (0.003)	0.524 (0.002)	0.529 (0.002)	0.530 (0.004)	0.530 (0.002)
Trimmed alternation (supplement), $\alpha = 0.10$	0.907 (0.002)	0.911 (0.002)	0.886 (0.012)	0.535 (0.004)	0.524 (0.002)	0.529 (0.002)	0.530 (0.004)	0.530 (0.002)
Trimmed alternation (supplement), $\alpha = 0.15$	0.907 (0.002)	0.910 (0.002)	0.909 (0.003)	0.629 (0.019)	0.537 (0.005)	0.529 (0.002)	0.530 (0.004)	0.530 (0.002)
Trimmed alternation (supplement), $\alpha = 0.25$	0.908 (0.002)	0.910 (0.002)	0.908 (0.003)	0.908 (0.003)	0.845 (0.013)	0.629 (0.013)	0.621 (0.018)	0.541 (0.004)
Trimmed alternation (supplement), $\alpha = 0.40$	0.900 (0.008)	0.911 (0.002)	0.909 (0.003)	0.908 (0.003)	0.909 (0.002)	0.909 (0.002)	0.908 (0.003)	0.908 (0.002)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.907 (0.002)	0.864 (0.018)	0.522 (0.002)	0.543 (0.004)	0.533 (0.003)	0.536 (0.003)	0.540 (0.005)	0.533 (0.003)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.907 (0.002)	0.911 (0.002)	0.838 (0.022)	0.530 (0.003)	0.526 (0.002)	0.535 (0.003)	0.534 (0.004)	0.531 (0.002)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.907 (0.002)	0.911 (0.002)	0.908 (0.003)	0.547 (0.011)	0.528 (0.004)	0.529 (0.002)	0.530 (0.004)	0.531 (0.002)

Table S10: Design D8: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)	0.999 (0.000)
ESF alone	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)	0.999 (0.000)
Earlier version, no covariate screen (ablation)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)
TCLUST-REG, $\alpha = 0.05$	0.999 (0.000)	0.960 (0.019)	0.522 (0.002)	0.539 (0.004)	0.543 (0.004)	0.549 (0.004)	0.547 (0.006)	0.532 (0.003)
TCLUST-REG, $\alpha = 0.10$	0.999 (0.000)	0.999 (0.000)	0.969 (0.017)	0.528 (0.003)	0.529 (0.002)	0.533 (0.003)	0.533 (0.004)	0.531 (0.003)
TCLUST-REG, $\alpha = 0.15$	0.999 (0.000)	0.998 (0.000)	0.999 (0.000)	0.584 (0.022)	0.527 (0.002)	0.530 (0.002)	0.525 (0.003)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.25$	0.999 (0.000)	0.998 (0.000)	0.998 (0.000)	0.999 (0.000)	0.937 (0.016)	0.582 (0.015)	0.620 (0.027)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.40$	0.998 (0.000)	0.998 (0.000)	0.998 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.937 (0.016)	0.582 (0.015)	0.621 (0.027)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)	0.999 (0.000)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)	0.999 (0.000)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)	0.998 (0.000)
Trimmed CWM, $\alpha = 0.05$	0.865 (0.026)	0.851 (0.026)	0.632 (0.023)	0.575 (0.014)	0.606 (0.012)	0.658 (0.016)	0.642 (0.020)	0.671 (0.016)
Trimmed CWM, $\alpha = 0.10$	0.882 (0.024)	0.866 (0.025)	0.832 (0.027)	0.581 (0.017)	0.563 (0.009)	0.610 (0.013)	0.559 (0.010)	0.626 (0.013)
Trimmed CWM, $\alpha = 0.15$	0.869 (0.025)	0.869 (0.024)	0.850 (0.024)	0.667 (0.026)	0.561 (0.009)	0.556 (0.006)	0.563 (0.014)	0.578 (0.010)
Trimmed CWM, $\alpha = 0.25$	0.906 (0.020)	0.891 (0.024)	0.896 (0.022)	0.797 (0.024)	0.771 (0.018)	0.629 (0.015)	0.646 (0.025)	0.561 (0.009)
Trimmed CWM, $\alpha = 0.40$	0.919 (0.017)	0.903 (0.022)	0.901 (0.021)	0.881 (0.021)	0.833 (0.016)	0.801 (0.015)	0.831 (0.022)	0.813 (0.016)
TLE, $\alpha = 0.05$	0.789 (0.031)	0.735 (0.034)	0.543 (0.014)	0.584 (0.015)	0.606 (0.010)	0.639 (0.013)	0.613 (0.016)	0.654 (0.014)
TLE, $\alpha = 0.10$	0.841 (0.028)	0.804 (0.030)	0.733 (0.034)	0.546 (0.010)	0.536 (0.003)	0.554 (0.005)	0.541 (0.005)	0.570 (0.006)
TLE, $\alpha = 0.15$	0.887 (0.026)	0.855 (0.029)	0.828 (0.030)	0.543 (0.010)	0.529 (0.002)	0.531 (0.003)	0.531 (0.003)	0.534 (0.002)
TLE, $\alpha = 0.25$	0.919 (0.024)	0.885 (0.028)	0.896 (0.025)	0.802 (0.031)	0.732 (0.024)	0.555 (0.007)	0.567 (0.016)	0.539 (0.003)
TLE, $\alpha = 0.40$	0.948 (0.018)	0.932 (0.021)	0.933 (0.022)	0.929 (0.023)	0.877 (0.020)	0.910 (0.018)	0.878 (0.027)	0.859 (0.021)
CTLE	0.884 (0.028)	0.782 (0.048)	0.825 (0.041)	0.722 (0.030)	0.682 (0.018)	0.697 (0.017)	0.640 (0.022)	0.655 (0.016)
Bisquare mixture	0.934 (0.027)	0.907 (0.026)	0.861 (0.031)	0.803 (0.027)	0.783 (0.018)	0.779 (0.017)	0.717 (0.025)	0.753 (0.017)
Laplace mixture	0.961 (0.018)	0.961 (0.018)	0.962 (0.018)	0.961 (0.019)	0.951 (0.014)	0.956 (0.013)	0.979 (0.014)	0.940 (0.016)
Contaminated normal mixture	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.984 (0.008)	0.958 (0.013)	0.922 (0.025)	0.852 (0.022)
Noise-component mixture	0.898 (0.027)	0.813 (0.033)	0.808 (0.032)	0.828 (0.031)	0.736 (0.023)	0.607 (0.017)	0.585 (0.022)	0.534 (0.003)
Sequential RANSAC (true noise scale given)	0.989 (0.010)	0.989 (0.010)	0.973 (0.014)	0.989 (0.010)	0.999 (0.000)	0.994 (0.005)	0.989 (0.010)	0.982 (0.008)
Least squares (50 starts)	0.999 (0.000)	0.850 (0.031)	0.539 (0.010)	0.528 (0.003)	0.527 (0.002)	0.534 (0.003)	0.527 (0.003)	0.531 (0.003)
ESF, best of five runs	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)	0.999 (0.000)
Trimmed alternation (supplement), $\alpha = 0.05$	0.999 (0.000)	0.999 (0.000)	0.631 (0.026)	0.528 (0.003)	0.527 (0.002)	0.532 (0.003)	0.526 (0.003)	0.528 (0.002)
Trimmed alternation (supplement), $\alpha = 0.10$	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.528 (0.003)	0.527 (0.002)	0.530 (0.002)	0.525 (0.003)	0.528 (0.002)
Trimmed alternation (supplement), $\alpha = 0.15$	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.778 (0.032)	0.532 (0.005)	0.530 (0.002)	0.525 (0.003)	0.528 (0.002)
Trimmed alternation (supplement), $\alpha = 0.25$	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.987 (0.007)	0.734 (0.020)	0.737 (0.030)	0.547 (0.007)
Trimmed alternation (supplement), $\alpha = 0.40$	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.998 (0.000)	0.998 (0.000)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.999 (0.000)	0.960 (0.019)	0.523 (0.002)	0.546 (0.005)	0.541 (0.004)	0.545 (0.004)	0.542 (0.005)	0.530 (0.002)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.999 (0.000)	0.999 (0.000)	0.969 (0.017)	0.528 (0.003)	0.528 (0.002)	0.532 (0.002)	0.527 (0.003)	0.533 (0.003)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.999 (0.000)	0.999 (0.000)	0.999 (0.000)	0.584 (0.022)	0.527 (0.002)	0.530 (0.002)	0.525 (0.003)	0.528 (0.002)

Table S11: Design D5: mean accuracy on clean units (Monte Carlo standard error).

Method	0.00	0.05	0.10	0.20	0.25	0.30	0.30 (part 1)	0.35
ESF	0.910 (0.003)	0.912 (0.002)	0.910 (0.002)	0.852 (0.021)	0.639 (0.018)	0.538 (0.007)	0.543 (0.010)	0.527 (0.002)
ESF alone	0.910 (0.003)	0.912 (0.002)	0.910 (0.002)	0.852 (0.021)	0.642 (0.018)	0.538 (0.007)	0.544 (0.010)	0.527 (0.002)
Earlier version, no covariate screen (ablation)	0.910 (0.003)	0.911 (0.002)	0.863 (0.018)	0.528 (0.003)	0.525 (0.002)	0.528 (0.002)	0.530 (0.003)	0.527 (0.002)
TCLUST-REG, $\alpha = 0.05$	0.910 (0.003)	0.911 (0.002)	0.679 (0.026)	0.541 (0.010)	0.538 (0.007)	0.529 (0.004)	0.538 (0.009)	0.527 (0.002)
TCLUST-REG, $\alpha = 0.10$	0.910 (0.003)	0.911 (0.002)	0.820 (0.023)	0.547 (0.012)	0.536 (0.006)	0.527 (0.002)	0.533 (0.006)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.15$	0.909 (0.003)	0.912 (0.003)	0.902 (0.008)	0.572 (0.017)	0.534 (0.005)	0.527 (0.002)	0.531 (0.006)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.25$	0.908 (0.003)	0.911 (0.003)	0.910 (0.002)	0.835 (0.023)	0.601 (0.015)	0.534 (0.005)	0.547 (0.010)	0.528 (0.002)
TCLUST-REG, $\alpha = 0.40$	0.908 (0.003)	0.910 (0.003)	0.909 (0.002)	0.866 (0.017)	0.766 (0.018)	0.622 (0.015)	0.630 (0.022)	0.546 (0.007)
TCLUST-REG, $\alpha = 0.25$, reweighted	0.909 (0.003)	0.912 (0.002)	0.911 (0.002)	0.834 (0.023)	0.599 (0.015)	0.535 (0.005)	0.552 (0.011)	0.527 (0.002)
TCLUST-REG, $\alpha = 0.40$, reweighted	0.910 (0.003)	0.912 (0.002)	0.911 (0.002)	0.869 (0.018)	0.765 (0.018)	0.623 (0.016)	0.631 (0.023)	0.545 (0.007)
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.910 (0.003)	0.912 (0.002)	0.911 (0.002)	0.869 (0.018)	0.769 (0.018)	0.610 (0.015)	0.618 (0.022)	0.543 (0.007)
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.910 (0.003)	0.912 (0.002)	0.912 (0.002)	0.881 (0.016)	0.775 (0.018)	0.619 (0.016)	0.628 (0.023)	0.540 (0.007)
TCLUST-REG*, $\alpha = 0.10$, reweighted	0.910 (0.003)	0.912 (0.002)	0.903 (0.008)	0.573 (0.017)	0.549 (0.009)	0.545 (0.008)	0.544 (0.009)	0.530 (0.004)
TCLUST-REG*, $\alpha = 0.25$, reweighted	0.910 (0.003)	0.912 (0.002)	0.912 (0.002)	0.857 (0.020)	0.603 (0.015)	0.551 (0.009)	0.543 (0.009)	0.530 (0.004)
Trimmed CWM, $\alpha = 0.05$	0.872 (0.015)	0.875 (0.015)	0.757 (0.024)	0.668 (0.019)	0.664 (0.014)	0.626 (0.013)	0.641 (0.016)	0.634 (0.013)
Trimmed CWM, $\alpha = 0.10$	0.862 (0.017)	0.886 (0.013)	0.836 (0.020)	0.683 (0.020)	0.664 (0.012)	0.648 (0.011)	0.653 (0.017)	0.660 (0.013)
Trimmed CWM, $\alpha = 0.15$	0.878 (0.014)	0.900 (0.009)	0.872 (0.015)	0.703 (0.020)	0.693 (0.012)	0.672 (0.013)	0.684 (0.017)	0.693 (0.014)
Trimmed CWM, $\alpha = 0.25$	0.861 (0.016)	0.888 (0.010)	0.886 (0.012)	0.803 (0.021)	0.737 (0.014)	0.699 (0.013)	0.671 (0.016)	0.675 (0.013)
Trimmed CWM, $\alpha = 0.40$	0.764 (0.024)	0.820 (0.020)	0.797 (0.022)	0.854 (0.017)	0.828 (0.011)	0.758 (0.015)	0.759 (0.022)	0.704 (0.014)
TLE, $\alpha = 0.05$	0.910 (0.003)	0.804 (0.024)	0.545 (0.011)	0.525 (0.003)	0.526 (0.002)	0.527 (0.002)	0.527 (0.003)	0.527 (0.002)
TLE, $\alpha = 0.10$	0.910 (0.003)	0.824 (0.023)	0.534 (0.009)	0.526 (0.003)	0.526 (0.002)	0.527 (0.002)	0.530 (0.003)	0.528 (0.002)
TLE, $\alpha = 0.15$	0.909 (0.003)	0.794 (0.025)	0.540 (0.008)	0.529 (0.003)	0.530 (0.004)	0.528 (0.002)	0.530 (0.003)	0.528 (0.002)
TLE, $\alpha = 0.25$	0.908 (0.003)	0.786 (0.022)	0.542 (0.009)	0.527 (0.003)	0.531 (0.004)	0.531 (0.003)	0.546 (0.010)	0.530 (0.002)
TLE, $\alpha = 0.40$	0.899 (0.008)	0.796 (0.021)	0.571 (0.017)	0.531 (0.003)	0.528 (0.003)	0.534 (0.004)	0.533 (0.003)	0.533 (0.004)
CTLE	0.874 (0.026)	0.911 (0.002)	0.911 (0.002)	0.729 (0.028)	0.581 (0.015)	0.525 (0.009)	0.483 (0.021)	0.507 (0.011)
Bisquare mixture	0.910 (0.003)	0.873 (0.013)	0.677 (0.026)	0.546 (0.013)	0.527 (0.007)	0.526 (0.002)	0.535 (0.007)	0.528 (0.004)
Laplace mixture	0.910 (0.003)	0.883 (0.011)	0.808 (0.020)	0.578 (0.017)	0.549 (0.009)	0.530 (0.004)	0.530 (0.003)	0.531 (0.004)
Contaminated normal mixture	0.910 (0.003)	0.907 (0.003)	0.883 (0.003)	0.831 (0.018)	0.767 (0.016)	0.678 (0.017)	0.703 (0.024)	0.606 (0.014)
Noise-component mixture	0.910 (0.003)	0.801 (0.024)	0.532 (0.008)	0.527 (0.003)	0.525 (0.002)	0.528 (0.002)	0.527 (0.003)	0.527 (0.002)
Sequential RANSAC (true noise scale given)	0.909 (0.003)	0.903 (0.008)	0.738 (0.027)	0.544 (0.010)	0.536 (0.006)	0.531 (0.004)	0.527 (0.003)	0.530 (0.002)
Least squares (50 starts)	0.910 (0.003)	0.527 (0.007)	0.525 (0.003)	0.528 (0.003)	0.525 (0.002)	0.528 (0.002)	0.530 (0.003)	0.527 (0.002)
ESF, best of five runs	0.910 (0.003)	0.912 (0.002)	0.911 (0.002)	0.906 (0.008)	0.693 (0.019)	0.538 (0.007)	0.559 (0.014)	0.528 (0.002)
Trimmed alternation (supplement), $\alpha = 0.05$	0.910 (0.003)	0.889 (0.013)	0.532 (0.008)	0.528 (0.003)	0.525 (0.002)	0.528 (0.002)	0.530 (0.003)	0.526 (0.002)
Trimmed alternation (supplement), $\alpha = 0.10$	0.909 (0.003)	0.912 (0.002)	0.783 (0.026)	0.529 (0.003)	0.525 (0.002)	0.527 (0.002)	0.531 (0.003)	0.527 (0.002)
Trimmed alternation (supplement), $\alpha = 0.15$	0.910 (0.003)	0.911 (0.002)	0.903 (0.008)	0.571 (0.017)	0.525 (0.002)	0.528 (0.002)	0.530 (0.003)	0.527 (0.002)
Trimmed alternation (supplement), $\alpha = 0.25$	0.910 (0.003)	0.912 (0.002)	0.912 (0.002)	0.865 (0.019)	0.606 (0.016)	0.531 (0.004)	0.544 (0.010)	0.528 (0.002)
Trimmed alternation (supplement), $\alpha = 0.40$	0.902 (0.007)	0.912 (0.002)	0.912 (0.002)	0.879 (0.016)	0.783 (0.017)	0.635 (0.016)	0.635 (0.023)	0.545 (0.008)
TCLUST-REG, $\alpha = 0.05$, reweighted	0.910 (0.003)	0.911 (0.002)	0.676 (0.026)	0.545 (0.010)	0.539 (0.007)	0.531 (0.004)	0.544 (0.011)	0.527 (0.002)
TCLUST-REG, $\alpha = 0.10$, reweighted	0.910 (0.003)	0.912 (0.002)	0.821 (0.023)	0.551 (0.013)	0.535 (0.006)	0.528 (0.002)	0.536 (0.006)	0.527 (0.002)
TCLUST-REG, $\alpha = 0.15$, reweighted	0.910 (0.003)	0.911 (0.002)	0.903 (0.008)	0.577 (0.018)	0.534 (0.006)	0.527 (0.002)	0.535 (0.006)	0.527 (0.002)
TCLUST-REG*, $\alpha = 0.10$	0.909 (0.003)	0.912 (0.003)	0.903 (0.008)	0.567 (0.017)	0.548 (0.009)	0.544 (0.008)	0.538 (0.009)	0.531 (0.004)

Table S12: Sensitivity of ESF alone to the subsample size m : mean accuracy on clean units on the data of the first part (50 data sets per cell; a different random seed from Tables S4–S11). Dashes: size not run for that design. The sizes used in the paper are $m = 8$ (D1, D6), 12 (D2) and 10 (D3).

Design	ε	$m = 6$	$m = 8$	$m = 9$	$m = 10$	$m = 12$	$m = 16$	$m = 20$
D1	0.00	0.903	0.911	–	0.911	0.911	0.911	–
D1	0.10	0.913	0.913	–	0.914	0.913	0.913	–
D1	0.20	0.909	0.910	–	0.910	0.910	0.910	–
D2	0.00	–	–	0.903	–	0.898	0.905	0.905
D2	0.10	–	–	0.892	–	0.906	0.906	0.905
D2	0.20	–	–	0.899	–	0.898	0.906	0.905
D3	0.00	0.875	0.896	–	0.904	0.910	0.911	–
D3	0.10	0.849	0.899	–	0.894	0.907	0.907	–
D3	0.20	0.869	0.891	–	0.906	0.907	0.900	–
D6	0.00	0.912	0.912	–	0.912	0.912	0.912	–
D6	0.10	0.912	0.912	–	0.912	0.912	0.912	–
D6	0.20	0.898	0.891	–	0.898	0.898	0.891	–

Table S16: The flagged set of ESF: fraction of clean units flagged, fraction of outliers flagged (recall) and fraction of flagged units that are outliers (precision), means over data sets (first part for $\varepsilon \leq 0.20$, second part beyond).

Design	ε	Clean flagged	Recall	Precision
D1	0.00	0.019	–	0.000
D1	0.05	0.011	1.000	0.825
D1	0.10	0.009	1.000	0.929
D1	0.20	0.002	1.000	0.991
D1	0.25	0.001	1.000	0.995
D1	0.30	0.000	1.000	0.999
D1	0.35	0.000	1.000	1.000
D2	0.00	0.006	–	0.000
D2	0.05	0.004	1.000	0.931
D2	0.10	0.001	0.993	0.991
D2	0.20	0.001	1.000	0.997
D2	0.25	0.000	0.912	0.999
D2	0.30	0.000	0.723	0.999
D2	0.35	0.000	0.317	0.999
D3	0.00	0.020	–	0.000
D3	0.05	0.012	1.000	0.835
D3	0.10	0.009	1.000	0.928
D3	0.20	0.003	1.000	0.986
D3	0.25	0.003	0.995	0.991
D3	0.30	0.009	0.960	0.975
D3	0.35	0.013	0.835	0.963
D4	0.00	0.033	–	0.000
D4	0.05	0.026	1.000	0.666
D4	0.10	0.021	1.000	0.843
D4	0.20	0.013	1.000	0.949
D4	0.25	0.010	1.000	0.971
D4	0.30	0.007	0.993	0.984
D4	0.35	0.006	0.975	0.989
D6	0.00	0.018	–	0.000
D6	0.05	0.011	1.000	0.825
D6	0.10	0.007	0.996	0.940
D6	0.20	0.003	0.893	0.986

Table S16, continued

Design	ε	Clean flagged	Recall	Precision
D6	0.25	0.001	0.621	0.991
D6	0.30	0.000	0.262	0.998
D6	0.35	0.000	0.056	1.000
D7	0.00	0.028	–	0.000
D7	0.05	0.020	1.000	0.724
D7	0.10	0.013	1.000	0.900
D7	0.20	0.007	1.000	0.973
D7	0.25	0.007	1.000	0.982
D7	0.30	0.005	0.995	0.988
D7	0.35	0.005	0.969	0.991
D8	0.00	0.014	–	0.000
D8	0.05	0.009	1.000	0.847
D8	0.10	0.006	1.000	0.948
D8	0.20	0.001	1.000	0.996
D8	0.25	0.000	1.000	0.999
D8	0.30	0.000	1.000	1.000
D8	0.35	0.000	1.000	1.000
D5	0.00	0.019	–	0.000
D5	0.05	0.013	1.000	0.802
D5	0.10	0.009	1.000	0.927
D5	0.20	0.004	0.919	0.977
D5	0.25	0.008	0.473	0.853
D5	0.30	0.012	0.150	0.583
D5	0.35	0.013	0.040	0.215

Table S20: Mean flagged fraction of the three methods that estimate the contamination: ESF, the contaminated normal mixture (posterior probability of a bad point above one half) and the noise-component mixture (posterior probability of the noise component above one half).

Design	ε	ESF	Contaminated normal	Noise component
D1	0.00	0.019	0.047	0.000
D1	0.05	0.060	0.055	0.050
D1	0.10	0.106	0.101	0.100
D1	0.20	0.211	0.217	0.214
D1	0.25	0.248	0.256	0.239
D1	0.30	0.306	0.317	0.245
D1	0.35	0.347	0.358	0.264
D2	0.00	0.006	0.019	0.000
D2	0.05	0.050	0.045	0.048
D2	0.10	0.101	0.087	0.097
D2	0.20	0.199	0.092	0.176
D2	0.25	0.230	0.044	0.175
D2	0.30	0.213	0.029	0.140
D2	0.35	0.108	0.018	0.139
D3	0.00	0.020	0.029	0.000
D3	0.05	0.065	0.056	0.055
D3	0.10	0.112	0.108	0.106
D3	0.20	0.201	0.206	0.204
D3	0.25	0.248	0.257	0.259
D3	0.30	0.293	0.314	0.338
D3	0.35	0.302	0.374	0.409
D4	0.00	0.033	0.020	0.000
D4	0.05	0.076	0.062	0.051

Table S20, continued

Design	ε	ESF	Contaminated normal	Noise component
D4	0.10	0.118	0.103	0.102
D4	0.20	0.207	0.207	0.201
D4	0.25	0.257	0.262	0.246
D4	0.30	0.304	0.317	0.249
D4	0.35	0.347	0.370	0.292
D5	0.00	0.019	0.031	0.001
D5	0.05	0.062	0.046	0.021
D5	0.10	0.111	0.004	0.003
D5	0.20	0.182	0.005	0.003
D5	0.25	0.119	0.002	0.002
D5	0.30	0.051	0.005	0.001
D5	0.35	0.021	0.005	0.002
D6	0.00	0.018	0.032	0.000
D6	0.05	0.058	0.069	0.055
D6	0.10	0.106	0.146	0.119
D6	0.20	0.170	0.302	0.251
D6	0.25	0.151	0.297	0.331
D6	0.30	0.074	0.163	0.406
D6	0.35	0.018	0.040	0.460
D7	0.00	0.028	0.042	0.000
D7	0.05	0.068	0.051	0.050
D7	0.10	0.115	0.107	0.106
D7	0.20	0.208	0.211	0.206
D7	0.25	0.257	0.248	0.231
D7	0.30	0.304	0.280	0.249
D7	0.35	0.344	0.315	0.278
D8	0.00	0.014	0.037	0.000
D8	0.05	0.057	0.056	0.048
D8	0.10	0.105	0.103	0.100
D8	0.20	0.204	0.215	0.198
D8	0.25	0.250	0.262	0.235
D8	0.30	0.301	0.310	0.246
D8	0.35	0.350	0.330	0.268
D1, $n = 100$	0.00	0.024	0.061	0.003
D1, $n = 100$	0.10	0.108	0.107	0.100
D1, $n = 100$	0.20	0.208	0.211	0.207
D1, $n = 100$	0.30	0.296	0.312	0.242
D1, $n = 1000$	0.00	0.018	0.020	0.000
D1, $n = 1000$	0.10	0.106	0.102	0.102
D1, $n = 1000$	0.20	0.201	0.207	0.204
D1, $n = 1000$	0.30	0.301	0.314	0.260
D3, $n = 1000$	0.00	0.017	0.016	0.000
D3, $n = 1000$	0.10	0.104	0.100	0.099
D3, $n = 1000$	0.20	0.201	0.207	0.203
D3, $n = 1000$	0.30	0.298	0.312	0.365
four groups	0.00	0.005	0.029	0.000
four groups	0.10	0.099	0.103	0.096
four groups	0.20	0.189	0.164	0.145
four groups	0.30	0.129	0.095	0.051
D7, dependent	0.00	0.059	0.034	0.000
D7, dependent	0.10	0.132	0.098	0.097
D7, dependent	0.20	0.230	0.206	0.201
D7, dependent	0.30	0.325	0.273	0.233
D1, moderate	0.05	0.050	0.050	0.036
D1, moderate	0.10	0.081	0.097	0.078
D1, moderate	0.20	0.143	0.216	0.166
D1, moderate	0.30	0.193	0.335	0.195
D3, moderate	0.05	0.052	0.048	0.039

Table S20, continued

Design	ε	ESF	Contaminated normal	Noise component
D3, moderate	0.10	0.077	0.094	0.074
D3, moderate	0.20	0.143	0.217	0.158
D3, moderate	0.30	0.169	0.336	0.145
groups 80/20	0.00	0.020	0.029	0.050
groups 80/20	0.10	0.108	0.103	0.103
groups 80/20	0.20	0.203	0.207	0.204
groups 80/20	0.30	0.284	0.315	0.320
groups 85/15	0.00	0.026	0.036	0.102
groups 85/15	0.10	0.110	0.103	0.103
groups 85/15	0.20	0.203	0.209	0.207
groups 85/15	0.30	0.256	0.311	0.302
clustered	0.00	0.018	0.022	0.001
clustered	0.10	0.109	0.118	0.107
clustered	0.20	0.190	0.028	0.017
clustered	0.30	0.072	0.009	0.000
D1, t_3	0.00	0.065	0.161	0.029
D1, t_3	0.10	0.143	0.138	0.125
D1, t_3	0.20	0.226	0.236	0.225
D3, t_3	0.00	0.064	0.147	0.032
D3, t_3	0.10	0.144	0.134	0.124
D3, t_3	0.20	0.221	0.229	0.223

Table S13: Sensitivity of ESF alone to the bound c_b of the scoring loss: mean accuracy on clean units (50 data sets per cell, from the part of the study shown; a different random seed from the main tables). The value used in the paper is $c_b = 3$. A larger bound recovers small groups but lowers the breakdown point in the other designs.

Part	Design	ε	$c_b = 3$	$c_b = 4$	$c_b = 5$
1	D1	0.10	0.913	0.913	0.913
1	D1	0.20	0.910	0.910	0.910
1	D2	0.10	0.902	0.901	0.900
1	D2	0.20	0.899	0.881	0.793
1	D3	0.10	0.900	0.907	0.907
1	D3	0.20	0.899	0.914	0.906
1	D5	0.10	0.911	0.911	0.911
1	D5	0.20	0.853	0.781	0.724
1	D6	0.10	0.912	0.912	0.912
1	D6	0.20	0.905	0.883	0.862
2	D1	0.30	0.911	0.911	0.860
2	D1	0.35	0.907	0.907	0.815
2	D4	0.30	0.907	0.907	0.907
2	D4	0.35	0.911	0.911	0.875
2	D7	0.30	0.904	0.890	0.860
2	D7	0.35	0.875	0.799	0.739
2	D8	0.30	0.999	0.999	0.989
2	D8	0.35	0.999	0.999	0.904
3	D1, groups 80/20	0.00	0.831	0.885	0.903
3	D1, groups 80/20	0.10	0.823	0.899	0.910
3	D1, groups 80/20	0.20	0.675	0.753	0.815
3	D1, groups 80/20	0.30	0.744	0.828	0.815
3	D1, groups 85/15	0.00	0.616	0.784	0.859
3	D1, groups 85/15	0.10	0.600	0.772	0.807
3	D1, groups 85/15	0.20	0.596	0.677	0.766
3	D1, groups 85/15	0.30	0.745	0.812	0.855
3	D1, clustered outliers	0.00	0.913	0.913	0.913
3	D1, clustered outliers	0.10	0.911	0.911	0.904
3	D1, clustered outliers	0.20	0.694	0.604	0.552
3	D1, clustered outliers	0.30	0.529	0.529	0.529

Table S14: ESF alone with the exact subsample fit replaced by one start of hard alternation on the subsample, by ten starts, or by elemental fits (d units drawn for each group and fitted exactly): mean accuracy on clean units (50 data sets per cell) and median time per data set in seconds. Everything else is unchanged; the random seed differs from the main tables.

Part	Design	ε	Exact	Alternation, 1 start	Alternation, 10 starts	Elemental
1	D1	0.10	0.905	0.914	0.913	0.907
1	D1	0.20	0.910	0.910	0.910	0.910
1	D2	0.10	0.905	0.887	0.902	0.817
1	D2	0.20	0.901	0.869	0.906	0.828
1	D3	0.10	0.876	0.878	0.901	0.770
1	D3	0.20	0.886	0.901	0.892	0.841
1	D6	0.10	0.912	0.912	0.912	0.912
1	D6	0.20	0.906	0.898	0.905	0.890
1	D8	0.10	0.999	0.999	0.999	0.999
1	D8	0.20	0.999	0.999	0.999	0.999
	time (s)		0.03	0.05	0.10	0.02
2	D1	0.30	0.911	0.911	0.911	0.904
2	D1	0.35	0.905	0.906	0.914	0.914
2	D7	0.30	0.911	0.875	0.910	0.853
2	D7	0.35	0.867	0.832	0.869	0.777
	time (s)		0.41	0.28	0.34	0.27
3	four parallel lines	0.10	0.956	0.878	0.935	0.739
3	four parallel lines	0.20	0.960	0.849	0.916	0.712
	time (s)		1.50	0.13	0.28	0.03

Table S15: ESF alone with the same subsample size $m = 12$ in every design, against the design-specific sizes of the main tables: mean accuracy on clean units (first part: 50 data sets per cell; second part: 100). The random seed differs from the main tables.

Design	m (main)	ε						
		0.00	0.05	0.10	0.20	0.25	0.30	0.35
D1	8	0.911	0.907	0.913	0.910	0.910	0.911	0.910
	12	0.911	0.907	0.913	0.910	0.910	0.911	0.899
D2	12	0.904	0.901	0.889	0.901	0.865	0.765	0.555
	12	0.898	0.906	0.906	0.906	0.855	0.785	0.536
D3	10	0.898	0.913	0.901	0.885	0.888	0.881	0.840
	12	0.911	0.906	0.908	0.899	0.864	0.873	0.811
D4	8	0.911	0.909	0.910	0.904	0.911	0.907	0.899
	12	0.911	0.909	0.909	0.904	0.911	0.910	0.896
D6	8	0.912	0.910	0.912	0.895	0.828	0.684	0.585
	12	0.912	0.909	0.912	0.898	0.807	0.652	0.551
D7	12	0.908	0.910	0.907	0.908	0.910	0.906	0.877
	12	0.908	0.911	0.908	0.900	0.906	0.902	0.860
D8	8	0.999	0.999	0.999	0.999	0.999	0.999	0.999
	12	0.999	0.999	0.999	0.999	0.999	0.999	0.975
D5	8	0.910	0.912	0.910	0.852	0.642	0.538	0.527
	12	0.910	0.912	0.911	0.857	0.614	0.528	0.527

Table S17: The recovery step applied after TCLUS-REG at levels 0.25 and 0.40 (300 starts, 10 refinement steps; the long search is in Table S40) with reweighting: mean accuracy on clean units without and with the recovery step, in the designs where it changes the mean by more than 0.01.

Design	ε	$\alpha = 0.25$		$\alpha = 0.40$	
		without	with	without	with
D3	0.00	0.888	0.911	0.666	0.910
D3	0.05	0.912	0.912	0.708	0.912
D3	0.10	0.907	0.907	0.762	0.907
D3	0.20	0.915	0.915	0.868	0.910
D5	0.30	0.535	0.533	0.623	0.610
D6	0.30	0.891	0.857	0.908	0.876
D6	0.35	0.778	0.699	0.898	0.740
D3, $n = 1000$	0.00	0.913	0.913	0.630	0.911
D3, $n = 1000$	0.10	0.911	0.911	0.810	0.910
four groups	0.00	0.827	0.847	0.746	0.766
D3, moderate	0.05	0.907	0.911	0.717	0.911
D3, moderate	0.10	0.911	0.911	0.667	0.906
D3, moderate	0.20	0.906	0.906	0.850	0.888
groups 80/20	0.00	0.634	0.910	0.583	0.910
groups 80/20	0.10	0.880	0.910	0.602	0.910
groups 80/20	0.20	0.914	0.914	0.699	0.903
groups 85/15	0.00	0.596	0.903	0.566	0.910
groups 85/15	0.10	0.731	0.909	0.578	0.904
groups 85/15	0.20	0.901	0.901	0.670	0.879
groups 85/15	0.30	0.852	0.852	0.813	0.824
clustered outliers	0.30	0.545	0.585	0.676	0.703

Table S18: Choice of K by the Bayesian information criterion of the Gaussian-lines-plus-noise model at the fit of ESF, among $K_0 - 1$, K_0 and $K_0 + 1$ (K_0 the true number): fraction of data sets for which K_0 is chosen (20 data sets per cell; 10 for four groups; descriptive).

Design	K_0	$\varepsilon = 0$	0.10	0.20
D1	2	1.00	1.00	1.00
D2	3	0.50	0.75	0.80
D3	2	1.00	1.00	1.00
D4	2	1.00	0.95	1.00
D5	2	1.00	1.00	0.90
D6	2	1.00	1.00	0.90
D7	2	1.00	0.95	1.00
D8	2	1.00	1.00	1.00
four groups	4	1.00	1.00	0.90
groups 80/20	2	1.00	0.90	0.95
groups 85/15	2	1.00	0.95	0.95
clustered outliers	2	1.00	0.10	0.05

Table S19: Median coefficient error $\min_{\text{relabelling}} \max_k \|\hat{\theta}_k - \theta_k^0\|$ over the data sets of each cell (parts one and two; $\varepsilon = 0.30$ from the second part).

Design	ε	ESF	TCLUST-REG 0.25 rw	TCLUST-REG 0.40 rw	Trimmed CWM 0.25	CTLE	Bisquare	Laplace	Contam. normal	Noise comp.
D1	0.00	0.11		0.11		0.26	0.11	0.11	0.12	0.11
D1	0.10	0.12		0.11		0.19	0.11	0.11	0.17	0.10
D1	0.20	0.13		0.13		0.18	0.12	0.33	0.23	0.12
D1	0.30	0.12		22		19	0.17	1.48	0.21	0.12
D2	0.00	0.19		0.36		1.69	0.47	1.43	1.49	0.36
D2	0.10	0.20		0.24		1.98	0.46	2.90	2.64	13
D2	0.20	0.24		0.28		2.31	14	3.02	2.70	20
D2	0.30	0.42		20		14	19	10	6.73	20
D3	0.00	0.11		0.16		2.85	0.11	0.13	0.16	0.12
D3	0.10	0.12		0.13		2.31	0.12	0.18	0.19	0.11
D3	0.20	0.12		0.12		0.23	0.30	0.74	0.25	0.11
D3	0.30	0.15		22		15	1.59	1.57	0.29	0.15
D4	0.00	0.14		0.15		0.42	0.11	0.10	0.13	0.09
D4	0.10	0.15		0.14		0.30	0.11	0.11	0.17	0.12
D4	0.20	0.16		0.15		0.31	0.11	0.22	0.19	0.13
D4	0.30	0.15		22		14	0.46	1.49	0.24	0.13
D6	0.00	0.11		0.11		0.26	0.11	0.10	0.12	0.09
D6	0.10	0.11		0.12		0.19	0.12	0.14	0.15	0.13
D6	0.20	0.16		0.16		0.24	0.30	0.31	0.23	0.16
D6	0.30	3.37		0.42		1.56	1.25	1.07	0.22	0.97
D7	0.00	0.12		0.12		0.20	0.11	0.11	0.14	0.10
D7	0.10	0.13		0.14		0.21	0.13	0.14	0.18	0.12
D7	0.20	0.15		0.14		0.56	0.14	0.25	0.22	0.14
D7	0.30	0.15		22		14	0.17	1.74	0.24	0.24
D8	0.00	0.09		0.09		0.40	0.18	0.09	0.11	0.08
D8	0.10	0.10		0.10		2.07	0.18	0.14	0.26	0.10
D8	0.20	0.11		0.11		2.60	2.94	3.01	0.40	0.11
D8	0.30	0.11		19		8.59	3.18	2.95	0.38	0.11
D5	0.00	0.12		0.12		0.26	0.12	0.10	0.14	0.10
D5	0.10	0.13		0.13		0.19	0.13	2.23	1.25	1.27
D5	0.20	0.11		0.11		0.21	0.32	2.60	2.19	1.38
D5	0.30	2.45		2.51		2.12	2.45	2.52	2.24	2.19

Table S21: Mixtures with more random starts: mean accuracy on clean units of the noise-component mixture with 10 and 100 starts (the latter half from random partitions, half from elemental lines), of the contaminated normal mixture with 50 elemental and 100 mixed starts, and of the Laplace mixture with 20 and 100 starts (`mixLp`, argument `nit`). The last two were run with more starts in the designs with three and four groups only.

Design	ε	Noise, 10	Noise, 100	CN, 50	CN, 100	Laplace, 20	Laplace, 100
D1	0.00	0.912	0.912	0.912	—	0.912	—
D1	0.05	0.907	0.907	0.907	—	0.906	—
D1	0.10	0.914	0.914	0.914	—	0.914	—
D1	0.20	0.910	0.910	0.910	—	0.906	—
D2	0.00	0.836	0.867	0.862	0.863	0.661	0.669
D2	0.05	0.843	0.694	0.824	0.803	0.605	0.632
D2	0.10	0.790	0.589	0.720	0.732	0.633	0.632
D2	0.20	0.748	0.390	0.523	0.497	0.607	0.613
D3	0.00	0.911	0.911	0.911	—	0.912	—
D3	0.05	0.913	0.913	0.913	—	0.910	—
D3	0.10	0.907	0.907	0.907	—	0.907	—
D3	0.20	0.915	0.915	0.915	—	0.914	—
D4	0.00	0.911	0.911	0.911	—	0.911	—
D4	0.05	0.911	0.911	0.911	—	0.911	—
D4	0.10	0.911	0.911	0.911	—	0.911	—
D4	0.20	0.896	0.905	0.906	—	0.906	—
D5	0.00	0.910	0.910	0.910	—	0.910	—
D5	0.05	0.801	0.909	0.907	—	0.883	—
D5	0.10	0.532	0.884	0.883	—	0.808	—
D5	0.20	0.527	0.822	0.831	—	0.578	—
D6	0.00	0.912	0.912	0.912	—	0.912	—
D6	0.05	0.909	0.909	0.910	—	0.909	—
D6	0.10	0.911	0.911	0.911	—	0.910	—
D6	0.20	0.907	0.907	0.906	—	0.906	—
D7	0.00	0.908	0.908	0.909	—	0.908	—
D7	0.05	0.911	0.911	0.911	—	0.911	—
D7	0.10	0.907	0.907	0.907	—	0.909	—
D7	0.20	0.901	0.909	0.909	—	0.907	—
D8	0.00	0.898	0.999	0.999	—	0.961	—
D8	0.05	0.813	0.999	0.999	—	0.961	—
D8	0.10	0.808	0.825	0.999	—	0.962	—
D8	0.20	0.828	0.528	0.999	—	0.961	—
D2	0.25	0.585	0.369	0.412	0.406	0.560	0.527
D2	0.30	0.455	0.367	0.388	0.374	0.532	0.478
D2	0.35	0.399	0.368	0.375	0.381	0.481	0.440
four groups	0.00	0.885	0.851	0.814	0.802	0.676	0.723
four groups	0.10	0.849	0.628	0.757	0.754	0.625	0.673
four groups	0.20	0.653	0.532	0.654	0.665	0.603	0.626
four groups	0.30	0.464	0.431	0.475	0.495	0.534	0.568

Table S22: Cost of the exact fit of one subsample by our branch and bound, without the admissibility constraint and on other subsamples than Section 2.2 of the paper: nodes visited and time in milliseconds (median and 90th percentile over 200 subsamples of ten simulated data sets with 10% of outliers), and the number $K^m/K!$ of labellings that plain enumeration would list.

Design	K	d	m	Nodes, median	Nodes, 90%	Time (ms), median	$K^m/K!$
D1	2	2	8	91	133	0.03	128
D1	2	2	12	309	634	0.08	2.0×10^3
D1	2	2	16	1.0×10^3	6.9×10^3	0.22	3.3×10^4
D2	3	2	12	1.7×10^3	2.9×10^3	0.43	8.9×10^4
D2	3	2	16	8.1×10^3	2.2×10^4	1.71	7.2×10^6
four groups	4	2	16	3.0×10^4	6.4×10^4	7.25	1.8×10^8
four groups	4	2	20	1.6×10^5	6.7×10^5	31.50	4.6×10^{10}
D7	2	4	12	606	815	0.57	2.0×10^3

Table S23: Sensitivity of the recovery step to its constants: mean accuracy of ESF alone and of ESF, and the change in the mean accuracy of ESF when one constant is moved (peak weight w from $\frac{1}{2}$ to 0.4 or 0.6; window from 3 to 2 or 4 cut-offs; minimum support from $0.03n$ to $0.02n$ or $0.05n$; restriction factor from 12 to 8 or 20; margin halved or doubled; limit on the flagged fraction from $\frac{1}{3}$ to 0.25 or 0.40), in the cells where some change exceeds 0.005. All data sets of the study.

Design	ε	ESF alone	ESF	$w=0.4$	$w=0.6$	window=2	window=4	support=0.02	support=0.05	restr.=8	restr.=20	margin/2	margin*2	limit=0.25	limit=0.40
D2	0.25	0.865	0.862	+0.000	+0.003	+0.000	+0.003	+0.000	+0.000	+0.003	-0.002	-0.006	+0.003	+0.000	+0.000
D2	0.35	0.555	0.552	-0.008	+0.002	+0.000	-0.004	+0.000	+0.000	+0.002	+0.000	-0.003	+0.000	+0.000	+0.000
D3	0.00	0.898	0.903	+0.000	+0.000	+0.000	+0.000	+0.008	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
D3	0.20	0.885	0.900	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.014	+0.000
D3	0.25	0.888	0.901	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.013	+0.000
D6	0.35	0.585	0.576	+0.000	+0.000	+0.000	+0.007	-0.004	+0.004	+0.000	+0.000	-0.004	+0.006	+0.000	+0.000
D1, $n = 100$	0.20	0.901	0.908	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.007	+0.000
D3, $n = 1000$	0.30	0.901	0.901	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000	+0.008
four groups	0.30	0.734	0.724	+0.000	+0.003	+0.000	+0.000	-0.000	+0.000	+0.004	-0.008	+0.000	+0.000	+0.000	+0.000
groups 80/20	0.00	0.819	0.910	+0.000	+0.000	-0.016	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
groups 80/20	0.10	0.836	0.910	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.007	+0.000
groups 80/20	0.20	0.727	0.882	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.155	+0.000
groups 80/20	0.30	0.748	0.748	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.054
groups 85/15	0.00	0.628	0.909	+0.000	-0.007	-0.013	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
groups 85/15	0.10	0.605	0.912	+0.000	-0.015	-0.030	+0.000	+0.000	-0.015	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
groups 85/15	0.20	0.619	0.881	+0.000	+0.000	+0.000	+0.000	+0.000	-0.026	+0.000	+0.000	+0.000	-0.016	-0.198	+0.000
groups 85/15	0.30	0.732	0.747	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.015	+0.022
clustered	0.20	0.700	0.781	+0.000	+0.000	+0.000	+0.000	+0.008	-0.016	+0.016	+0.000	+0.000	+0.000	+0.000	+0.000
clustered	0.30	0.536	0.550	+0.000	+0.000	+0.000	+0.000	+0.008	-0.007	+0.007	+0.000	+0.000	+0.000	+0.000	+0.000

Table S24: The best-of-five rule: mean accuracy of ESF averaged over five runs with different seeds, and of the fit of highest likelihood among the five, in the cells where they differ by more than 0.005. All data sets of the study.

Design	ε	One run	Best of five	Difference
D1	0.35	0.908	0.914	+0.006
D2	0.35	0.546	0.450	-0.096
D3	0.20	0.909	0.915	+0.006
D3	0.25	0.886	0.906	+0.020
D3	0.30	0.875	0.903	+0.028
D3	0.35	0.828	0.878	+0.050
D5	0.20	0.868	0.906	+0.037
D5	0.25	0.635	0.693	+0.058
D6	0.20	0.899	0.905	+0.006
D6	0.25	0.853	0.887	+0.034
D6	0.30	0.698	0.752	+0.054
D6	0.35	0.574	0.582	+0.008
D7	0.30	0.904	0.910	+0.005
D7	0.35	0.878	0.909	+0.031
D3, $n = 1000$	0.10	0.904	0.911	+0.007
D3, $n = 1000$	0.30	0.895	0.914	+0.019
four groups	0.20	0.949	0.963	+0.014
four groups	0.30	0.729	0.737	+0.008
D1, moderate	0.30	0.891	0.910	+0.019
D3, moderate	0.30	0.883	0.911	+0.028
groups 80/20	0.10	0.905	0.911	+0.006
groups 80/20	0.20	0.905	0.915	+0.010
groups 80/20	0.30	0.744	0.841	+0.097
groups 85/15	0.20	0.881	0.892	+0.011
groups 85/15	0.30	0.762	0.829	+0.066
clustered	0.20	0.786	0.851	+0.065
clustered	0.30	0.562	0.572	+0.010
D3, t_3	0.20	0.923	0.931	+0.009

Table S25: Further designs (third part of the study, descriptive, 50 data sets per cell): mean accuracy on clean units of ESF, of ESF alone, of the best competitor chosen afterwards in each cell, and of TCLUST-REG with reweighting at the two conservative levels (300 starts); $\hat{\alpha}$ is the mean flagged fraction of ESF.

Design	ε	ESF	ESF alone	Best competitor	TCR _{0.25+rw}	TCR _{0.40+rw}	$\hat{\alpha}$	
D1, $n = 100$	0.00	0.912	0.913	Contaminated normal mixture	0.912	0.910	0.902	0.024
	0.10	0.918	0.918	TCLUST-REG (long search), $\alpha = 0.25$, reweighted	0.919	0.918	0.912	0.108
	0.20	0.908	0.901	TCLUST-REG (long search), $\alpha = 0.40$, reweighted, then the recovery step	0.907	0.906	0.900	0.208
	0.30	0.908	0.908	Contaminated normal mixture	0.912	0.662	0.911	0.296
D1, $n = 1000$	0.00	0.910	0.910	TLE, $\alpha = 0.25$	0.910	0.910	0.910	0.018
	0.10	0.911	0.911	Noise-component mixture	0.911	0.911	0.911	0.106
	0.20	0.911	0.911	Laplace mixture	0.912	0.911	0.911	0.201
	0.30	0.902	0.902	TCLUST-REG (long search), $\alpha = 0.40$, reweighted	0.910	0.522	0.910	0.301
D3, $n = 1000$	0.00	0.913	0.913	Sequential RANSAC (true noise scale given)	0.914	0.913	0.630	0.017
	0.10	0.911	0.911	Sequential RANSAC (true noise scale given)	0.912	0.911	0.810	0.104
	0.20	0.903	0.903	Sequential RANSAC (true noise scale given)	0.912	0.911	0.911	0.201
	0.30	0.901	0.901	Sequential RANSAC (true noise scale given)	0.915	0.736	0.913	0.298
four parallel lines, $n = 400$	0.00	0.958	0.958	TCLUST-REG, $\alpha = 0.05$, reweighted	0.964	0.827	0.746	0.005
	0.10	0.965	0.965	TCLUST-REG (long search), $\alpha = 0.25$, reweighted	0.934	0.837	0.746	0.099
	0.20	0.949	0.949	TCLUST-REG (long search), $\alpha = 0.25$, reweighted, then the recovery step	0.913	0.765	0.686	0.189
	0.30	0.724	0.734	TCLUST-REG (long search), $\alpha = 0.40$, reweighted	0.761	0.689	0.658	0.129
D7, dependent covariates	0.00	0.909	0.909	Noise-component mixture	0.910	0.908	0.909	0.059
	0.10	0.915	0.915	Noise-component mixture	0.917	0.915	0.914	0.132
	0.20	0.909	0.909	CTLE	0.908	0.907	0.907	0.230
	0.30	0.910	0.910	TCLUST-REG (long search), $\alpha = 0.40$, reweighted	0.911	0.552	0.903	0.325
D1, moderate outliers	0.05	0.907	0.907	Sequential RANSAC (true noise scale given)	0.908	0.907	0.907	0.050
	0.10	0.910	0.910	TCLUST-REG, $\alpha = 0.15$, reweighted	0.911	0.911	0.910	0.081
	0.20	0.914	0.914	TCLUST-REG, $\alpha = 0.10$	0.916	0.913	0.913	0.143
	0.30	0.901	0.901	TCLUST-REG, $\alpha = 0.25$, reweighted	0.910	0.910	0.910	0.193
D3, moderate outliers	0.05	0.910	0.910	TCLUST-REG, $\alpha = 0.15$	0.912	0.907	0.717	0.052
	0.10	0.910	0.910	Contaminated normal mixture	0.911	0.911	0.667	0.077
	0.20	0.904	0.904	Sequential RANSAC (true noise scale given)	0.907	0.906	0.850	0.143
	0.30	0.874	0.874	TCLUST-REG, $\alpha = 0.25$	0.911	0.910	0.906	0.169
D1, groups of 80% and 20%	0.00	0.910	0.819	Contaminated normal mixture	0.912	0.634	0.583	0.020
	0.10	0.910	0.836	Sequential RANSAC (true noise scale given)	0.911	0.880	0.602	0.108
	0.20	0.882	0.727	Contaminated normal mixture	0.915	0.914	0.699	0.203
	0.30	0.748	0.748	Sequential RANSAC (true noise scale given)	0.904	0.820	0.866	0.284
D1, groups of 85% and 15%	0.00	0.909	0.628	Contaminated normal mixture	0.912	0.596	0.566	0.026
	0.10	0.912	0.605	Sequential RANSAC (true noise scale given)	0.916	0.731	0.578	0.110
	0.20	0.881	0.619	TCLUST-REG (long search), $\alpha = 0.25$	0.909	0.901	0.670	0.203
	0.30	0.747	0.732	Sequential RANSAC (true noise scale given)	0.891	0.852	0.813	0.256
D1, clustered outliers	0.00	0.914	0.914	Bisquare mixture	0.914	0.913	0.913	0.018
	0.10	0.910	0.910	Sequential RANSAC (true noise scale given)	0.911	0.911	0.911	0.109
	0.20	0.781	0.700	TCLUST-REG, $\alpha = 0.40$, reweighted	0.905	0.897	0.905	0.190
	0.30	0.550	0.536	TCLUST-REG (long search), $\alpha = 0.40$, reweighted, then the recovery step	0.703	0.545	0.676	0.072
D1, t_3 errors	0.00	0.927	0.927	TCLUST-REG, $\alpha = 0.40$	0.928	0.928	0.927	0.065
	0.10	0.934	0.934	Laplace mixture	0.936	0.935	0.935	0.143
	0.20	0.931	0.931	TCLUST-REG, $\alpha = 0.25$	0.932	0.931	0.931	0.226
D3, t_3 errors	0.00	0.933	0.933	TCLUST-REG, $\alpha = 0.10$	0.933	0.933	0.853	0.064
	0.10	0.930	0.930	Noise-component mixture	0.931	0.930	0.919	0.144
	0.20	0.926	0.926	TCLUST-REG, $\alpha = 0.25$	0.932	0.931	0.930	0.221

Table S26: Exact solution of the admissible subsample problem, further sizes (synthetic data sets of 300 units with 10% of outliers, 30 subsamples per row): median time in milliseconds of our depth-first search without and with an incumbent from alternation, of repetitive branch and bound as in the paper and with the incumbents of Brusco (2006) (each nested problem started from the optimum of the next smaller one, the full problem from alternation), and of Gurobi. A superscript counts the solves that reached the limit (60 seconds or 2×10^8 nodes).

Design	K	d	m	Ours	With incumbent	Repetitive	Repetitive, incumbents	Gurobi
as D1	2	2	12	0.10	0.89	0.41	1.27	—
as D1	2	2	20	0.77	1.39	3.05	2.65	—
as D1	2	2	28	19.4	4.43	47.5	21.9	—
as D2	3	2	12	0.80	1.72	1.93	2.52	—
as D2	3	2	20	10.5	5.61	30.9	13.1	—
five parallel lines	5	2	20	189	159	509	329	60 000 ²⁵
two groups, four covariates	2	5	12	2.21	3.26	4.51	5.74	67.7
two groups, four covariates	2	5	16	3.53	4.18	16.2	15.5	117
two groups, four covariates	2	5	20	6.84	5.93	36.0	31.2	206

Table S27: Mixtures given one extra start from the partition of least squares with 50 random starts, in the designs where they fail: mean accuracy on clean units of the noise-component and contaminated normal mixtures with their usual starts and with the extra start (the fit of highest likelihood after the short runs is continued, as before).

Design	ε	Noise, usual	Noise, with LS start	CN, usual	CN, with LS start
D2	0.00	0.836	0.890	0.862	0.887
D2	0.05	0.843	0.459	0.824	0.795
D2	0.10	0.790	0.365	0.720	0.632
D2	0.20	0.748	0.369	0.523	0.406
D8	0.00	0.898	0.999	0.999	0.999
D8	0.05	0.813	0.854	0.999	0.999
D8	0.10	0.808	0.533	0.999	0.999
D8	0.20	0.828	0.538	0.999	0.999
D2	0.25	0.585	0.369	0.412	0.375
D2	0.30	0.455	0.367	0.388	0.367
D2	0.35	0.399	0.368	0.375	0.368
D8	0.25	0.736	0.527	0.984	0.984
D8	0.30	0.607	0.530	0.958	0.958
D8	0.35	0.534	0.528	0.852	0.852
four groups	0.00	0.885	0.953	0.814	0.947
four groups	0.10	0.849	0.524	0.757	0.741
four groups	0.20	0.653	0.512	0.654	0.583
four groups	0.30	0.464	0.494	0.475	0.505

Table S28: Small-group designs with one subsample size, $m = 12$, instead of the size set from the smallest group proportion: mean accuracy on clean units of ESF with the size used in the paper, and of ESF alone and ESF with $m = 12$ (parts three and Plan 4).

Design	ε	m in the paper	ESF	ESF alone, $m = 12$	ESF, $m = 12$
groups 80/20	0.00	16	0.910	0.760	0.911
groups 80/20	0.10	16	0.910	0.802	0.896
groups 80/20	0.20	16	0.882	0.709	0.902
groups 80/20	0.30	16	0.748	0.697	0.705
groups 85/15	0.00	20	0.909	0.578	0.909
groups 85/15	0.10	20	0.912	0.594	0.900
groups 85/15	0.20	20	0.881	0.601	0.876
groups 85/15	0.30	20	0.747	0.656	0.677
groups 80/20 (fresh)	0.00	16	0.912	0.804	0.911
groups 80/20 (fresh)	0.10	16	0.905	0.759	0.907
groups 80/20 (fresh)	0.20	16	0.889	0.762	0.912
groups 85/15 (fresh)	0.00	20	0.910	0.579	0.910
groups 85/15 (fresh)	0.10	20	0.895	0.582	0.890
groups 85/15 (fresh)	0.20	20	0.882	0.589	0.875
80/20, three covariates	0.00	20	0.909	0.675	0.911
80/20, three covariates	0.10	20	0.912	0.650	0.912
80/20, three covariates	0.20	20	0.830	0.651	0.831
85/15, parallel lines	0.00	20	0.998	0.784	0.998
85/15, parallel lines	0.10	20	0.999	0.691	0.999
85/15, parallel lines	0.20	20	0.880	0.686	0.856

Table S29: ESF with $B = 500$ replicates (ten stages of 50) instead of $B = 100$: mean accuracy on clean units, in the cells where the two differ by more than 0.005 (80 cells of parts one to three and Plan 4 examined; new random seeds for $B = 500$).

Design	ε	$B = 100$	$B = 500$	Difference
D2	0.10	0.889	0.899	+0.010
D2	0.20	0.901	0.911	+0.010
D2	0.30	0.765	0.782	+0.017
D2	0.35	0.552	0.564	+0.013
D3	0.00	0.903	0.910	+0.007
D3	0.10	0.901	0.907	+0.007
D3	0.20	0.900	0.914	+0.014
D3	0.30	0.881	0.887	+0.006
D3	0.35	0.840	0.847	+0.006
D4	0.35	0.899	0.905	+0.005
D5	0.20	0.852	0.875	+0.023
D5	0.25	0.639	0.618	-0.021
D5	0.30	0.538	0.531	-0.007
D6	0.20	0.891	0.906	+0.015
D6	0.25	0.828	0.856	+0.028
D6	0.30	0.677	0.696	+0.019
D6	0.35	0.576	0.569	-0.007
D7	0.35	0.877	0.897	+0.020
groups 80/20	0.10	0.910	0.891	-0.018
groups 80/20	0.20	0.882	0.900	+0.018
groups 80/20	0.30	0.748	0.704	-0.044
groups 85/15	0.30	0.747	0.677	-0.070
clustered	0.20	0.781	0.808	+0.027
clustered	0.30	0.550	0.557	+0.007
85/15, parallel lines	0.20	0.880	0.874	-0.006

S4 Exact solvers for the subsample problem

We compared four exact solvers on the same subsamples, 30 for each design and size, drawn from simulated data sets with 10% of outliers (Section 4.1 of the paper), each solver on one core and all on the problem that Algorithm 1 needs: (2) with every group of at least $d + 1$ units. The first is the depth-first search of Section S7, which also prunes a partial labelling when the units left cannot bring every group to $d + 1$ units, without and with an incumbent from ten starts of hard alternation. The second is repetitive branch and bound: the unconstrained problems on the last $1, 2, \dots, m - 1$ units are solved in turn, and the optimal value of the units not yet labelled is added to the bound, which is valid because the residual sum of squares of a group is superadditive over disjoint sets of its units. The third is the mixed logical-quadratic programme of Carbonneau et al. (2011), with binary labels z_{ik} , free coefficients, residuals e_i with $z_{ik} = 1 \Rightarrow e_i = y_i - x_i^\top \theta_k$ as indicator constraints, the quadratic objective $\sum_i e_i^2$, group sizes $\sum_i z_{ik} \geq d + 1$ and the symmetry-breaking constraints $z_{ik} = 0$ for $k > i$, solved by Gurobi 13.0 (Gurobi Optimization, 2026). The fourth is the same programme with big- M constraints, which need bounded coefficients ($|\theta_{kj}| \leq 50$, $M_i = |y_i| + 50 \sum_j |x_{ij}|$), and the objective written as an epigraph constraint, solved by the open-source SCIP 10.0 through PySCIPOpt (Maher et al., 2016). The limit was 60 seconds for the programmes and 2×10^8 nodes, about a minute, for the searches. The probabilistic branch and bound of Fois et al. (2026) targets whole samples without outliers, and we did not run it.

Table S30: Exact solution of the admissible subsample problem: median time in milliseconds over 30 subsamples, and the median number of nodes of our search. A superscript counts the solves that reached the limit.

Design	K	m	Branch and bound			MIQP		Nodes, ours
			ours	with incumbent	repetitive	Gurobi	SCIP	
D1	2	8	0.09	1.56	0.23	7.60	426	87
D1	2	12	0.20	1.63	0.70	17.9	698	356
D1	2	16	0.39	1.59	1.49	29.3	942	916
D1	2	20	1.26	1.84	4.57	44.5	1 617	4 142
D1	2	24	2.80	2.65	10.2	72.0	2 846	10 179
D1	2	28	8.94	4.66	21.6	140	3 783	33 713
D1	2	32	70.5 ¹	35.5 ¹	171 ¹	176	5 353	273 510
D1	2	40	736 ⁵	146 ⁴	1 473 ⁴	317	10 496	3 168 871
D2	3	12	0.84	2.53	2.24	95.0	2 670	1 952
D2	3	16	3.46	3.83	10.1	320	9 774	10 761
D2	3	20	21.0	20.9	44.9	666	23 163	75 720
D2	3	24	66.0	54.1	171	736	21 211 ¹	346 562
D7	2	12	0.69	1.56	2.03	29.9	478	621
D7	2	16	1.43	2.14	6.00	77.4	1 590	2 052
four lines	4	16	11.2	9.46	27.7	1 507	13 855	47 284
four lines	4	20	37.2	25.1	120	6 527	35 404 ²	166 304

Whenever two solvers finished, their optimal values agreed to a relative 2×10^{-6} , except that SCIP returned a larger value on 42 of the 477 subsamples it solved: there the optimum has a group fitted by a line with a coefficient beyond ± 50 , which the bound of the big- M programme excludes. Table S30 gives the times. At the sizes that ESF uses, $m \leq 20$, the depth-first search was 30 to 180 times faster than Gurobi and 700 to 4,700 times faster than SCIP; the SCIP programme differs in its big- M form and epigraph objective, so this factor reflects the formulation as well as the solver. The advantage shrinks as m grows, because the search is exponential in m while the programme's relaxations prune better on larger problems: with two groups, Gurobi was 16 times slower at $m = 28$, 2.5 times slower at $m = 32$, and faster from $m = 40$, where the search reached its node limit on 5 of 30 subsamples. An incumbent from alternation pays from $m = 24$ with two groups and from $m = 16$ with four, and the repetitive bound made the search two to four times slower at every size: solving the nested subproblems costs more than the bound prunes. With about 110 solves per data set, a mixed-integer solver would turn the milliseconds of Algorithm 1 into seconds or minutes. Table S26

extends the comparison. With four covariates the search was 30 to 33 times faster than Gurobi; with five groups and $m = 20$ it took a median of 0.19 seconds, while Gurobi reached its time limit on 25 of 30 subsamples; and starting each nested problem of the repetitive scheme from the optimum of the next smaller one, as Brusco (2006) does, made it faster at the larger sizes but never as fast as the plain search. The subsample size of Algorithm 1 thus lies where the simple search is the right tool, and a larger m , which Section 2.5 of the paper shows does not help, would call for a mixed-integer solver.

S5 Further tables of the paper

Table S31: Mean flagged fraction $\hat{\alpha}$ of ESF, by design and fraction of outliers ε .

Design	0.00	0.05	0.10	0.20	0.25	0.30	0.35
D1	0.019	0.060	0.106	0.211	0.248	0.306	0.347
D2	0.006	0.050	0.101	0.199	0.230	0.213	0.108
D3	0.020	0.065	0.112	0.201	0.248	0.293	0.302
D4	0.033	0.076	0.118	0.207	0.257	0.304	0.347
D6	0.018	0.058	0.106	0.170	0.151	0.074	0.018
D7	0.028	0.068	0.115	0.208	0.257	0.304	0.344
D8	0.014	0.057	0.105	0.204	0.250	0.301	0.350
D5	0.019	0.062	0.111	0.182	0.119	0.051	0.021

Table S32: Computing time per data set in seconds, the first two parts of the study pooled (without the $\varepsilon = 0.30$ cells of the first part, which the second part replaces). TCLUST-REG ran under GNU Octave, which is slower than MATLAB, so its times overstate its cost and are indicative only.

Method	Median	90th percentile
ESF	0.08	1.78
ESF alone	0.07	1.66
TCLUST-REG	3.32	5.11
TCLUST-REG, second trimming	176.44	240.00
Trimmed CWM	4.05	5.42
TLE	10.42	27.16
CTLE	14.63	36.71
Bisquare mixture	3.76	9.13
Laplace mixture	4.46	9.65
Contaminated normal mixture	0.46	0.76
Noise-component mixture	0.02	0.04

Table S33: Plan 5: mean accuracy on clean units of ESF alone and of ESF with the recovery step, the best competitor chosen after the run and the paired difference (standard error in brackets), and the mean fractions of units flagged by their residuals and by the covariate screen only (50 data sets per cell).

Design	ε	ESF alone	ESF	Best competitor	Difference (s.e.)	Flagged, residual	Flagged, covariates
skewed errors	0.00	0.923	0.923	TCLUST-REG (0.15): 0.923	-0.001 (0.001)	0.052	0.000
skewed errors	0.10	0.917	0.917	noise comp.: 0.918	-0.001 (0.001)	0.131	0.000
skewed errors	0.20	0.923	0.923	TLE(0.25): 0.924	-0.001 (0.001)	0.211	0.000
70/30, three covariates	0.00	0.908	0.908	bisquare: 0.911	-0.003 (0.001)	0.020	0.007
70/30, three covariates	0.10	0.884	0.906	TCLUST-REG (0.15): 0.910	-0.004 (0.001)	0.109	0.005
70/30, three covariates	0.20	0.883	0.903	TCLUST-REG (0.25): 0.909	-0.006 (0.005)	0.204	0.005
scale growing with $ x $	0.00	0.913	0.913	TCLUST-REG (0.05): 0.914	-0.001 (0.001)	0.059	0.000
scale growing with $ x $	0.10	0.913	0.913	contam. normal: 0.915	-0.002 (0.001)	0.130	0.000
scale growing with $ x $	0.20	0.909	0.909	TLE(0.25): 0.910	-0.001 (0.001)	0.213	0.000

S5.1 Detail from Section 4 of the paper

Solver comparison. All solvers of Section S4 returned the same optimal values when they finished, apart from subsamples whose optimum lies outside the coefficient bounds of the SCIP formulation.

Ablation of the exact fit. With the exact subsample fit replaced by one start of hard alternation on the subsample, ESF alone lost up to 0.04 with three groups or three covariates and 0.11 with four groups; with elemental fits, d units per group fitted exactly as in RANSAC, it lost up to 0.11 with two unequal groups, 0.09 with three groups or three covariates and 0.25 with four groups; ten starts

of alternation were ahead of the exact fit by 0.025 in one cell of D3 and behind by 0.02 and 0.04 with four groups (Table S14).

Constants of ESF. The 0.975 quantile of the covariate screen is the usual choice for reweighting the minimum covariance determinant; with one covariate the robust distance uses the median and the median absolute deviation.

Settings of the competitors added after Plan 1. The mixture with Laplace errors of Song et al. (2014) was run with 20 random starts. In the mixture of contaminated normal regressions each group has a proportion of at least one half of good points and the rest with an inflated scale, both estimated (50 starts). The noise-component mixture has a component uniform over the range of the response enlarged by 10% (10 starts). Sequential RANSAC extracts the K lines one at a time, each the elemental fit among 500 random draws with the most remaining units within 2.5 noise scales, refitted three times on those units, which are then removed; in D4 it is given the mean of the two noise scales. The contaminated normal and noise-component mixtures flag a unit when its posterior probability of being bad, or of the uniform component, exceeds one half. In D5 and D6 the runs of TCLUST-REG first planned were also made with 5% second trimming in the covariates. FSDA offers no automatic choice of the trimming level for regression.

Failed fits. CTLE failed on 36 of the 1,600 data sets of the first part used in Section 4.2 of the paper and on 154 of the 2,400 of the second part; counted over its successful fits only, its worst mean accuracy up to 20% of outliers is still 0.59 in D2 and 0.72 in D8.

Variation between seeds in the comparison of Plan 1. In D3 at $\varepsilon = 0.20$, where the largest shortfall of ESF alone against the best competitor occurs, ten runs of ESF alone with different seeds or constants on the same 50 data sets gave mean accuracies between 0.885 and 0.914.

Coefficient error. The coefficient error, the largest distance between a fitted and a true coefficient vector after relabelling, tells the same story as the accuracy (Table S19). Where a method keeps its accuracy near the ceiling, its median error is between 0.1 and 0.4; where it fails, the error is of the order of the distance between the lines or larger.

Flagged fraction. At low contamination the excess of the flagged fraction of the procedure over ε is largest in D4, where the noisier group overlaps the other most. It falls well below ε from $\varepsilon = 0.25$ in D5 and D6 and from $\varepsilon = 0.30$ in D2. After TCLUST-REG at either conservative level, reweighting and the recovery step, the flagged fraction differs from that of ESF by at most 0.036 in any cell with $\varepsilon \leq 0.20$.

Competitor effort. With 100 starts (Table S21) the contaminated normal and Laplace mixtures stayed, with three groups, at 0.86 and 0.67 without outliers and at about 0.5 and 0.61 at $\varepsilon = 0.20$; with four groups the Laplace mixture stayed below 0.73 at every level, against about 0.95 for ESF. The noise-component mixture with 100 starts became worse with three groups and in D8 from $\varepsilon = 0.20$, presumably because its fits of higher likelihood absorb the outliers into a wide Gaussian component, and better with high-leverage points. One extra start from the least-squares partition (Table S27) repaired the contaminated normal and noise-component mixtures without outliers (three groups: 0.89; four groups: 0.95), but not with outliers, where it improved them by at most 0.04 and in places made them worse.

Cost. The competitors run from published software took about three minutes with second trimming in the covariates. Our implementations of the contaminated normal and noise-component mixtures, plain ECM and EM fits, took 0.46 and 0.02 seconds per data set. With $n = 1000$ the running time of ESF alone grows from 0.07 to about 0.1 seconds, against two to about thirty seconds for the published competitors.

Outliers of the designs. Uniform outliers (D6) are redrawn while within three noise scales of a true line, so that every outlier is a genuine one; the high-leverage outliers of D5 have responses far from both lines at their covariate values.

Third part of the study. The clustered outliers of the third part sit in a tight cloud at covariate value about 2 and response about 20 noise scales. With dependent covariates the covariate screen flags two to three percentage points more of the clean units than with independent covariates, because the joint distribution of the covariates is not elliptical. With groups of 80% and 20% or 85% and 15% at $\varepsilon = 0.30$ the first step flags 0.26 to 0.28 of the units on average. With t_3 errors and outliers the flagged fraction exceeds ε by two to five percentage points while still flagging every outlier (Table S25); the flagged fraction estimates the contamination only relative to Gaussian errors. In Plan 5 the excess of the flagged fraction over ε arose mostly without outliers, and the covariate screen flagged under 1% of the units (Table S33).

Confirmation of the recovery step under Plan 3. The step changed the fit on 3.4% of the data sets, and lowered a cell mean by 0.007 to 0.010 only where ESF had already broken down.

Repeated runs. In the five runs with different seeds, the fraction of data sets on which the accuracies differed by more than 0.05 was at most 15% in any one design with up to 20% of outliers, the design with clustered outliers. The losses with $B = 500$ presumably have the cause described in Section 4.5 of the paper: the selection criterion of ESF alone can prefer two lines on the large group to one line on each group, so a wider search finds such fits more often, and with a small group the first step is biased.

Sensitivity of the recovery step to the limit on the flagged fraction. A limit of 0.40 instead of $\frac{1}{3}$ gained up to 0.05 in the two small-group designs at $\varepsilon = 0.30$ and moved no other cell by more than 0.01 (Table S23).

Tone perception data. About 60% of the 85 trials nearest the rising line lie within 0.01 of it and the others spread up to 0.5.

The recovery step after TCLUST-REG at level 0.40. Apart from the uniform-noise design, the step harms the level by at most 0.016, in D5 at 30% of outliers. With uniform noise at 35%, moving each constant of the step one step up or down, and a limit of 0.25 instead of $\frac{1}{3}$ on the flagged fraction, did not prevent the loss; the restriction factor 8 recovered 0.026 of it.

Fishery data, other methods. CTLE agrees with TCLUST-REG at low levels, and the trimmed cluster-weighted model varies with its starts at every level. With $K = 3$, ESF alone gave no stable answer for any subsample size from $m = 12$ to $m = 20$.

Fishery data, TCLUST-REG at level 0.15. Between the levels that treat the cheap flows as a group and those that separate the two dearer price levels, at level 0.15, four of the ten starts of TCLUST-REG return two lines with the same intercept and different slopes.

Taxi data. The trips treated as contamination use other tariffs (suburban destinations, Newark airport, negotiated fares) or are invalid records (a fare, distance or duration of zero or less, or a duration above three hours). The data and the script that draws the twenty samples are in the repository. We did not add noise to the flat fares, since the exact group is the structure of interest.

Fishery data, dependence on the seed. Among twenty disjoint sets of five seeds, keeping the fit of highest likelihood under the model of the recovery step returned the cheap group and the dearer line (intercepts 2.16 to 2.18 and 2.78) every time with $K = 2$, and the three price levels (intercepts 1.90 to 1.91, 2.49 to 2.54 and 2.81 to 2.84) every time with $K = 3$. More replicates also help: in a second set of twenty seeds, these fits were returned by 17, 18 and 19 seeds with $B = 100, 500$ and 1000 for $K = 2$, and by 14, 18 and 19 for $K = 3$.

Table S34: Analysis plans and the SHA-256 digests of their files (first 16 hexadecimal digits; the files and the full digests are in the repository). The digest of Plan 5 is that of its file without the last line, a note on an unused optional argument added after the plan was hashed.

Plan	Written before	Concerns	SHA-256
1	any run of the study	first part	b87770ca12fbc8e1
2	the second part was run	second part	5e37bad8706336c6
3	the recovery step was applied to the stored fits	the recovery step	7b7ce8b0f8aa67fd
4	the fresh data were generated	fresh data	a98284b7dfc2b6e9
5	the new designs were run	new designs	8e9255c2a28a3548

Table S35: Tone perception data (Section 5.2 of the paper), $K = 2$, over 20 seeds or 10 random starts, on the original data and with the ten points at $(0, 4)$ of Bai et al. (2012) added: the flat line (intercept, slope; range in brackets when it exceeds 0.005), the number of seeds with other fits, and the trimmed or flagged fraction. A fit counts as “other” unless it has a nearly flat line near 2 (intercept between 1.8 and 2.1, slope below 0.1 in absolute value) and a line of slope within 0.1 of one; the fractions are over the remaining seeds, and the Laplace mixture flags no units. On the original data the rising line has intercept between 0 and 0.07 and slope between 0.97 and 1.00; the trimmed cluster-weighted model returns other lines from 3 of 10 starts at level 0.10 and 6 of 10 at level 0.25 with the points added, and the flagged fraction of ESF then rises to 0.29. Per-seed flags are in the repository.

Method	Original data ($n = 150$)			With ten points at $(0, 4)$	
	Flat line	Other fits	Fraction	Other fits	Fraction
ESF	1.97, 0.02	0/20	0.25 [0.25, 0.27]	0/20	0.29 [0.29, 0.31]
Contaminated normal mixture	1.95, 0.03	0/20	0.23	0/20	0.14
Noise-component mixture	1.92, 0.04	0/20	0.07	7/20	0.13
TCLUST-REG, $\alpha = 0.02$	1.93, 0.04	0/10	0.02	10/10	–
TCLUST-REG, $\alpha = 0.05$	1.91, 0.04	0/10	0.05	10/10	–
TCLUST-REG, $\alpha = 0.10$	1.93, 0.04	0/10	0.10	0/10	0.10
TCLUST-REG, $\alpha = 0.15$	1.95, 0.03	0/10	0.15	0/10	0.15
TCLUST-REG, $\alpha = 0.25$	1.98, 0.02	0/10	0.25	0/10	0.25
TCLUST-REG, $\alpha = 0.40$	2.01, 0.00	0/10	0.40	0/10	0.40
CTLE	1.93, 0.04	0/10	0.09	0/10	0.14 [0.14, 0.14]
Laplace mixture	1.96 [1.94, 1.96], 0.02 [0.02, 0.03]	0/10	–	10/10	–
Trimmed CWM, $\alpha = 0.05$	1.92, 0.04	0/10	0.05	10/10	–
Trimmed CWM, $\alpha = 0.10$	1.93, 0.04	0/10	0.10	3/10	0.10
Trimmed CWM, $\alpha = 0.25$	1.95 [1.95, 1.97], 0.03 [0.02, 0.03]	2/10	0.25	6/10	0.25

Table S40: Long-search TCLUST-REG (1000 starts, 50 refinement steps) with reweighting, alone and followed by the recovery step, at levels 0.25 and 0.40, cell by cell: mean accuracy on clean units, and the paired difference of ESF from the rows with the recovery step (standard error in brackets). Part: 1, 2 and 3 of the study, P4 and P5: Plans 4 and 5.

Part	Design	ε	ESF	0.25, rw	0.25, rw, rec. (diff.)	0.40, rw	0.40, rw, rec. (diff.)
1	D1-K2cross	0.00	0.911	0.911	0.911 -0.000 (0.000)	0.911	0.911 -0.001 (0.000)
1	D1-K2cross	0.05	0.907	0.907	0.907 +0.000 (0.000)	0.907	0.907 +0.000 (0.001)
1	D1-K2cross	0.10	0.913	0.913	0.913 +0.000 (0.000)	0.914	0.914 -0.000 (0.000)
1	D1-K2cross	0.20	0.910	0.910	0.910 +0.000 (0.000)	0.910	0.910 -0.000 (0.000)
1	D2-K3par	0.00	0.904	0.892	0.892 +0.012 (0.003)	0.828	0.831 +0.073 (0.012)
1	D2-K3par	0.05	0.901	0.896	0.896 +0.005 (0.006)	0.856	0.856 +0.045 (0.013)
1	D2-K3par	0.10	0.889	0.903	0.903 -0.014 (0.009)	0.870	0.870 +0.019 (0.014)
1	D2-K3par	0.20	0.901	0.893	0.893 +0.008 (0.009)	0.867	0.867 +0.034 (0.013)
1	D3-unequal	0.00	0.903	0.886	0.911 -0.008 (0.008)	0.664	0.910 -0.007 (0.008)
1	D3-unequal	0.05	0.913	0.912	0.912 +0.001 (0.000)	0.689	0.912 +0.000 (0.000)
1	D3-unequal	0.10	0.901	0.907	0.907 -0.006 (0.007)	0.746	0.907 -0.007 (0.007)

Table S40, continued

Part	Design	ε	ESF	0.25, rw	0.25, rw, rec. (diff.)	0.40, rw	0.40, rw, rec. (diff.)
1	D3-unequal	0.20	0.900	0.915	0.915 -0.015 (0.010)	0.872	0.914 -0.015 (0.010)
1	D4-hetero	0.00	0.911	0.911	0.911 -0.000 (0.001)	0.911	0.911 -0.001 (0.001)
1	D4-hetero	0.05	0.909	0.908	0.908 +0.000 (0.001)	0.909	0.909 -0.000 (0.001)
1	D4-hetero	0.10	0.910	0.910	0.910 +0.000 (0.000)	0.910	0.910 -0.000 (0.001)
1	D4-hetero	0.20	0.904	0.904	0.904 -0.000 (0.000)	0.903	0.903 +0.000 (0.000)
1	D5-leverage	0.00	0.910	0.909	0.909 +0.001 (0.000)	0.910	0.910 +0.000 (0.000)
1	D5-leverage	0.05	0.912	0.912	0.912 -0.000 (0.000)	0.912	0.912 +0.000 (0.000)
1	D5-leverage	0.10	0.910	0.911	0.911 -0.001 (0.001)	0.912	0.912 -0.001 (0.001)
1	D5-leverage	0.20	0.852	0.834	0.834 +0.018 (0.026)	0.869	0.869 -0.017 (0.024)
1	D6-uniform	0.00	0.912	0.911	0.911 +0.001 (0.000)	0.912	0.912 +0.000 (0.001)
1	D6-uniform	0.05	0.910	0.909	0.909 +0.000 (0.001)	0.909	0.909 +0.001 (0.001)
1	D6-uniform	0.10	0.912	0.911	0.911 +0.001 (0.000)	0.911	0.911 +0.001 (0.000)
1	D6-uniform	0.20	0.891	0.906	0.906 -0.015 (0.010)	0.905	0.905 -0.014 (0.011)
1	D7-p3	0.00	0.908	0.907	0.908 +0.000 (0.001)	0.907	0.907 +0.001 (0.001)
1	D7-p3	0.05	0.910	0.910	0.910 +0.000 (0.001)	0.910	0.910 +0.000 (0.001)
1	D7-p3	0.10	0.907	0.909	0.909 -0.001 (0.001)	0.908	0.908 -0.001 (0.001)
1	D7-p3	0.20	0.908	0.908	0.908 +0.000 (0.001)	0.908	0.908 +0.000 (0.001)
1	D8-K2par-sep	0.00	0.999	0.999	0.999 +0.000 (0.000)	0.999	0.999 +0.000 (0.000)
1	D8-K2par-sep	0.05	0.999	0.999	0.999 +0.000 (0.000)	0.999	0.999 +0.000 (0.000)
1	D8-K2par-sep	0.10	0.999	0.999	0.999 +0.000 (0.000)	0.999	0.999 +0.000 (0.000)
1	D8-K2par-sep	0.20	0.999	0.999	0.999 +0.000 (0.000)	0.999	0.999 +0.000 (0.000)
2	D1-K2cross	0.25	0.910	0.848	0.848 +0.062 (0.014)	0.910	0.910 -0.000 (0.000)
2	D1-K2cross	0.30	0.911	0.561	0.561 +0.350 (0.011)	0.911	0.911 +0.000 (0.000)
2	D1-K2cross	0.35	0.910	0.528	0.528 +0.383 (0.005)	0.910	0.910 -0.000 (0.006)
2	D2-K3par	0.25	0.862	0.761	0.761 +0.101 (0.014)	0.874	0.874 -0.012 (0.012)
2	D2-K3par	0.30	0.765	0.618	0.612 +0.153 (0.020)	0.858	0.858 -0.093 (0.017)
2	D2-K3par	0.35	0.552	0.506	0.506 +0.046 (0.020)	0.761	0.758 -0.206 (0.021)
2	D3-unequal	0.25	0.901	0.882	0.882 +0.019 (0.009)	0.909	0.909 -0.008 (0.005)
2	D3-unequal	0.30	0.881	0.751	0.755 +0.126 (0.012)	0.905	0.905 -0.024 (0.008)
2	D3-unequal	0.35	0.840	0.703	0.703 +0.137 (0.013)	0.872	0.872 -0.031 (0.009)
2	D4-hetero	0.25	0.911	0.877	0.877 +0.034 (0.011)	0.911	0.911 -0.000 (0.000)
2	D4-hetero	0.30	0.907	0.619	0.619 +0.288 (0.017)	0.910	0.910 -0.004 (0.003)
2	D4-hetero	0.35	0.899	0.531	0.531 +0.369 (0.008)	0.904	0.904 -0.005 (0.005)
2	D5-leverage	0.25	0.639	0.596	0.597 +0.042 (0.019)	0.747	0.744 -0.105 (0.021)
2	D5-leverage	0.30	0.538	0.536	0.533 +0.005 (0.007)	0.623	0.607 -0.069 (0.016)
2	D5-leverage	0.35	0.527	0.528	0.528 -0.001 (0.001)	0.539	0.533 -0.006 (0.004)
2	D6-uniform	0.25	0.828	0.907	0.907 -0.079 (0.016)	0.907	0.907 -0.079 (0.016)
2	D6-uniform	0.30	0.677	0.891	0.857 -0.180 (0.019)	0.909	0.873 -0.196 (0.019)
2	D6-uniform	0.35	0.576	0.778	0.695 -0.119 (0.016)	0.898	0.724 -0.149 (0.018)
2	D7-p3	0.25	0.910	0.810	0.810 +0.100 (0.017)	0.910	0.910 -0.000 (0.000)
2	D7-p3	0.30	0.906	0.568	0.568 +0.338 (0.012)	0.909	0.909 -0.002 (0.003)
2	D7-p3	0.35	0.877	0.530	0.530 +0.347 (0.011)	0.904	0.904 -0.027 (0.010)
2	D8-K2par-sep	0.25	0.999	0.937	0.937 +0.061 (0.016)	0.999	0.999 -0.000 (0.000)
2	D8-K2par-sep	0.30	0.999	0.582	0.582 +0.417 (0.015)	0.999	0.999 +0.000 (0.000)
2	D8-K2par-sep	0.35	0.999	0.528	0.528 +0.471 (0.002)	0.999	0.999 +0.000 (0.000)
3	E1-K2cross-n100	0.00	0.912	0.909	0.909 +0.003 (0.002)	0.901	0.908 +0.004 (0.003)
3	E1-K2cross-n100	0.10	0.918	0.919	0.919 -0.000 (0.001)	0.911	0.918 +0.001 (0.001)
3	E1-K2cross-n100	0.20	0.908	0.906	0.906 +0.002 (0.001)	0.900	0.907 +0.001 (0.001)
3	E1-K2cross-n100	0.30	0.908	0.662	0.662 +0.246 (0.025)	0.911	0.911 -0.003 (0.006)
3	E2-K2cross-n1000	0.00	0.910	0.910	0.910 -0.000 (0.000)	0.910	0.910 -0.000 (0.000)
3	E2-K2cross-n1000	0.10	0.911	0.911	0.911 +0.000 (0.000)	0.911	0.911 +0.000 (0.000)
3	E2-K2cross-n1000	0.20	0.911	0.911	0.911 +0.000 (0.000)	0.911	0.911 +0.000 (0.000)
3	E2-K2cross-n1000	0.30	0.902	0.522	0.522 +0.380 (0.011)	0.910	0.910 -0.008 (0.008)
3	E3-unequal-n1000	0.00	0.913	0.913	0.913 -0.000 (0.000)	0.598	0.913 -0.001 (0.000)
3	E3-unequal-n1000	0.10	0.911	0.912	0.912 -0.000 (0.000)	0.737	0.911 +0.000 (0.000)
3	E3-unequal-n1000	0.20	0.903	0.911	0.911 -0.008 (0.008)	0.911	0.911 -0.008 (0.008)
3	E3-unequal-n1000	0.30	0.901	0.735	0.735 +0.166 (0.014)	0.913	0.913 -0.012 (0.009)
3	E4-K4par	0.00	0.958	0.934	0.936 +0.021 (0.011)	0.818	0.841 +0.117 (0.016)
3	E4-K4par	0.10	0.965	0.934	0.934 +0.031 (0.010)	0.819	0.818 +0.147 (0.015)
3	E4-K4par	0.20	0.949	0.907	0.913 +0.036 (0.015)	0.808	0.814 +0.136 (0.016)
3	E4-K4par	0.30	0.724	0.742	0.742 -0.018 (0.025)	0.761	0.761 -0.037 (0.025)
3	E5-p3corr	0.00	0.909	0.908	0.907 +0.001 (0.001)	0.909	0.909 +0.000 (0.001)
3	E5-p3corr	0.10	0.915	0.915	0.915 -0.000 (0.001)	0.915	0.915 -0.001 (0.001)
3	E5-p3corr	0.20	0.909	0.907	0.907 +0.002 (0.001)	0.907	0.907 +0.002 (0.001)
3	E5-p3corr	0.30	0.910	0.555	0.555 +0.356 (0.013)	0.911	0.911 -0.001 (0.001)
3	F1-K2cross-moderate	0.05	0.907	0.907	0.907 -0.000 (0.000)	0.907	0.907 -0.000 (0.000)
3	F1-K2cross-moderate	0.10	0.910	0.911	0.911 -0.000 (0.001)	0.910	0.910 +0.000 (0.001)
3	F1-K2cross-moderate	0.20	0.914	0.914	0.914 +0.001 (0.001)	0.914	0.914 +0.000 (0.001)
3	F1-K2cross-moderate	0.30	0.901	0.910	0.910 -0.010 (0.009)	0.910	0.910 -0.009 (0.008)
3	F2-unequal-moderate	0.05	0.910	0.907	0.911 -0.001 (0.001)	0.668	0.910 -0.000 (0.000)
3	F2-unequal-moderate	0.10	0.910	0.910	0.910 -0.000 (0.001)	0.659	0.906 +0.004 (0.005)
3	F2-unequal-moderate	0.20	0.904	0.906	0.906 -0.002 (0.001)	0.853	0.895 +0.009 (0.005)
3	F2-unequal-moderate	0.30	0.874	0.910	0.910 -0.037 (0.013)	0.907	0.907 -0.033 (0.014)
3	G1-small20	0.00	0.910	0.631	0.910 +0.001 (0.001)	0.567	0.910 +0.000 (0.000)

Table S40, continued

Part	Design	ε	ESF	0.25, rw	0.25, rw, rec. (diff.)	0.40, rw	0.40, rw, rec. (diff.)
3	G1-small20	0.10	0.910	0.881	0.910 +0.000 (0.001)	0.588	0.910 -0.000 (0.001)
3	G1-small20	0.20	0.882	0.914	0.914 -0.032 (0.014)	0.678	0.910 -0.028 (0.014)
3	G1-small20	0.30	0.748	0.820	0.820 -0.072 (0.022)	0.859	0.865 -0.117 (0.020)
3	G2-small15	0.00	0.909	0.600	0.902 +0.007 (0.007)	0.567	0.904 +0.006 (0.007)
3	G2-small15	0.10	0.912	0.705	0.909 +0.004 (0.004)	0.573	0.912 -0.000 (0.001)
3	G2-small15	0.20	0.881	0.902	0.902 -0.020 (0.012)	0.643	0.876 +0.005 (0.013)
3	G2-small15	0.30	0.747	0.857	0.857 -0.110 (0.021)	0.813	0.818 -0.070 (0.025)
3	G3-cluster	0.00	0.914	0.913	0.913 +0.001 (0.000)	0.913	0.913 +0.001 (0.000)
3	G3-cluster	0.10	0.910	0.911	0.911 -0.001 (0.000)	0.911	0.911 -0.001 (0.000)
3	G3-cluster	0.20	0.781	0.897	0.897 -0.117 (0.026)	0.905	0.905 -0.125 (0.026)
3	G3-cluster	0.30	0.550	0.545	0.585 -0.035 (0.018)	0.676	0.703 -0.154 (0.026)
3	H1-K2cross-t3	0.00	0.927	0.928	0.928 -0.000 (0.000)	0.927	0.927 +0.000 (0.000)
3	H1-K2cross-t3	0.10	0.934	0.935	0.935 -0.001 (0.000)	0.934	0.934 -0.000 (0.000)
3	H1-K2cross-t3	0.20	0.931	0.931	0.931 +0.000 (0.000)	0.931	0.931 -0.000 (0.000)
3	H2-unequal-t3	0.00	0.933	0.933	0.933 -0.000 (0.000)	0.856	0.933 +0.000 (0.000)
3	H2-unequal-t3	0.10	0.930	0.930	0.930 +0.000 (0.000)	0.919	0.930 -0.000 (0.000)
3	H2-unequal-t3	0.20	0.926	0.931	0.931 -0.005 (0.004)	0.930	0.930 -0.004 (0.004)
P4	G1f-small20	0.00	0.912	0.641	0.912 +0.000 (0.001)	0.563	0.905 +0.007 (0.006)
P4	G1f-small20	0.10	0.905	0.829	0.909 -0.004 (0.004)	0.579	0.910 -0.005 (0.004)
P4	G1f-small20	0.20	0.889	0.912	0.912 -0.024 (0.013)	0.680	0.896 -0.007 (0.016)
P4	G2f-small15	0.00	0.910	0.605	0.910 -0.000 (0.001)	0.559	0.903 +0.007 (0.007)
P4	G2f-small15	0.10	0.895	0.705	0.912 -0.016 (0.011)	0.575	0.896 -0.000 (0.001)
P4	G2f-small15	0.20	0.882	0.904	0.904 -0.022 (0.012)	0.604	0.865 +0.017 (0.010)
P4	G4-small20-p3	0.00	0.909	0.702	0.911 -0.001 (0.001)	0.611	0.903 +0.006 (0.006)
P4	G4-small20-p3	0.10	0.912	0.842	0.911 +0.001 (0.002)	0.633	0.910 +0.002 (0.002)
P4	G4-small20-p3	0.20	0.830	0.906	0.906 -0.076 (0.021)	0.696	0.851 -0.021 (0.026)
P4	G5-small15-par	0.00	0.998	0.672	0.998 +0.000 (0.000)	0.586	0.998 +0.000 (0.000)
P4	G5-small15-par	0.10	0.999	0.926	0.995 +0.004 (0.003)	0.600	0.989 +0.010 (0.010)
P4	G5-small15-par	0.20	0.880	0.969	0.969 -0.088 (0.022)	0.730	0.896 -0.016 (0.028)
P5	S1-skew	0.00	0.923	0.923	0.922 +0.000 (0.001)	0.923	0.923 -0.001 (0.001)
P5	S1-skew	0.10	0.917	0.917	0.917 +0.001 (0.001)	0.916	0.916 +0.001 (0.001)
P5	S1-skew	0.20	0.923	0.907	0.907 +0.015 (0.011)	0.924	0.924 -0.001 (0.001)
P5	S2-p3-unequal	0.00	0.908	0.884	0.908 +0.000 (0.001)	0.784	0.909 -0.001 (0.001)
P5	S2-p3-unequal	0.10	0.906	0.907	0.907 -0.001 (0.001)	0.830	0.893 +0.013 (0.010)
P5	S2-p3-unequal	0.20	0.903	0.907	0.907 -0.004 (0.005)	0.904	0.904 -0.000 (0.001)
P5	S3-xhet	0.00	0.913	0.913	0.913 -0.000 (0.001)	0.913	0.914 -0.000 (0.001)
P5	S3-xhet	0.10	0.913	0.914	0.914 -0.001 (0.001)	0.913	0.913 +0.000 (0.001)
P5	S3-xhet	0.20	0.909	0.910	0.910 -0.000 (0.001)	0.909	0.909 -0.000 (0.001)

Table S36: Plan 3 (Section 4.6 of the paper): mean accuracy on clean units of ESF alone and of the procedure, and the paired difference with its standard error, in the cells where the recovery step changed the mean by more than 0.005 (50 data sets per cell, 100 in the second part).

Design	ε	ESF alone	ESF	Difference	SE
D2 (part 1)	0.30	0.747	0.740	-0.007	0.007
D3	0.00	0.898	0.903	+0.005	0.005
D3	0.20	0.885	0.900	+0.014	0.010
D3	0.25	0.888	0.901	+0.013	0.006
D6	0.30	0.684	0.677	-0.007	0.005
D6	0.35	0.585	0.576	-0.010	0.006
D1, $n = 100$	0.20	0.901	0.908	+0.007	0.007
four groups	0.30	0.734	0.724	-0.010	0.006
groups 80/20	0.00	0.819	0.910	+0.091	0.022
groups 80/20	0.10	0.836	0.910	+0.074	0.021
groups 80/20	0.20	0.727	0.882	+0.155	0.026
groups 85/15	0.00	0.628	0.909	+0.281	0.023
groups 85/15	0.10	0.605	0.912	+0.308	0.020
groups 85/15	0.20	0.619	0.881	+0.262	0.023
groups 85/15	0.30	0.732	0.747	+0.015	0.011
clustered outliers	0.20	0.700	0.781	+0.080	0.022
clustered outliers	0.30	0.536	0.550	+0.013	0.009

Table S37: Competitor settings as first planned (TCLUST-REG with 300 random starts and 10 refinement steps): worst mean accuracy on clean units (the lowest of the cell means) over $\varepsilon \in \{0, 0.05, 0.10, 0.20\}$, by design, first part of the study. Values below 0.85 in italics. Columns: the seven designs without high-leverage points, then D5. *With 5% second trimming in the covariates (run in D5 and D6 only). The Monte Carlo standard error of a cell mean is about 0.003 near the ceiling and up to 0.05 where a method fails (Tables S4–S11).

Method	D1	D2	D3	D4	D6	D7	D8	D5
ESF	0.91	0.89	0.90	0.90	0.89	0.91	1.00	0.85
ESF alone	0.91	0.89	0.89	0.90	0.90	0.91	1.00	0.85
Earlier version, no covariate screen (ablation)	0.90	0.89	0.90	0.90	0.87	0.90	1.00	<i>0.53</i>
TCLUST-REG, $\alpha = 0.05$	<i>0.53</i>	<i>0.37</i>	<i>0.69</i>	<i>0.55</i>	0.86	<i>0.52</i>	<i>0.52</i>	<i>0.54</i>
TCLUST-REG, $\alpha = 0.10$	<i>0.53</i>	<i>0.46</i>	<i>0.70</i>	<i>0.52</i>	0.91	<i>0.53</i>	<i>0.53</i>	<i>0.55</i>
TCLUST-REG, $\alpha = 0.15$	<i>0.54</i>	<i>0.60</i>	<i>0.78</i>	<i>0.65</i>	0.91	<i>0.55</i>	<i>0.58</i>	<i>0.57</i>
TCLUST-REG, $\alpha = 0.25$	0.91	<i>0.84</i>	0.89	0.91	0.91	0.90	1.00	<i>0.83</i>
TCLUST-REG, $\alpha = 0.40$	0.90	<i>0.76</i>	<i>0.68</i>	0.90	0.91	0.90	1.00	0.87
TCLUST-REG, $\alpha = 0.25$, reweighted	0.91	0.87	0.89	0.90	0.91	0.91	1.00	<i>0.83</i>
TCLUST-REG, $\alpha = 0.40$, reweighted	0.90	<i>0.80</i>	<i>0.67</i>	0.90	0.91	0.91	1.00	0.87
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.91	<i>0.80</i>	0.91	0.90	0.91	0.91	1.00	0.87
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.91	0.87	<i>0.83</i>	0.86	0.91	0.91	1.00	0.88
TCLUST-REG*, $\alpha = 0.10$, reweighted	–	–	–	–	0.91	–	–	<i>0.57</i>
TCLUST-REG*, $\alpha = 0.25$, reweighted	–	–	–	–	0.91	–	–	0.86
Trimmed CWM, $\alpha = 0.05$	<i>0.56</i>	<i>0.43</i>	<i>0.70</i>	<i>0.56</i>	<i>0.82</i>	<i>0.54</i>	<i>0.58</i>	<i>0.67</i>
Trimmed CWM, $\alpha = 0.10$	<i>0.54</i>	<i>0.51</i>	<i>0.71</i>	<i>0.54</i>	<i>0.85</i>	<i>0.54</i>	<i>0.58</i>	<i>0.68</i>
Trimmed CWM, $\alpha = 0.15$	<i>0.58</i>	<i>0.54</i>	<i>0.77</i>	<i>0.63</i>	0.87	<i>0.56</i>	<i>0.67</i>	<i>0.70</i>
Trimmed CWM, $\alpha = 0.25$	0.87	<i>0.65</i>	<i>0.61</i>	<i>0.84</i>	0.86	<i>0.81</i>	<i>0.80</i>	<i>0.80</i>
Trimmed CWM, $\alpha = 0.40$	<i>0.74</i>	<i>0.62</i>	<i>0.59</i>	<i>0.73</i>	<i>0.78</i>	<i>0.81</i>	0.88	<i>0.76</i>
TLE, $\alpha = 0.05$	<i>0.55</i>	<i>0.46</i>	<i>0.83</i>	<i>0.58</i>	0.89	<i>0.53</i>	<i>0.54</i>	<i>0.53</i>
TLE, $\alpha = 0.10$	<i>0.53</i>	<i>0.55</i>	<i>0.82</i>	<i>0.53</i>	0.90	<i>0.54</i>	<i>0.55</i>	<i>0.53</i>
TLE, $\alpha = 0.15$	<i>0.56</i>	<i>0.57</i>	0.88	<i>0.65</i>	0.91	<i>0.59</i>	<i>0.54</i>	<i>0.53</i>
TLE, $\alpha = 0.25$	0.90	<i>0.72</i>	<i>0.66</i>	0.91	0.91	0.89	<i>0.80</i>	<i>0.53</i>
TLE, $\alpha = 0.40$	0.90	<i>0.67</i>	<i>0.58</i>	0.89	0.86	<i>0.83</i>	0.93	<i>0.53</i>
CTLE	0.91	<i>0.59</i>	0.87	0.85	0.87	0.89	<i>0.72</i>	<i>0.73</i>
Bisquare mixture	0.89	<i>0.56</i>	0.88	0.90	0.90	0.90	<i>0.80</i>	<i>0.55</i>
Laplace mixture	0.91	<i>0.61</i>	0.91	0.91	0.91	0.91	0.96	<i>0.58</i>
Contaminated normal mixture	0.91	<i>0.52</i>	0.91	0.91	0.91	0.91	1.00	<i>0.83</i>
Noise-component mixture	0.91	<i>0.75</i>	0.91	0.90	0.91	0.90	<i>0.81</i>	<i>0.53</i>
Sequential RANSAC (true noise scale given)	0.91	<i>0.69</i>	0.91	0.90	0.91	0.91	0.97	<i>0.54</i>
Least squares (50 starts)	<i>0.53</i>	<i>0.37</i>	<i>0.70</i>	<i>0.52</i>	<i>0.58</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>

Table S38: Competitor settings as first planned: worst mean accuracy on clean units (the lowest of the cell means) over $\varepsilon \in \{0.25, 0.30, 0.35\}$ (second part of the study, new data), by design. Values below 0.85 in italics. Columns: the seven designs without high-leverage points, then D5. *With 5% second trimming in the covariates (run in D5 and D6 only). Standard errors as in Table S37.

Method	D1	D2	D3	D4	D6	D7	D8	D5
ESF	0.91	<i>0.55</i>	<i>0.84</i>	0.90	<i>0.58</i>	0.88	1.00	<i>0.53</i>
ESF alone	0.91	<i>0.56</i>	<i>0.84</i>	0.90	<i>0.59</i>	0.88	1.00	<i>0.53</i>
Earlier version, no covariate screen (ablation)	0.87	<i>0.52</i>	<i>0.81</i>	0.88	<i>0.56</i>	<i>0.75</i>	1.00	<i>0.53</i>
TCLUST-REG, $\alpha = 0.05$	<i>0.54</i>	<i>0.37</i>	<i>0.68</i>	<i>0.54</i>	<i>0.56</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>
TCLUST-REG, $\alpha = 0.10$	<i>0.53</i>	<i>0.37</i>	<i>0.69</i>	<i>0.53</i>	<i>0.58</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>
TCLUST-REG, $\alpha = 0.15$	<i>0.53</i>	<i>0.38</i>	<i>0.70</i>	<i>0.53</i>	<i>0.64</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>
TCLUST-REG, $\alpha = 0.25$	<i>0.53</i>	<i>0.54</i>	<i>0.71</i>	<i>0.53</i>	<i>0.79</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>
TCLUST-REG, $\alpha = 0.40$	0.91	<i>0.73</i>	0.91	0.91	0.90	0.86	1.00	<i>0.55</i>
TCLUST-REG, $\alpha = 0.25$, reweighted	<i>0.53</i>	<i>0.54</i>	<i>0.70</i>	<i>0.53</i>	<i>0.78</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>
TCLUST-REG, $\alpha = 0.40$, reweighted	0.91	<i>0.70</i>	0.87	0.90	0.90	0.86	1.00	<i>0.55</i>
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.91	<i>0.70</i>	0.87	0.90	<i>0.74</i>	0.85	1.00	<i>0.54</i>
Trimmed alternation, $\alpha = 0.40$, reweighted, then the recovery step	0.91	<i>0.81</i>	<i>0.84</i>	0.90	<i>0.75</i>	0.91	1.00	<i>0.54</i>
TCLUST-REG*, $\alpha = 0.10$, reweighted	–	–	–	–	<i>0.58</i>	–	–	<i>0.53</i>
TCLUST-REG*, $\alpha = 0.25$, reweighted	–	–	–	–	<i>0.80</i>	–	–	<i>0.53</i>
Trimmed CWM, $\alpha = 0.05$	<i>0.58</i>	<i>0.41</i>	<i>0.68</i>	<i>0.58</i>	<i>0.67</i>	<i>0.55</i>	<i>0.61</i>	<i>0.63</i>
Trimmed CWM, $\alpha = 0.10$	<i>0.55</i>	<i>0.41</i>	<i>0.68</i>	<i>0.55</i>	<i>0.65</i>	<i>0.54</i>	<i>0.56</i>	<i>0.65</i>
Trimmed CWM, $\alpha = 0.15$	<i>0.54</i>	<i>0.44</i>	<i>0.70</i>	<i>0.54</i>	<i>0.67</i>	<i>0.54</i>	<i>0.56</i>	<i>0.67</i>
Trimmed CWM, $\alpha = 0.25$	<i>0.55</i>	<i>0.53</i>	<i>0.71</i>	<i>0.55</i>	<i>0.68</i>	<i>0.54</i>	<i>0.56</i>	<i>0.67</i>
Trimmed CWM, $\alpha = 0.40$	<i>0.85</i>	<i>0.62</i>	<i>0.64</i>	<i>0.83</i>	<i>0.72</i>	<i>0.68</i>	<i>0.80</i>	<i>0.70</i>
TLE, $\alpha = 0.05$	<i>0.58</i>	<i>0.38</i>	<i>0.81</i>	<i>0.60</i>	<i>0.73</i>	<i>0.55</i>	<i>0.61</i>	<i>0.53</i>
TLE, $\alpha = 0.10$	<i>0.54</i>	<i>0.39</i>	<i>0.77</i>	<i>0.55</i>	<i>0.80</i>	<i>0.53</i>	<i>0.54</i>	<i>0.53</i>
TLE, $\alpha = 0.15$	<i>0.53</i>	<i>0.39</i>	<i>0.76</i>	<i>0.55</i>	<i>0.81</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>
TLE, $\alpha = 0.25$	<i>0.53</i>	<i>0.50</i>	<i>0.80</i>	<i>0.56</i>	0.89	<i>0.54</i>	<i>0.54</i>	<i>0.53</i>
TLE, $\alpha = 0.40$	0.91	<i>0.71</i>	<i>0.68</i>	0.91	0.88	0.89	0.86	<i>0.53</i>
CTLE	<i>0.65</i>	<i>0.40</i>	<i>0.65</i>	<i>0.47</i>	<i>0.67</i>	<i>0.55</i>	<i>0.66</i>	<i>0.51</i>
Bisquare mixture	<i>0.73</i>	<i>0.42</i>	<i>0.82</i>	<i>0.76</i>	<i>0.83</i>	<i>0.63</i>	<i>0.75</i>	<i>0.53</i>
Laplace mixture	0.91	<i>0.48</i>	0.90	0.91	0.90	0.90	0.94	<i>0.53</i>
Contaminated normal mixture	0.90	<i>0.37</i>	0.91	0.90	<i>0.67</i>	<i>0.65</i>	0.85	<i>0.61</i>
Noise-component mixture	<i>0.55</i>	<i>0.40</i>	<i>0.72</i>	<i>0.55</i>	<i>0.87</i>	<i>0.54</i>	<i>0.53</i>	<i>0.52</i>
Sequential RANSAC (true noise scale given)	0.91	<i>0.64</i>	0.91	0.91	<i>0.73</i>	0.91	0.98	<i>0.53</i>
Least squares (50 starts)	<i>0.53</i>	<i>0.37</i>	<i>0.70</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>

Table S39: Further competitors and checks decided after the analysis plans (descriptive; Section 4.4 of the paper): mean over cells of the mean accuracy on clean units, the number of cells with a mean below 0.85, and the worst cell. rw: the reweighting step; recovery: followed by the recovery step. A superscript gives the number of cells with results when it is smaller than the total.

Method	$\varepsilon \leq 0.20$ (89 cells)			$\varepsilon \geq 0.25$ (34 cells)		
	Mean	Below 0.85	Worst	Mean	Below 0.85	Worst
ESF	0.914	2	0.78	0.820	13	0.53
ESF, $m = 12$ in every design	0.912	2	0.75	0.813	13	0.53
ESF alone	0.874	18	0.60	0.820	13	0.53
ESF alone, $m = 12$ in every design	0.864	19	0.56	0.811	13	0.53
TCLUST-REG 0.40, rw, recovery (main rival)	0.909	4	0.81	0.863	8	0.53
TCLUST-REG 0.25, rw, recovery (main rival)	0.917	1	0.83	0.693	26	0.51
TCLUST-REG 0.30, rw, recovery	0.915	0	0.87	0.794	17	0.53
TCLUST-REG 0.35, rw, recovery	0.912	4	0.83	0.851	10	0.53
TCLUST-REG 0.25, equal weights, rw, recovery	0.919	0	0.86	0.769	24	0.53
TCLUST-REG 0.40, equal weights, rw	0.788	34	0.52	0.861	9	0.55
TCLUST-REG 0.40, equal weights, rw, recovery	0.907	4	0.79	0.856	10	0.54
TCLUST-REG 0.25, adaptive second trimming, rw, recovery	0.899	1	0.80 ⁽¹²⁾	0.687	8	0.53 ⁽⁹⁾
TCLUST-REG 0.40, adaptive second trimming, rw, recovery	0.893	1	0.73 ⁽¹²⁾	0.765	4	0.54 ⁽⁹⁾
TCLUST-REG 0.40, equal scales, rw, recovery	0.836	3	0.77 ⁽⁷⁾	0.800	2	0.66 ⁽⁴⁾
TCLUST-REG, level from the monitoring path (first stable pair), rw, recovery	0.809	36	0.37	0.622	29	0.37
TCLUST-REG, level from the monitoring path (stable from there on), rw, recovery	0.906	7	0.76	0.856	10	0.54
TCLUST-REG 0.40, incremental reweighting	0.836	32	0.55	0.835	13	0.54
TCLUST-REG 0.40, incremental reweighting, recovery	0.876	22	0.55	0.823	15	0.54
Trimmed CWM 0.25, rw	0.804	39	0.56	0.682	32	0.53
Trimmed CWM 0.25, rw, recovery	0.882	12	0.66	0.679	31	0.53
Trimmed CWM 0.40, rw	0.745	66	0.54	0.786	24	0.61
Trimmed CWM 0.40, rw, recovery	0.871	20	0.64	0.785	24	0.62
Mixture of t regressions	0.887	12	0.38	0.710	25	0.37
RLGA, trimming 0.25, rw, recovery	0.859	24	0.63	0.679	32	0.53
RLGA, trimming 0.40, rw, recovery	0.845	31	0.60	0.712	33	0.54
TCLUST-REG 0.40, rw, recovery with robust noise range	0.885	15	0.60	0.863	8	0.53
ESF with robust noise range in the recovery step	0.892	10	0.62	0.820	13	0.53

Table S41: Plan 4 (Section 4.6 of the paper), fresh data: mean accuracy on clean units of ESF alone and of ESF, the paired gain of ESF, the best competitor chosen after the run, and the paired difference between ESF and that competitor (50 data sets per cell; standard errors in brackets).

Design	ε	ESF alone	ESF	Gain (s.e.)	Best competitor	Difference (s.e.)
groups 80/20	0.00	0.821	0.912	+0.091 (0.022)	contam. normal: 0.914	-0.002 (0.001)
groups 80/20	0.10	0.817	0.905	+0.088 (0.023)	TCLUST-REG (0.15): 0.910	-0.005 (0.004)
groups 80/20	0.20	0.730	0.889	+0.158 (0.026)	TCLUST-REG (0.25): 0.914	-0.026 (0.013)
groups 85/15	0.00	0.663	0.910	+0.247 (0.025)	contam. normal: 0.912	-0.002 (0.001)
groups 85/15	0.10	0.617	0.895	+0.278 (0.023)	contam. normal: 0.913	-0.018 (0.011)
groups 85/15	0.20	0.616	0.882	+0.266 (0.024)	noise comp.: 0.908	-0.026 (0.012)
80/20, three covariates	0.00	0.773	0.909	+0.136 (0.025)	bisquare: 0.913	-0.003 (0.001)
80/20, three covariates	0.10	0.732	0.912	+0.180 (0.025)	contam. normal: 0.915	-0.002 (0.001)
80/20, three covariates	0.20	0.675	0.830	+0.155 (0.025)	contam. normal: 0.908	-0.078 (0.021)
85/15, parallel lines	0.00	0.860	0.998	+0.139 (0.028)	TLE(0.10): 0.999	-0.000 (0.000)
85/15, parallel lines	0.10	0.721	0.999	+0.278 (0.029)	contam. normal: 0.999	-0.000 (0.000)
85/15, parallel lines	0.20	0.697	0.880	+0.183 (0.030)	TCLUST-REG (0.25): 0.999	-0.118 (0.026)

S6 Other settings of TCLUST-REG, further competitors, repeated runs, and the third part of the study

This section holds the detail behind Sections 4.4 to 4.6 and 5 of the paper. Throughout, ESF is the flagging procedure of Algorithm 1 of the paper, “rw” the reweighting rule of the paper (the map of its Section 3 applied twice, $c = 2.5$, a median absolute deviation per group), and the recovery step is Algorithm 2 of the paper.

S6.1 Other settings of TCLUST-REG

With three and four groups the level 0.40 loses part of a group. With equal group weights in the assignment (option `equalweights` of FSDA), the long search, reweighting and the recovery step, run on every data set, TCLUST-REG at this level reached 0.87 in D2 and 0.86 with four groups up to $\varepsilon = 0.20$, against 0.83 and 0.81 with estimated weights, and at level 0.25 it reached 0.90 and 0.96 (ESF: 0.89 and 0.95). Part of the loss thus comes from the estimated weights. Over all cells, equal weights changed the level 0.40 little and made the level 0.25 the most accurate pipeline up to 20% of outliers, but not beyond (Table 2 of the paper; Table S39). With equal scales (restriction factor 1; 300 starts, D2 and the four-group design only) the level 0.40 reached 0.86 and 0.77. Adaptive second-level trimming in the covariates (FSDA’s option $\alpha_X = 0.99$, run with 300 starts in D5 to D7 of the first two parts; the Octave run needed one more one-line substitute, for `any(x, 'all')`) did not help with high-leverage points: at $\varepsilon = 0.20$ it lowered the accuracy of TCLUST-REG with reweighting and the recovery step from 0.83 and 0.87 to 0.80 and 0.73 at levels 0.25 and 0.40, and from 25% of outliers it stayed at or below 0.63; in D6 it moved the level 0.25 by -0.025 at 30% and $+0.031$ at 35%, and in D7 it lowered the level 0.40 at 35% from 0.90 to 0.86. FSDA’s option $\alpha_X = 1$, which fits the cluster-weighted model, was not run; the trimmed cluster-weighted model of RobMixReg represents that model in the study. Nor did the monitoring path give a TCLUST-REG without a level. We fitted it at the levels 0, 0.05, $\dots$, 0.40 (300 starts) and compared consecutive classifications by the adjusted Rand index. Taking the first level at which two consecutive classifications agree (index at least 0.9) chose a level below the contamination (a median of 0.05 to 0.15 for $\varepsilon \geq 0.10$), and with reweighting and the recovery step fell below 0.85 in 36 of the 89 cells with $\varepsilon \leq 0.20$; taking the smallest level from which all later consecutive classifications agree nearly always chose 0.40 and behaved like that level (7 cells below 0.85, the worst 0.76 with four groups).

S6.2 Further competitors and checks

The following analyses were decided after the plans and run on every data set, and are descriptive (Table S39). The mixture of t regressions of Yao et al. (2014), with a common number of degrees of freedom and 20 starts, fails like the other mixtures with three and four groups (0.38 and 0.51 up to $\varepsilon = 0.20$) and with clustered outliers (0.54). Robust linear grouping (García-Escudero et al., 2009), which draws random subsamples as ESF does but fits orthogonal hyperplanes at a fixed level, often returned a nearly vertical hyperplane: at level 0.25 with reweighting and the recovery step its accuracy fell below 0.85 in 24 of the 89 cells with $\varepsilon \leq 0.20$. The incremental reweighting of Dotto et al. (2018), which lowers the trimming level from 0.40 to 0 in steps of 0.02 and refits at each step, did not do better than the single reweighting step after TCLUST-REG at level 0.40: it recovered part of the small groups (0.57–0.98 against 0.56–0.73 up to $\varepsilon = 0.20$), gained up to 0.03 with three and four groups without outliers and lost up to 0.13 with them at $\varepsilon = 0.20$, and fell below 0.85 in 32 cells, and in 22 after the recovery step against four. Two changes to our own constants were also tried. Taking the noise range of the recovery step over the unflagged units only, so that the outliers do not set it, lost small groups (down to 0.62 with groups of 85% and 15% at $\varepsilon = 0.20$) and did not prevent the harm with uniform noise described in Section 4.6 of the paper. Taking the scale of the scoring loss of ESF as the 10% quantile of the replicates’ median squared residuals instead of their minimum, on parts one and three, lowered the accuracy by up to 0.05 with high-leverage points and uniform noise at $\varepsilon = 0.20$, and did not remove the loss with $B = 500$.

S6.3 The comparison of Plan 1, the loss cap, repeated runs and the best-of-five rule

The comparison of Plan 1 used the seven designs without high-leverage points and the four levels $\varepsilon \leq 0.20$, 28 cells, and no cell failed: the largest shortfall of ESF alone against the best competitor was 0.029 (standard error 0.014), in D3 at $\varepsilon = 0.20$, and in 23 of the 28 cells it was below 0.005 (Table S1); for ESF with the recovery step (Plan 3) it was 0.017, in D3 at $\varepsilon = 0.20$ (0.016 in D6), with the long-search rows and the later competitors included, and that of ESF alone then becomes 0.031, against sequential RANSAC in D3 at $\varepsilon = 0.20$, of the order of the variation between random seeds. The second criterion of Plan 1 failed for TCLUST-REG at level 0.25 with reweighting, never more than 0.033 below ESF alone (D2, $\varepsilon = 0$; at most 0.012 with the long search), so this competitor was tested again, on new data, at higher contamination.

A larger loss cap is no cure for the small groups that ESF alone flags: with $c_b = 5$ the small groups are largely recovered, but the accuracy falls from 0.91 to 0.82 in D1 and from 0.88 to 0.74 in D7 at $\varepsilon = 0.35$, and from 0.90 to 0.79 in D2 and from 0.85 to 0.72 in D5 at $\varepsilon = 0.20$ (Table S13).

In five runs with different seeds on each data set of the study, the accuracies of ESF differed by more than 0.05 on 3.5% of the data sets with up to 20% of outliers and on 13% of all data sets, most of them where ESF breaks down. More replicates do not remove the small-group failure of ESF alone: with $B = 500$ the largest losses were with small groups at 30% of outliers (Table S29), because a wider search finds the fits with two lines on the large group more often, which the scoring criterion allows (Section 2.3 of the paper); with a small group, the recovery step is a necessity, not an option.

A default for real data (Section 2.5 of the paper). The study used $B = 100$ and one seed per data set. On the fishery data (Section 5.1 of the paper) the answer depended on the seed, and two remedies were examined there. Keeping, among five seeds, the fit of highest likelihood under the model of the recovery step returned the modal fit every time over twenty disjoint sets of seeds; on the simulated data it raised cell means by more than 0.005 in 27 of 102 cells and never lowered one by more than 0.001 up to 20% of outliers, but beyond it lowered one cell, three groups at 35%, by 0.096 (Table S24). More replicates also reduced the dependence on the fishery data, where with $B = 100$, 500 and 1000 the modal fit was returned by 17, 18 and 19 of 20 seeds with $K = 2$ and by 14, 18 and 19 with $K = 3$, but on the simulated data $B = 500$ moved cell means by -0.07 to $+0.03$ (Section 2.3 of the paper; Table S29). The default for real data is therefore what the study evaluated, $B = 100$ with the best of five seeds by that likelihood (a rule decided after the analysis plans), except when a group is fitted exactly, as the flat tariff of Section 5.3 of the paper: the likelihood then rewards any line through tied points, the rule chose a wrong fit once, and the seeds should be compared by eye. A larger B is an option that helped on the fishery data and not uniformly in the simulation. The flagged fraction $\hat{\alpha}$ is the check on the result. A value well below any plausible contamination is the sign that the outliers have been absorbed into a fitted line and the fit has broken down (Section 4.3 of the paper); the remedy is a trimming level above the contamination with reweighting, followed by the recovery step. A value well above the plausible contamination, as on the tone and taxi data, means that the errors are not close to Gaussian or that a group's covariates are compact: the fitted lines were unaffected in both cases, but $\hat{\alpha}$ is then not an estimate of the contamination.

S6.4 The third part of the study and Plan 5, detail

The third, descriptive part of the study (2,300 data sets, 50 per cell) varied what the first two held fixed: the sample size (D1 with $n = 100$ and $n = 1000$, D3 with $n = 1000$); the number of groups (four parallel lines four noise scales apart, $n = 400$, $m = 16$); the covariates (D7 with dependent covariates); the outliers (D1 and D3 with moderate outliers shifted by 4 to 8 noise scales; D1 with the outliers in a tight cloud); the group sizes (D1 with groups of 80% and 20%, $m = 16$, and of 85% and 15%, $m = 20$); and the errors (D1 and D3 with t_3 errors scaled to unit variance, at $\varepsilon \in \{0, 0.10, 0.20\}$, added after Plan 3). Table S25 gives the results cell by cell. The sample size changes little: with $n = 100$ and $n = 1000$ ESF stays within 0.014 of the best competitor chosen afterwards in every cell up to $\varepsilon = 0.30$. With four groups and outliers it is the best method with estimated group weights,

0.96 and 0.95 at $\varepsilon = 0.10$ and 0.20 against 0.93 and 0.91 for TCLUST-REG at level 0.25, while at level 0.40 TCLUST-REG stays between 0.81 and 0.84 even without outliers; at $\varepsilon = 0.30$ it breaks down (0.72), where no method with estimated weights does better than 0.77 and TCLUST-REG at level 0.40 with equal weights, reweighting and the recovery step reaches 0.81. Dependent covariates make no difference. Moderate outliers are harder to flag, so the flagged fraction falls short of ε , by up to thirteen percentage points at $\varepsilon = 0.30$, but ESF stays within 0.01 of the best competitor everywhere except D3 at $\varepsilon = 0.30$, where it trails TCLUST-REG at level 0.25 by 0.04.

Small groups are what the recovery step is for. ESF alone reaches only 0.82 without outliers and 0.73 at $\varepsilon = 0.20$ with groups of 80% and 20%, and between 0.60 and 0.73 at every ε with groups of 85% and 15%; its failed fits are two nearly parallel lines that share the large group, with most of the small group flagged, which the bounded loss allows, since a unit of the small group costs at most c_b^2 squared scales. A larger cap is no cure: with $c_b = 5$ the small groups are largely recovered, but the accuracy falls by up to 0.13 elsewhere (Table S13; Section S6.3). With the recovery step ESF reaches 0.88 to 0.91 with either small group up to $\varepsilon = 0.20$, within 0.035 of the best competitor in every such cell; at $\varepsilon = 0.30$ the first step flags more than a quarter of the units, the step rarely applies, and the accuracy stays at 0.75.

Heavy-tailed errors leave the accuracy intact: with t_3 errors ESF is within 0.006 of the best competitor in all six cells, but the flagged fraction then measures the tails of the errors as well as the outliers, 6.5% of the units without outliers. Clustered outliers are the reverse case. At $\varepsilon = 0.10$ the cloud is flagged; at $\varepsilon = 0.20$, ESF alone fits the cloud with one of its two lines, and its accuracy falls to 0.70; the recovery step repairs 11 of the 27 fits that fail in this way and raises the mean to 0.78, against 0.90 for TCLUST-REG at a level above the contamination. At $\varepsilon = 0.30$ no method of Table 3 of the paper exceeds 0.72. Plan 5 added three designs at $\varepsilon \in \{0, 0.10, 0.20\}$, 50 data sets per cell: D1 with skewed errors (centred and scaled χ_3^2), groups of 70% and 30% with three covariates ($m = 16$), and D1 with an error scale growing with the covariate, from 0.5 at $x = 0$ to 2 at $|x| = 3$. All nine cells passed the comparison of Plan 1, ESF being within 0.006 of the best competitor everywhere (Table S33); the flagged fraction exceeded ε by 1 to 6 percentage points with skewed or heteroscedastic errors.

S6.5 The fishery figure, the fishery and taxi tables, and the constants

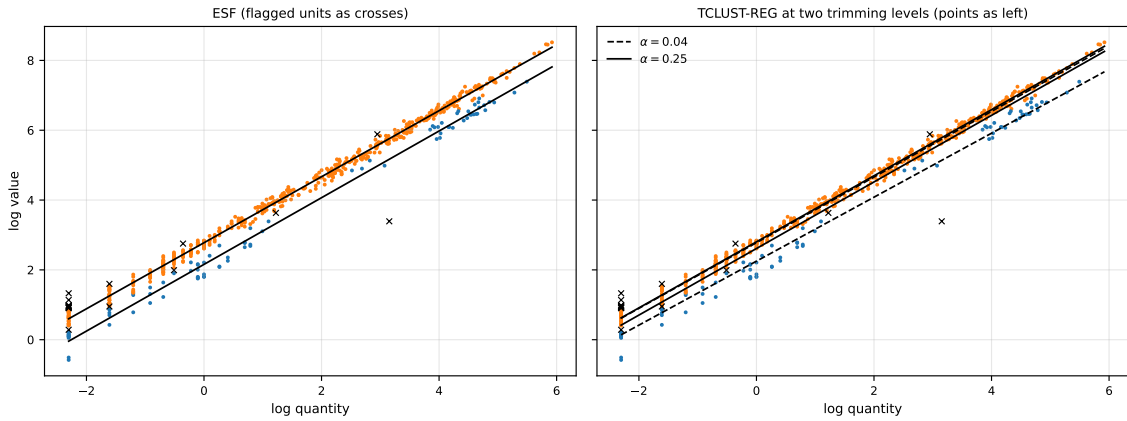


Figure S1: Fishery data (Section 5.1 of the paper) on the logarithmic scale. Left: the two lines of ESF with the recovery step (seed 0), with its flagged flows as crosses and the other flows coloured by group. Right: TCLUST-REG at trimming levels 0.04 (dashed) and 0.25 (solid), with the same points.

On the taxi data (Table S43), TLE at levels 0.10 and 0.15 and CTLE stopped with an error on 18, 19 and 13 of the 20 samples; when they returned a fit, it was as accurate as the best (0.98). The Laplace mixture fell below 0.7 on 14 samples, and least squares on all.

Table S42: Fishery data (Section 5.1 of the paper), logarithmic scale, $K = 2$: estimated intercepts and slopes, median over random seeds or starts, with the range in brackets when it exceeds 0.005, and the trimmed or flagged fraction.

Method	Group 1: intercept, slope		Group 2: intercept, slope		Trimmed or flagged
ESF (20 seeds)	2.16	[2.16, 2.58], 0.96 [0.95, 0.98]	2.78	[2.78, 2.85], 0.94	0.03 [0.02, 0.12]
ESF alone (20 seeds)	2.57	[2.56, 2.59], 0.98 [0.97, 0.98]	2.85	[2.84, 2.86], 0.95 [0.94, 0.95]	0.11 [0.10, 0.12]
TCLUST-REG, $\alpha = 0.02$ (10 seeds)		2.21, 0.93		2.78, 0.94	0.02
TCLUST-REG, $\alpha = 0.04$ (10 seeds)		2.25, 0.92		2.78, 0.94	0.04
TCLUST-REG, $\alpha = 0.06$ (10 seeds)		2.24, 0.92		2.78, 0.94	0.06
TCLUST-REG, $\alpha = 0.08$ (10 seeds)		2.22 [2.22, 2.24], 0.92		2.78, 0.94	0.08
TCLUST-REG, $\alpha = 0.10$ (10 seeds)	2.31	[1.91, 2.37], 0.94 [0.90, 0.99]		2.77, 0.95	0.10
TCLUST-REG, $\alpha = 0.15$ (10 seeds)	2.50	[1.81, 2.78], 0.94 [0.38, 0.95]	2.79	[2.78, 2.84], 0.94 [0.79, 0.95]	0.15
TCLUST-REG, $\alpha = 0.20$ (10 seeds)	2.62	[2.56, 2.78], 0.95 [0.81, 1.07]	2.78	[2.78, 2.80], 0.95 [0.81, 0.95]	0.20
TCLUST-REG, $\alpha = 0.25$ (10 seeds)	2.61	[2.55, 2.79], 0.96 [0.87, 1.01]	2.79	[2.79, 2.87], 0.94 [0.84, 0.95]	0.25
TCLUST-REG, $\alpha = 0.30$ (10 seeds)	2.63	[2.55, 2.67], 0.95 [0.93, 1.03]	2.80	[2.78, 2.82], 0.94 [0.94, 0.95]	0.30
CTLE (10 seeds)		2.34, 0.94		2.79, 0.94	0.02
Trimmed CWM, $\alpha = 0.04$ (10 seeds)	2.62	[2.28, 2.71], 0.89 [0.72, 0.97]	2.80	[2.78, 2.80], 0.94 [0.82, 0.94]	0.04
Trimmed CWM, $\alpha = 0.10$ (10 seeds)	2.65	[2.56, 2.75], 0.96 [0.90, 1.00]	2.78	[2.76, 2.85], 0.94 [0.83, 0.96]	0.10
Trimmed CWM, $\alpha = 0.25$ (10 seeds)	2.73	[2.57, 2.78], 0.96 [0.91, 1.00]	2.79	[2.78, 2.84], 0.95 [0.94, 0.96]	0.25

Table S43: Taxi trips from JFK airport (Section 5.3 of the paper), 20 random samples of 2,000 trips, $K = 2$: mean and worst accuracy on the trips with rate codes 1 and 2, median coefficient error against the flat fare (70, 0, 0) and a robust least-squares fit to all metered trips of the month, mean flagged or trimmed fraction, and median time. A failed fit counts as accuracy 0. “Reweighted” is the rule of the paper, not that of Dotto et al. (2018).

Method	Accuracy, mean	Accuracy, worst	Coefficient error	Flagged or trimmed	Time (s)
ESF	0.978	0.974	0.23	0.23	31
ESF alone	0.978	0.974	0.23	0.23	24
TCLUST-REG (long search), $\alpha = 0.25$, reweighted, then the recovery step	0.979	0.975	0.10	0.14	86
TCLUST-REG (long search), $\alpha = 0.40$, reweighted, then the recovery step	0.978	0.973	0.14	0.14	54
TCLUST-REG (long search), $\alpha = 0.25$, reweighted	0.979	0.975	0.10	0.14	86
TCLUST-REG (long search), $\alpha = 0.40$, reweighted	0.978	0.973	0.14	0.14	54
TCLUST-REG, $\alpha = 0.05$	0.972	0.959	0.98	0.05	7.30
TCLUST-REG, $\alpha = 0.10$	0.978	0.975	0.06	0.10	5.11
TCLUST-REG, $\alpha = 0.15$	0.978	0.975	0.09	0.15	4.35
TCLUST-REG, $\alpha = 0.25$	0.977	0.967	0.21	0.25	4.10
TCLUST-REG, $\alpha = 0.40$	0.975	0.965	0.48	0.40	3.02
TCLUST-REG, $\alpha = 0.25$, reweighted	0.979	0.975	0.10	0.14	4.10
TCLUST-REG, $\alpha = 0.40$, reweighted	0.978	0.970	0.13	0.14	3.02
Trimmed CWM, $\alpha = 0.05$	0.905	0.592	1.82	0.05	12
Trimmed CWM, $\alpha = 0.10$	0.978	0.975	0.06	0.10	12
Trimmed CWM, $\alpha = 0.15$	0.978	0.974	0.15	0.15	12
Trimmed CWM, $\alpha = 0.25$	0.978	0.970	0.20	0.25	11
Trimmed CWM, $\alpha = 0.40$	0.977	0.967	0.32	0.40	12
TLE, $\alpha = 0.05$	0.567	0.000	17.74	0.05	58
TLE, $\alpha = 0.10$	0.098	0.000	0.12	0.10	6.77
TLE, $\alpha = 0.15$	0.049	0.000	0.16	0.15	8.41
TLE, $\alpha = 0.25$	0.700	0.000	64.30	0.25	61
TLE, $\alpha = 0.40$	0.686	0.593	66.26	0.40	53
CTLE	0.342	0.000	0.24	0.09	80
Bisquare mixture	0.979	0.975	0.08	0.00	5.41
Laplace mixture	0.676	0.534	65.37	0.00	1136
Contaminated normal mixture	0.978	0.974	0.12	0.09	0.29
Noise-component mixture	0.978	0.974	0.23	0.08	0.04
Least squares (50 starts)	0.589	0.571	6.72	–	0.07

Table S44: Constants of ESF and of the recovery step (Section 2.5 of the paper), their values, the range examined, and the largest change in a cell’s mean accuracy over that range, up to 20% of outliers unless stated. For the loss cap the changes are relative to $c_b = 3$. S: table of this Supplement.

Constant	Value	Range examined	Change	Evidence
<i>ESF (Algorithm 1 of the paper)</i>				
subsample size m	$(d+1)/\pi_{\min}$ with room	up to 16–20; $m = 12$ everywhere	0.02; 0.035 from 25%; 0.025 (small groups); 0.07 with four groups ($m = 12$ for 16)	S12, S15, S28, S39
replicates B , stages L	100, 10	$B = 500$; 500, 1000 (fishery)	–0.02 to +0.03; –0.07 at 30%	S29; Section 5.1 of the paper
cut-off c	2.5	2 to 3	0.024; 0.04 at 30%	S3
loss cap c_b	3	3 to 5	0.13 lost at 5 (D5); 0.24 gained with a group of 15%	S13
vote weight λ	3	0, ∞	0.05 at $\lambda = 0$	S2
covariate quantile	0.975	0.95 to 0.99	0.007; 0.015 at 30%; 0.21 with leverage points	S3
minimum group for screen	max(10, 5p)	–	–	–
<i>Recovery step (Algorithm 2 of the paper)</i>				
peak weight	$\frac{1}{2}$	0.4, 0.6	0.015	S23
window	3c scales	2c, 4c	0.03	S23
minimum support	max{2(d+1), 0.03n}	0.02n, 0.05n	0.026	S23
restriction factor	12	8, 20	0.016	S23
margin	$\frac{d+1}{2} \log n$	halved, doubled	0.016	S23
limit on flagged fraction	$n/3$	0.25n, 0.40n	0.20 at 0.25n	S23
noise range, random lines, applications	1.1 range(y), 500, 3	–	–	–

Table S45: Criteria that count flagged or trimmed clean units (descriptive; Section 4.1 of the paper), over the same 123 cells as Table 2 of the paper: the adjusted Rand index on all units, with the outliers as a class of their own and every flagged or trimmed unit labelled as an outlier, and the misclassification rate of the clean units, counting a flagged clean unit as an error (the labels matched to the true groups by the best permutation). Flags are those of the reweighting rule ($c = 2.5$), with the covariate screen for ESF and without it for TCLUST-REG. Mean over cells of the cell means, and the worst cell. rw: the reweighting step; recovery: followed by the recovery step.

Method	$\varepsilon \leq 0.20$ (89 cells)				$\varepsilon \geq 0.25$ (34 cells)			
	ARI mean	ARI worst	Misc. mean	Misc. worst	ARI mean	ARI worst	Misc. mean	Misc. worst
ESF	0.719	0.54	0.102	0.29	0.660	0.01	0.183	0.48
TCLUST-REG, $\alpha = 0.25$, rw, recovery	0.725	0.57	0.096	0.17	0.519	0.00	0.308	0.49
TCLUST-REG, $\alpha = 0.40$, rw, recovery	0.715	0.56	0.104	0.19	0.714	0.01	0.138	0.48
TCLUST-REG, $\alpha = 0.25$, rw	0.694	0.26	0.127	0.45	0.521	0.00	0.306	0.49
TCLUST-REG, $\alpha = 0.40$, rw	0.624	0.24	0.201	0.52	0.720	0.02	0.133	0.47

Table S46: The recovery step after the trimmed cluster-weighted model (descriptive; Section 4.6 of the paper): mean over cells of the mean accuracy on clean units, the number of cells with a mean below 0.85, and the worst cell, over the same 123 cells as Table 2 of the paper. The model is fitted by RobMixReg at levels 0.25 and 0.40 (50 starts), then reweighted by our rule (rw) and followed by the recovery step.

Method	$\varepsilon \leq 0.20$ (89 cells)			$\varepsilon \geq 0.25$ (34 cells)		
	Mean	Below 0.85	Worst	Mean	Below 0.85	Worst
Trimmed CWM, $\alpha = 0.25$, alone	0.786	47	0.57	0.677	32	0.53
Trimmed CWM, $\alpha = 0.25$, rw	0.804	39	0.56	0.682	32	0.53
Trimmed CWM, $\alpha = 0.25$, rw, recovery	0.882	12	0.66	0.679	31	0.53
Trimmed CWM, $\alpha = 0.40$, alone	0.716	77	0.56	0.759	28	0.58
Trimmed CWM, $\alpha = 0.40$, rw	0.745	66	0.54	0.786	24	0.61
Trimmed CWM, $\alpha = 0.40$, rw, recovery	0.871	20	0.64	0.785	24	0.62

Table S47: The five analysis plans (Section 4.1 of the paper): what each fixed before the data it concerns were analysed (Plans 1 and 2) or generated (Plans 3 to 5), and the data it concerns. The third part of the study (2,300 data sets) is descriptive and had no plan.

Plan	What it fixed	Data it concerns
1	ESF alone against the best competitor chosen after the run in each cell: mean paired difference not below -0.03 , upper 95% bound not below -0.02 ; every fixed-level competitor to fall 0.05 below ESF alone somewhere	First part: D1 to D8, $\varepsilon \in \{0, 0.05, 0.10, 0.20, 0.30\}$, 50 data sets per cell, 2,000; the comparison concerns $\varepsilon \leq 0.20$
2	Whether TCLUST-REG at level 0.25 with reweighting fails above its level while ESF alone does not (gap of 0.05, interval excluding zero, in some design without high-leverage points)	Second part: $\varepsilon \in \{0.25, 0.30, 0.35\}$, new data, 100 per cell, 2,400
3	The recovery step, code frozen: gain of at least 0.05 in the small-group cells, no loss above 0.01 at $\varepsilon \leq 0.20$, the comparison of Plan 1 for ESF with the step	The stored fits of ESF alone on the 6,400 data sets of the first three parts
4	The same confirmation on fresh data	600 new data sets in four small-group designs
5	The comparison of Plan 1 in three designs outside the Gaussian family	450 new data sets

Table S48: Sensitivity of the recovery step to its noise range (descriptive; Section 4.6 of the paper): the range R , 1.1 times the range of the response, multiplied by 2 and 5, as if the most extreme unit lay two or five times farther out, and replaced by two ranges tied to the fitted scales (20 median group scales; the range of the unflagged units plus 10 median scales); mean over cells of the mean accuracy on clean units, the number of cells with a mean below 0.85, and the worst cell, over the same 123 cells as Table 2 of the paper. Fits and seeds as in the study.

Method	$\varepsilon \leq 0.20$ (89 cells)			$\varepsilon \geq 0.25$ (34 cells)		
	Mean	Below 0.85	Worst	Mean	Below 0.85	Worst
ESF, range R as in the paper	0.914	2	0.78	0.820	13	0.53
ESF, range $2R$	0.912	2	0.69	0.812	15	0.53
ESF, range $5R$	0.910	3	0.68	0.799	16	0.53
ESF, range 20 median scales	0.891	12	0.62	0.819	13	0.53
ESF, unflagged range plus 10 median scales	0.903	9	0.70	0.817	13	0.53
TCLUST-REG 0.40, rw, recovery, range R	0.909	4	0.81	0.863	8	0.53
TCLUST-REG 0.40, rw, recovery, range $2R$	0.908	5	0.81	0.846	10	0.54
TCLUST-REG 0.40, rw, recovery, range $5R$	0.906	7	0.75	0.830	12	0.54
TCLUST-REG 0.40, rw, recovery, 20 median scales	0.883	15	0.61	0.859	9	0.54
TCLUST-REG 0.40, rw, recovery, unflagged range plus 10 scales	0.899	10	0.73	0.853	9	0.54

Table S49: TCLUST-REG’s own assignment against the criterion of the paper (descriptive; Section 4.1 of the paper): mean accuracy on clean units of the long-search fits at levels 0.25 and 0.40 (without reweighting) when every clean unit takes the label of its nearest fitted line, as in the paper, and when it takes the label that TCLUST-REG assigns (`out.idx`), trimmed clean units then taking that of their nearest line; and the fraction of clean units trimmed. Same fits and data sets as the study.

Design	ε	$\alpha = 0.25$			$\alpha = 0.40$		
		Nearest line	Own assignment	Clean trimmed	Nearest line	Own assignment	Clean trimmed
D2	0.00	0.876	0.868	0.250	0.794	0.792	0.400
D2	0.05	0.883	0.874	0.213	0.830	0.829	0.371
D2	0.10	0.891	0.878	0.165	0.841	0.837	0.332
D2	0.20	0.883	0.861	0.065	0.840	0.832	0.251
D4	0.00	0.911	0.917	0.250	0.908	0.916	0.400
D4	0.05	0.908	0.914	0.210	0.905	0.911	0.368
D4	0.10	0.909	0.918	0.167	0.907	0.912	0.333
D4	0.20	0.907	0.925	0.066	0.905	0.911	0.252
D2	0.25	0.756	0.735	0.018	0.857	0.847	0.195
D2	0.30	0.618	0.604	0.005	0.859	0.838	0.141
D2	0.35	0.507	0.497	0.002	0.834	0.801	0.076
D4	0.25	0.875	0.896	0.013	0.911	0.917	0.199
D4	0.30	0.617	0.621	0.004	0.910	0.918	0.140
D4	0.35	0.530	0.531	0.003	0.911	0.926	0.072
$K = 4$	0.00	0.925	0.929	0.250	0.792	0.809	0.400
$K = 4$	0.10	0.932	0.932	0.168	0.794	0.809	0.334
$K = 4$	0.20	0.896	0.884	0.068	0.795	0.805	0.256
$K = 4$	0.30	0.748	0.749	0.004	0.758	0.756	0.150

Table S50: The eleven cells that decide the comparison up to 20% of outliers, with 200 data sets each: the 50 of the study and 150 further ones from the same generators (descriptive; Section 4.2 of the paper). Mean accuracy on clean units of ESF with its standard error, and, for TCLUST-REG with the long search, reweighting and the recovery step at level 0.25, at level 0.30 and at level 0.25 with equal weights, the mean and the paired difference ESF minus that pipeline with its standard error.

Design	ε	n_{rep}	ESF	0.25		0.30		0.25, equal weights	
			mean (s.e.)	mean	ESF – (s.e.)	mean	ESF – (s.e.)	mean	ESF – (s.e.)
D2	0.00	200	0.905 (0.001)	0.889	+0.016 (0.003)	0.876	+0.029 (0.005)	0.897	+0.008 (0.002)
D2	0.05	200	0.903 (0.002)	0.893	+0.010 (0.003)	0.886	+0.017 (0.004)	0.899	+0.004 (0.002)
D2	0.10	200	0.899 (0.003)	0.899	-0.000 (0.003)	0.889	+0.009 (0.004)	0.900	-0.002 (0.003)
D2	0.20	200	0.904 (0.003)	0.891	+0.012 (0.004)	0.901	+0.003 (0.003)	0.906	-0.003 (0.002)
$K = 4$	0.00	200	0.961 (0.002)	0.929	+0.032 (0.006)	0.901	+0.060 (0.007)	0.963	-0.002 (0.002)
$K = 4$	0.10	200	0.959 (0.002)	0.923	+0.036 (0.006)	0.898	+0.061 (0.007)	0.962	-0.003 (0.002)
$K = 4$	0.20	200	0.945 (0.005)	0.901	+0.044 (0.008)	0.888	+0.058 (0.008)	0.955	-0.010 (0.005)
80/20	0.20	200	0.883 (0.007)	0.910	-0.027 (0.007)	0.909	-0.026 (0.007)	0.909	-0.026 (0.007)
clustered	0.20	200	0.810 (0.012)	0.904	-0.094 (0.012)	0.909	-0.100 (0.012)	0.910	-0.100 (0.012)
80/20, $p = 3$	0.20	200	0.850 (0.009)	0.907	-0.057 (0.009)	0.896	-0.046 (0.010)	0.900	-0.050 (0.009)
85/15, parallel	0.20	200	0.880 (0.013)	0.967	-0.088 (0.011)	0.962	-0.083 (0.011)	0.964	-0.085 (0.011)

S7 Implementation of ESF

Exact subsample fits. Label vectors are enumerated depth first in restricted-growth form: unit j may receive any label already used by units $1, \dots, j-1$ or the smallest unused one. This lists each partition into at most K groups exactly once and removes the $K!$ relabellings. Each group keeps its sufficient statistics $(n_k, X_k^\top X_k, X_k^\top y_k, y_k^\top y_k)$, updated in $O(d^2)$ when a unit enters or leaves, and its residual sum of squares, recomputed by a Cholesky solve of size d . For rank-deficient groups the minimum-norm solution is used; its residual sum of squares is the minimum over all coefficient vectors, so the group criterion is exact in every case. A group’s residual sum of squares cannot decrease when a unit is added, so the sum over groups at a partial labelling is a lower bound for every completion, and a branch is abandoned when the bound reaches the best value found. The cost of the search is reported in Section 2.2 of the paper and in Table S30.

The code was checked against plain enumeration of all K^m label vectors, with residual sums of squares computed independently by a least-squares routine, on 200 instances with (K, d, m) equal to $(2, 1, 10)$, $(2, 2, 10)$, $(3, 2, 8)$ or $(2, 3, 9)$, a third of them with tied covariates.

Scoring scale. The scale s of the bounded loss (Algorithm 1 of the paper, line 5) is the smallest, over the replicates, of the median squared residual over all unflagged units divided by 0.4549, the median of χ_1^2 : for a replicate that fits the clean majority this is a rough estimate of the error variance. The minimum is biased downwards by design: the cap should follow the best replicate, not a typical one.

Alignment and vote. Replicates are aligned before voting by solving an assignment problem between their coefficients and those of a common reference, by enumerating the $K!$ relabellings for $K \leq 6$, which covers every experiment here (the Hungarian method (Kuhn, 1955) is used beyond). The reference is a pilot replicate, the best of ten preliminary replicates by the median of the squared residuals; it serves only for alignment and does not vote. After the vote, a group with fewer than d unflagged units keeps the coefficients of the best-scoring replicate b^* of the last stage. The final reweighting step recomputes the flags at the current fit and refits each group on its unflagged units.

Concentration steps. Every unit is assigned to its nearest surface, the h units with the smallest squared residuals are kept, and each group is refitted on its kept units; this is repeated until the trimmed sum of squares stops decreasing, at most 200 times.

Software. ESF and the recovery step are written in Python with `numpy`, `scipy` and `numba`, and the covariate screen uses the minimum covariance determinant of `scikit-learn` when there is more than one covariate. The code, the scripts that produce every table and figure, and the raw results are at <https://github.com/sorujov/eesp>.

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